

3.1 General properties of waves

Paper 2 past questions year by year, with figures and mark schemes. 0625 IGCSE and 5054 O Level. Only 3.1: energy transfer, wave terms, $v = f\lambda$, transverse / longitudinal, reflection, refraction, diffraction, ripple tank.

2004

Oct/Nov 5054/2 Theory

Question 5

- 5 Fig. 5.1 shows part of a long, thin spring used to demonstrate a transverse wave.

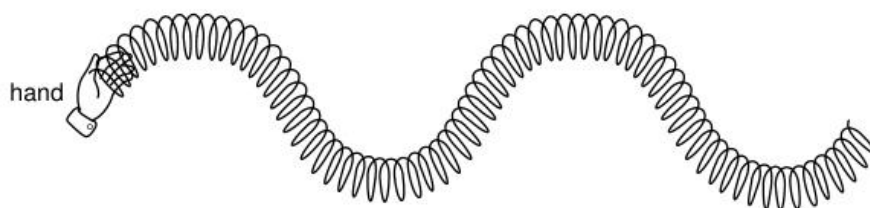


Fig. 5.1

The wave shown in Fig. 5.1 has a frequency of 4.0 Hz.

- (a) (i) On Fig. 5.1, mark the direction the hand must move to make a transverse wave.
(ii) Describe how the hand must move to make a transverse wave of frequency 4.0 Hz.

.....
.....
.....

[2]

- (b) The speed of the wave is 0.80 m/s. Calculate its wavelength.
State clearly the formula that you use.

wavelength = [3]

- (c) State what must be done to double the wavelength of the wave on the spring.

.....
.....
.....[1]

Mark scheme — MS not in pack

2005

Oct/Nov 5054/2 Theory

Question 3

- 3 A student produces wavefronts in a ripple tank to demonstrate refraction, as shown in Fig. 3.1. He places a sheet of glass under the water on the right-hand side of the tank. The arrows show the directions of movement of the wavefronts.

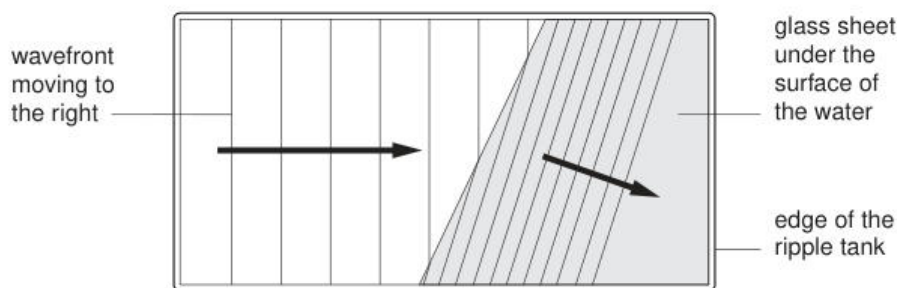


Fig. 3.1

- (a) State what is meant by a *wavefront*.

.....

 [1]

- (b) State what happens to each of the following quantities as the wavefronts change direction.

(i) wavelength
 [1]

(ii) speed
 [1]

(iii) frequency
 [1]

Mark scheme — MS not in pack

2015

Feb/March 0625/22 Theory

Question 1

- 1 A coconut falls from position X to position Y, as shown in Fig. 1.1.

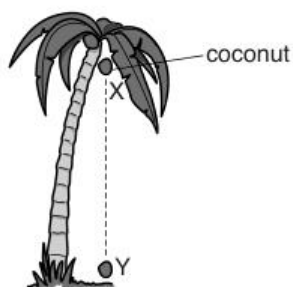


Fig. 1.1

- (a) State what causes the coconut to accelerate.

.....[1]

- (b) Compare the acceleration at positions X and Y. Tick one box.

☐

much greater at X than at Y

☐

much greater at Y than at X

☐

about the same at X as at Y

[1]

- (c) Using the axes in Fig. 1.2, sketch the graph showing how the speed of the coconut changes as it falls from X to Y. [2]



Fig. 1.2

- (d) State how the distance XY is determined from your graph in (c).

.....[1]

[Total: 5]

Question 6

- 6 Two students A and B are carrying out an experiment to determine the speed of sound.

They are standing side by side at a distance of 480 m from the school wall, as shown in Fig. 6.1.

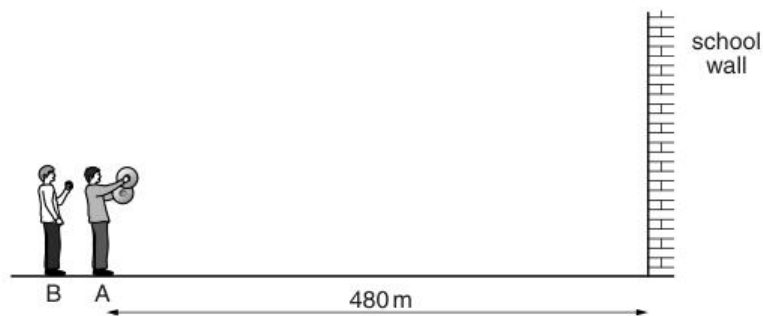


Fig. 6.1 (not to scale)

Student A makes a loud sound by banging two pan lids together once. A short time later, both students hear the sound again.

- (a) State what causes this second sound.

.....[1]

- (b) Underline the phrases that correctly complete the sentences below.

The second sound is $\left\{ \begin{array}{l} \text{louder than} \\ \text{equally as loud as} \\ \text{quieter than} \end{array} \right\}$ the original sound.

This is because the $\left\{ \begin{array}{l} \text{amplitude} \\ \text{frequency} \\ \text{speed} \\ \text{wavelength} \end{array} \right\}$ of the sound has $\left\{ \begin{array}{l} \text{changed.} \\ \text{not changed.} \end{array} \right\}$

[2]

(c) The time between the two sounds, as measured by student B, is 3.0 s.

(i) Suggest the instrument that student B used to measure the time.

.....[1]

(ii) Use the measurements to calculate the speed of sound.

speed = m/s [3]

[Total: 7]

Question 7

- 7 The electric circuit in Fig. 7.1 consists of a battery, two lamps and a switch. A voltmeter is connected across one of the lamps.

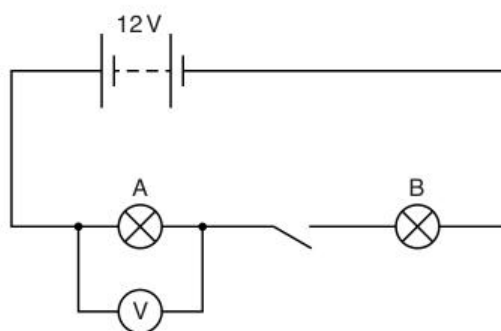


Fig. 7.1

- (a) Which quantity does the voltmeter measure? Tick one box.

- | | |
|--------------------------|----------------------|
| ☐ | charge |
| ☐ | current |
| ☐ | potential difference |
| ☐ | power |

[1]

- (b) The switch is closed so that there is a current in the lamps.

- (i) What flows in the circuit in order to create the current? Tick one box.

- | | |
|--------------------------|----------------------|
| ☐ | charge |
| ☐ | potential difference |
| ☐ | power |
| ☐ | resistance |

[1]

- (ii) The resistance of each lamp is 18Ω .

Calculate

1. the combined resistance of the lamps,

combined resistance = Ω [1]

2. the current in the circuit, stating the unit in your answer.

current = [4]

- (iii) Calculate the value of the reading on the voltmeter.

voltmeter reading = V [2]

- (c) Suggest one disadvantage of connecting lamps in series rather than in parallel.

.....

..... [1]

[Total: 10]

Mark scheme — See official MS

May/June 0625/23 Theory

Question 9

- 9 (a) Fig. 9.1 represents a wave in a water tank. Five distances are shown, labelled A, B, C, D and E.

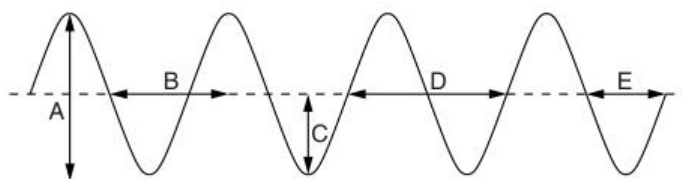


Fig. 9.1

- (i) 1. Which property of the wave is represented by distance C?
[1]

2. Which of the labelled distances represents the wavelength of the wave?
[1]

- (ii) Water waves are transverse waves.

State another example of a transverse wave.

.....[1]

- (iii) The speed of the wavefronts is 7.5 cm/s.

Calculate the distance moved by a wavefront in 4.0 s.

distance = cm [2]

(b) Fig. 9.2 shows, from above, a wave travelling on water.

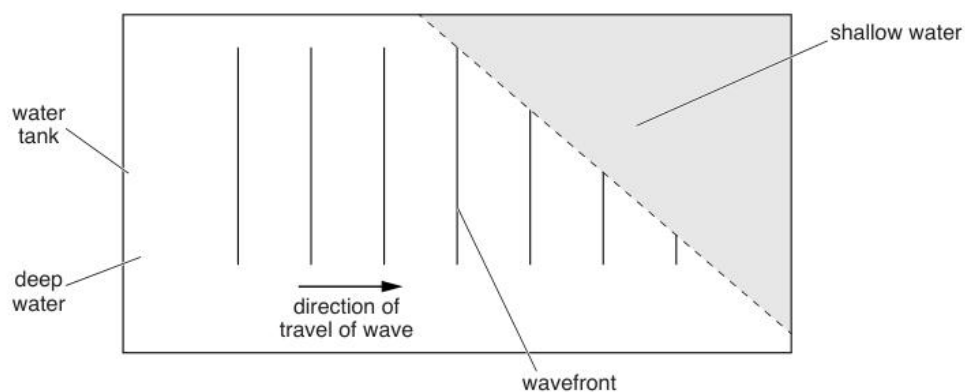


Fig. 9.2

The wave has reached an area of shallow water, where it travels more slowly.

(i) Complete the diagram to show the wavefronts in the shallow water. [3]

(ii) State the term used to describe what happens to the wave as it enters the shallow water.

.....[1]

[Total: 9]

Question 13



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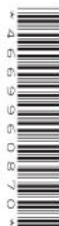
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NUMBER

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PHYSICS

Paper 2 Core

0625/23

May/June 2015

1 hour 15 minutes

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

Electronic calculators may be used.

You may lose marks if you do not show your working or if you do not use appropriate units.

Take the weight of 1 kg to be 10 N (i.e. acceleration of free fall = 10 m/s^2).

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

The syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **19** printed pages and **1** blank page.

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[Turn over]

Mark scheme — See official MS

May/June 5054/21 Theory

Question 4

- 4 Fig. 4.1 shows a wave on a string. The wave is travelling towards the right.

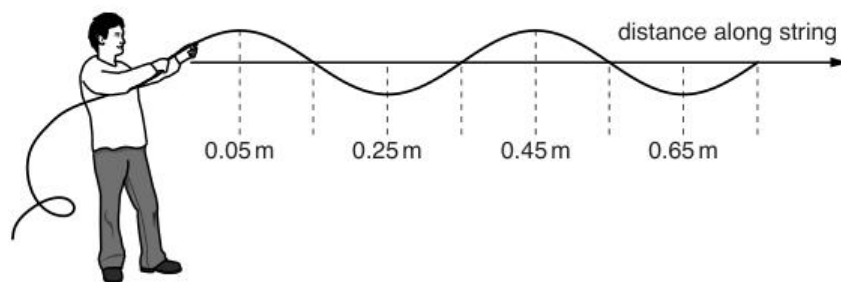


Fig. 4.1

- (a) Describe the movement of the particles of the string.

.....

.....

.....

.....[2]

- (b) Determine the wavelength of the wave.

wavelength =[1]

- (c) The speed of the wave is 2.0 m/s.

- (i) Calculate the frequency of the wave.

frequency =[2]

- (ii) On Fig. 4.1, draw the string at a time 0.10 s later than the time in Fig. 4.1. [1]

Question 9

- 9 Fig. 9.1 shows the wavefronts of a water wave in deep water in a ripple tank.

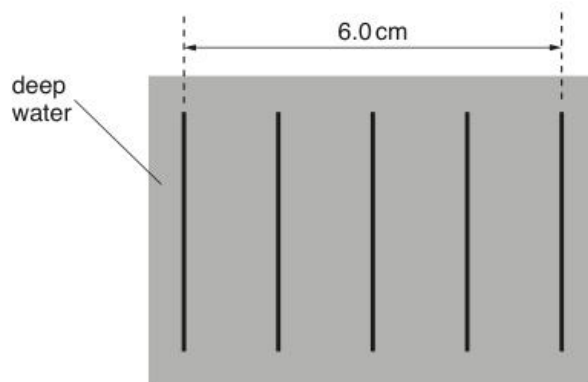


Fig. 9.1

The frequency of the wave in deep water is 5.0 Hz.

- (a) Explain what is meant by the *frequency* of a wave.

.....

 [2]

- (b) (i) Determine the wavelength of the wave in deep water.

..... [1]

- (ii) Calculate the speed of the wave in deep water.

speed = [2]

- (c) The wave passes from deep water into shallow water. The speed of the wave is less in shallower water.

- (i) State and explain how this affects the wavelength of the wave.

.....

 [2]

- (ii) The wave in deep water shown in Fig. 9.2 travels towards the right and enters the shallow water at an angle. The wave refracts.

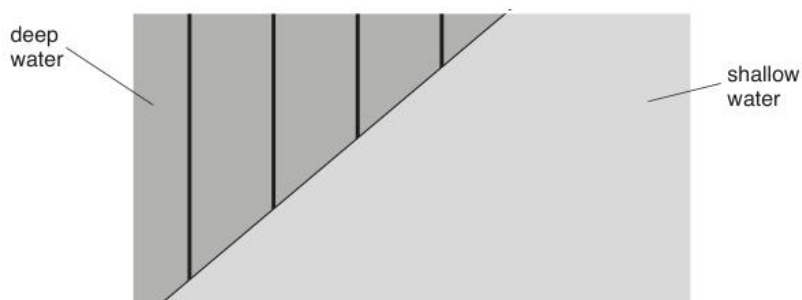


Fig. 9.2

On Fig. 9.2, draw the wavefronts in the shallow water.

[3]

- (d) Sound is also a wave.

- (i) Describe one difference between a sound wave and a water wave.

.....

 [2]

- (ii) The speed of sound in carbon dioxide gas is less than the speed of sound in air. Using this information, or otherwise, describe an experiment to show the refraction of sound waves. You may include a diagram of your apparatus.

.....

 [3]

Mark scheme — See official MS

Oct/Nov 0625/21 Theory

Question 4

- 4 A student has a mobile (cell) phone. The phone receives a signal from a transmitter and produces a ring tone.

(a) State two differences between the microwave signal received by the phone and the sound wave produced when the phone rings.

1.

2.

[2]

(b) Fig. 4.1 represents the waves emitted by the mobile phone. The waves interact with a wall, and a doorway, in the room.

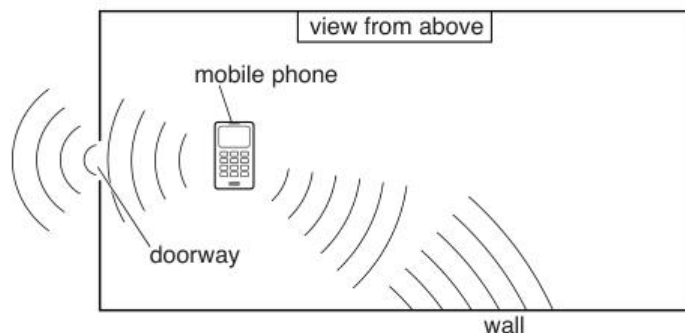


Fig. 4.1 (not to scale)

With reference to Fig. 4.1, complete each of the following sentences using a word from the box below.

diffracted	dispersed	reflected	refracted
------------	-----------	-----------	-----------

(i) When the waves hit the wall, the waves are [1]

(ii) When the waves pass through the doorway, the waves are [1]

[Total: 4]

Question 7

- 7 A battle re-enactment enables observers to see and hear an old cannon being fired.

Fig. 7.1 shows the battle site and the distant cliffs.

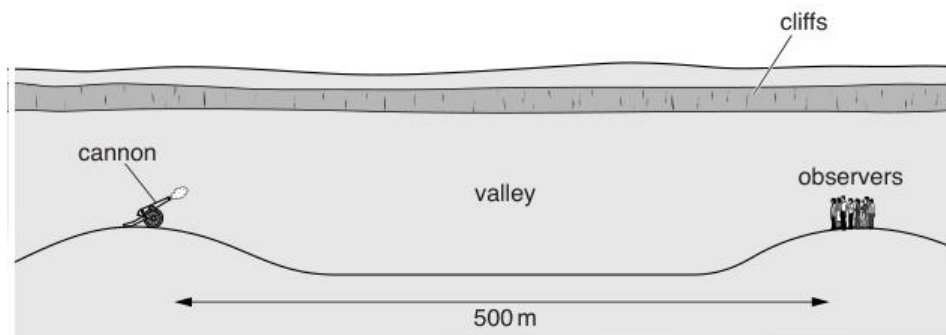


Fig. 7.1 (not to scale)

The cannon is fired. Observers see the smoke and then hear the bang.

- (a) (i) Explain why there is a short delay between seeing the smoke and hearing the bang of the cannon.

.....
.....
..... [2]

- (ii) An observer notices that, after the cannon is fired, she hears a loud bang and then a quieter bang a short time later.

Tick one box to identify the reason for the second bang.

- ☐ diffraction of the sound in the valley
☐ dispersion of the sound in the valley
☐ reflection of the sound from the cliff
☐ refraction of the sound from the cliff

[1]

- (b) Another observer is standing 500 m away from the cannon. He uses a stopwatch to measure the time delay between seeing the smoke and hearing the first bang. His timings are shown in the table.

measurement	time delay/s
1	1.9
2	1.5
3	1.3
4	1.4
5	1.7

Use the measurements in the table to calculate an accurate value for the speed of the sound produced by the cannon.

speed of sound = m/s [4]

[Total: 7]

Mark scheme — See official MS

Oct/Nov 0625/23 Theory

Question 4

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(a) State two differences between the microwave signal received by the phone and the sound wave produced when the phone rings.

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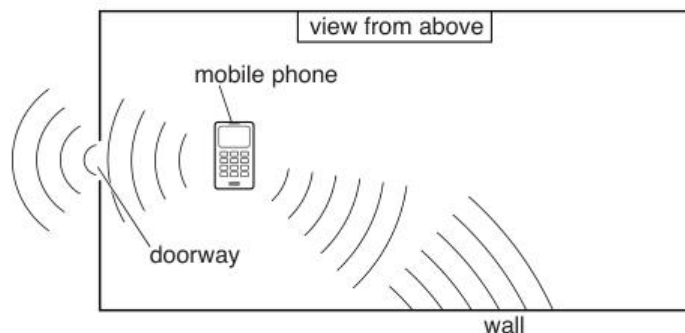


Fig. 4.1 (not to scale)

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diffracted	dispersed	reflected	refracted
------------	-----------	-----------	-----------

(i) When the waves hit the wall, the waves are [1]

(ii) When the waves pass through the doorway, the waves are [1]

[Total: 4]

Question 7

7 Audible sound waves and ultrasound waves both consist of compressions and rarefactions.

(a) Describe one difference between a compression and a rarefaction.

.....
.....
..... [1]

(b) In a particular medium, the speed of both audible sound and ultrasound waves is v .

(i) State the equation that relates the wavelength λ of a sound wave to its frequency f .

.....
..... [1]

(ii) Explain how, in this medium, the wavelength of an audible sound wave compares with the wavelength of an ultrasound wave.

.....
.....
.....
..... [1]

(c) Describe one use of ultrasound waves.

.....
.....
.....
.....
..... [2]

Question 21

21 Which is a unit of wavelength?

- A** hertz
- B** metre
- C** metre per second
- D** second

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0625/22/F/M/16

[Turn over

Mark scheme — Answer: B [1]

May/June 0625/21 MCQ

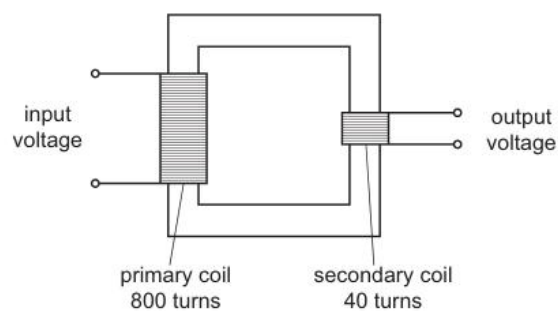
Question 20

20 Sound waves of frequency 2.0 kHz travel through a substance at a speed of 800 m/s.

What is the wavelength of the waves?

- A** 0.40 m **B** 2.5 m **C** 400 m **D** 1600 m

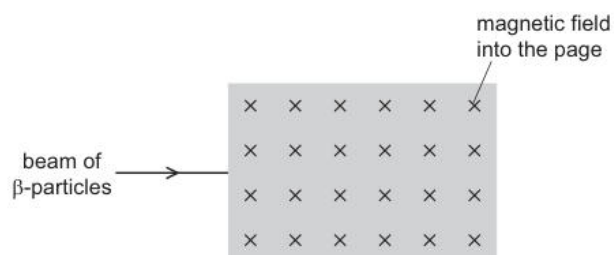
- 35 The diagram shows a transformer.



The input voltage is 240 V.

What is the output voltage?

- A** 6.0 V **B** 12 V **C** 20 V **D** 40 V
- 36 The diagram shows a shaded area where the direction of a magnetic field is into the page.
A beam of β -particles enters the field as shown.



In which direction is the beam of β -particles deflected as they enter the magnetic field?

- A** into the page
B out of the page
C down the page
D up the page

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Mark scheme — Answer: A [1]

Question 21

21 Which row shows the natures of light waves, sound waves and X-rays?

	light waves	sound waves	X-rays
A	longitudinal	longitudinal	transverse
B	longitudinal	transverse	longitudinal
C	transverse	longitudinal	transverse
D	transverse	transverse	longitudinal

Mark scheme — Answer: C [1]

May/June 0625/22 MCQ

Question 21

21 The frequency of a wave is doubled. The speed of the wave does not change.

What happens to the wavelength of the wave?

- A** It becomes four times as large.
- B** It does not change.
- C** It doubles.
- D** It halves.

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Mark scheme — Answer: D [1]

Question 22

22 Which row shows the natures of light waves, sound waves and X-rays?

	light waves	sound waves	X-rays
A	longitudinal	longitudinal	transverse
B	longitudinal	transverse	longitudinal
C	transverse	longitudinal	transverse
D	transverse	transverse	longitudinal

Mark scheme — Answer: C [1]

May/June 0625/23 MCQ

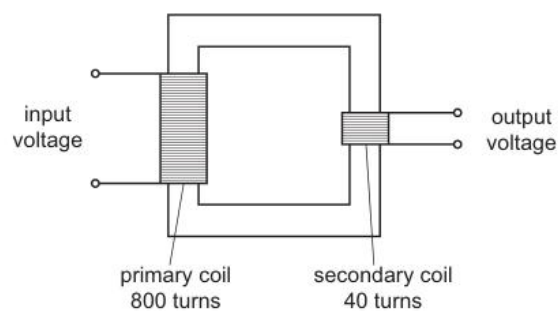
Question 20

20 A water wave passes into a region where the wave travels more slowly.

As it passes into the slow region, what happens to the frequency and what happens to the wavelength of the wave?

	frequency	wavelength
A	decreases	remains the same
B	increases	remains the same
C	remains the same	decreases
D	remains the same	increases

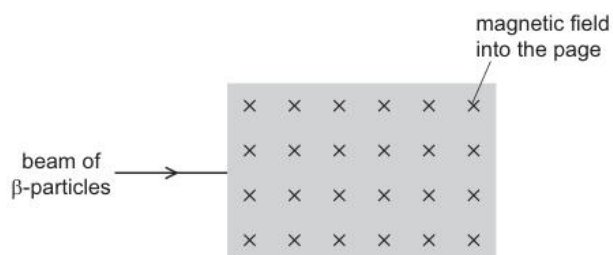
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Mark scheme — Answer: C [1]

Question 21

21 Which row shows the natures of light waves, sound waves and X-rays?

	light waves	sound waves	X-rays
A	longitudinal	longitudinal	transverse
B	longitudinal	transverse	longitudinal
C	transverse	longitudinal	transverse
D	transverse	transverse	longitudinal

Mark scheme — Answer: C [1]

May/June 5054/21 Theory

Question 5

- 5 (a) Light is a transverse wave. The direction of vibration is perpendicular to the direction of transfer of the energy.

Complete the table of Fig. 5.1 to show the direction of vibration and the type of wave associated with a sound wave and with a wave on the surface of water in a ripple tank.

wave	direction of vibration	type of wave
light	perpendicular to the direction of transfer of the energy	transverse
sound		
water		

[2]

Fig. 5.1

- (b) A wave travels along a rope. Fig. 5.2 shows how the height of one point on the rope varies with time as the wave passes.

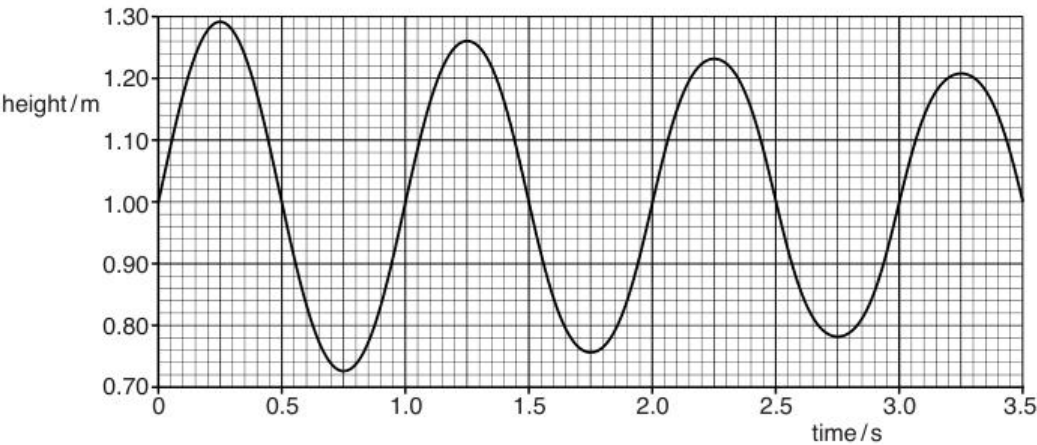
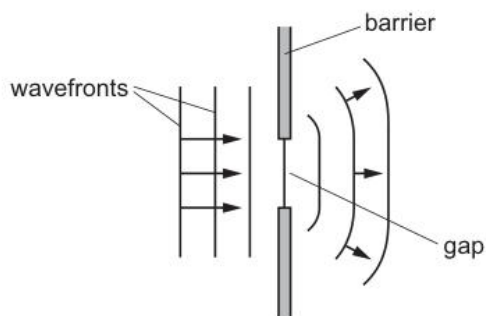


Fig. 5.2

- (i) Determine the amplitude of the vibration at time = 0.25 s.
.....[1]
- (ii) Describe how the graph shows that the frequency of the wave is constant.
.....
.....[1]

Question 21

21 The diagram represents plane wavefronts being diffracted by passing through a gap in a barrier.



Which pair of changes **must** increase the amount of diffraction that occurs?

- A decrease the wavelength and decrease the size of the gap
- B decrease the wavelength and increase the size of the gap
- C increase the wavelength and decrease the size of the gap
- D increase the wavelength and increase the size of the gap

Mark scheme — Answer: C [1]

Question 25

25 A girl notices that, when she shouts into a cave, she hears an echo.

Which wave property causes the echo?

- A diffraction
- B dispersion
- C reflection
- D refraction

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Mark scheme — Answer: C [1]

Oct/Nov 0625/22 MCQ

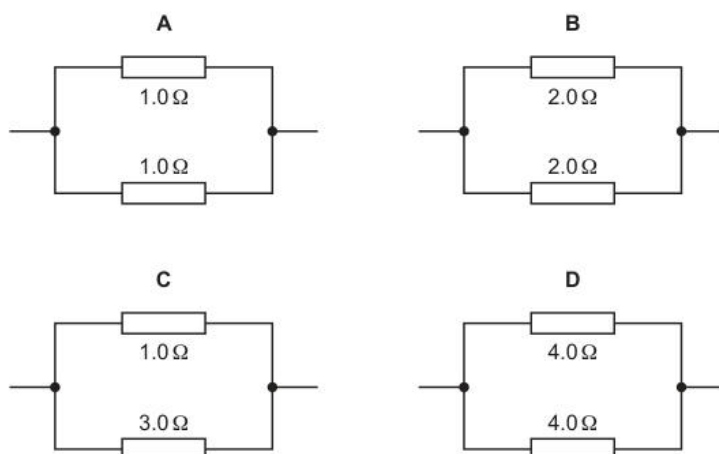
Question 10

10 Electricity can be generated using different energy resources.

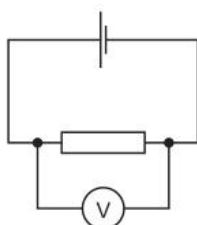
Which energy resource is used to generate electricity without needing any moving parts?

- A** geothermal
- B** hydroelectric
- C** solar
- D** water waves

- 32 Which combination of resistors in parallel has an effective resistance of $0.50\ \Omega$?

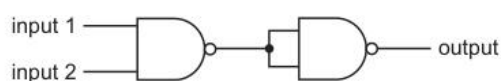


- 33 In the circuit shown, the voltmeter reads 2.0 V . A charge of 5.0 C passes through the resistor in a certain time.



How much energy is supplied to the resistor in this time?

- A 0.40 J B 2.5 J C 10 J D 20 J
- 34 Two NAND gates are connected together as shown.



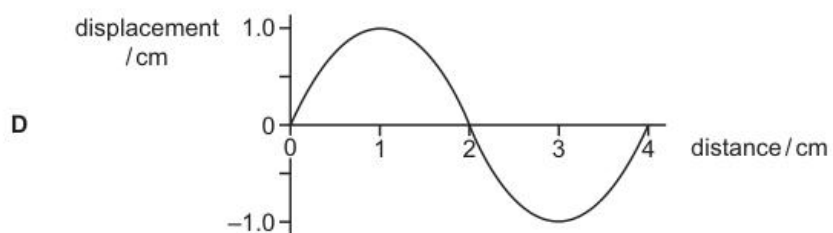
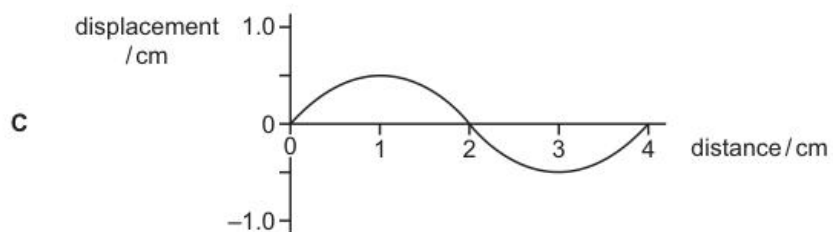
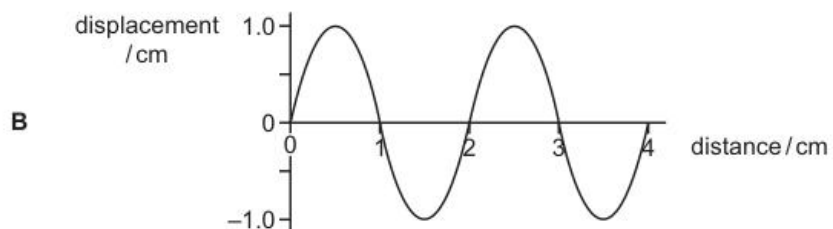
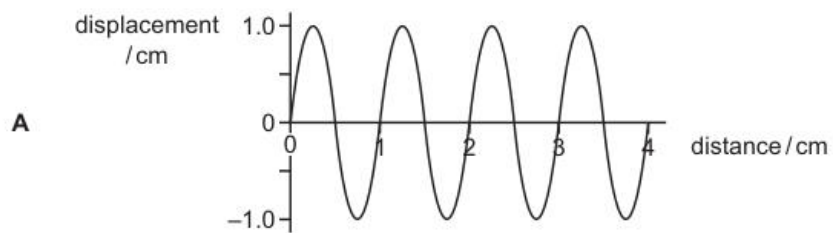
The arrangement acts like a single logic gate.

What is the name of this single logic gate?

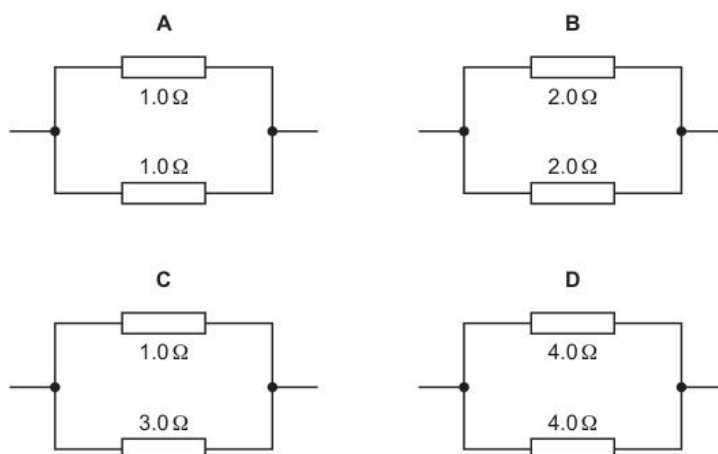
- A AND B NOR C NOT D OR

Question 20

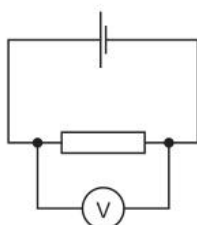
20 Which wave has an amplitude equal to half its wavelength?



- 32 Which combination of resistors in parallel has an effective resistance of $0.50\ \Omega$?

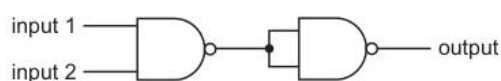


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Question 25

25 A girl notices that, when she shouts into a cave, she hears an echo.

Which wave property causes the echo?

- A diffraction
- B dispersion
- C reflection
- D refraction

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[Turn over

Mark scheme — Answer: C [1]

Oct/Nov 0625/23 MCQ

Question 11

11 Which list contains only energy resources that derive their energy from the Sun?

- A geothermal, nuclear, tidal
- B hydroelectric, water waves, wind
- C geothermal, hydroelectric, solar
- D nuclear, solar, water waves

Mark scheme — Answer: B [1]

Question 20

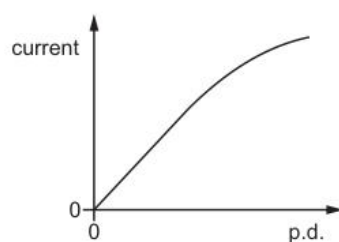
- 20** A man is talking at the side of a house. He can be heard by a woman at the front of the house even though she cannot see him.



What is the explanation for this?

- A** Sound waves are longitudinal and light waves are transverse.
- B** Sound waves are transverse and light waves are longitudinal.
- C** The sound waves have a long wavelength and the light waves have short wavelength.
- D** The sound waves have a short wavelength and the light waves have long wavelength.

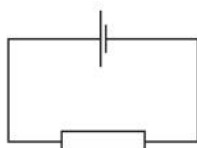
- 30 The graph shows the relationship between the current in a circuit component and the potential difference (p.d.) across it. The graph has a straight section and a curved section.



What happens to the resistance of the component in these two sections as the current increases?

	straight section	curved section
A	resistance increases	resistance decreases
B	resistance increases	resistance increases
C	no change in resistance	resistance decreases
D	no change in resistance	resistance increases

- 31 In the circuit shown there is a current of 10 A.



How much charge passes through the resistor in 2.0 minutes?

- A** 0.083 C **B** 5 C **C** 20 C **D** 1200 C

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Mark scheme — Answer: C [1]

Question 23

23 Which description applies to infra-red radiation?

- A longitudinal and electromagnetic
- B longitudinal but not electromagnetic
- C transverse and electromagnetic
- D transverse but not electromagnetic

Mark scheme — Answer: C [1]

Question 24

24 A girl notices that, when she shouts into a cave, she hears an echo.

Which wave property causes the echo?

- A diffraction
- B dispersion
- C reflection
- D refraction

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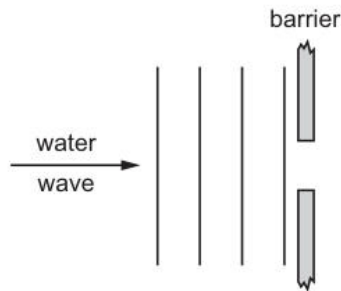
Mark scheme — Answer: C [1]

2017

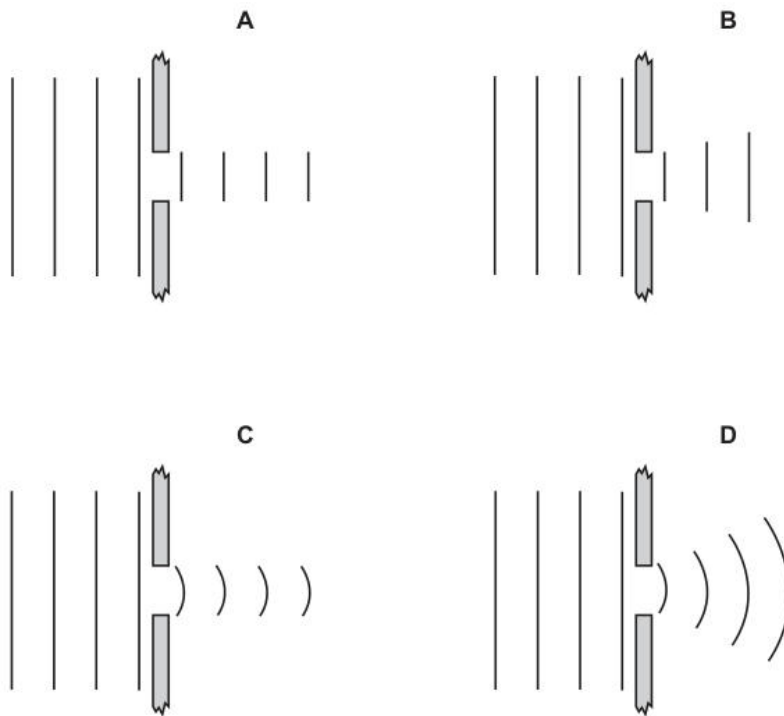
Feb/March 0625/22 MCQ

Question 20

20 In a shallow tank, a water wave moves towards a barrier with a narrow gap.



Which diagram shows the wave beyond the barrier?



Mark scheme — Answer: D [1]

Question 23

23 Which statement about sound waves is correct?

- A** They are able to travel through a vacuum.
- B** They are able to travel through solids.
- C** They are transverse waves.
- D** They travel at the same speed in all substances.

Mark scheme — Answer: B [1]

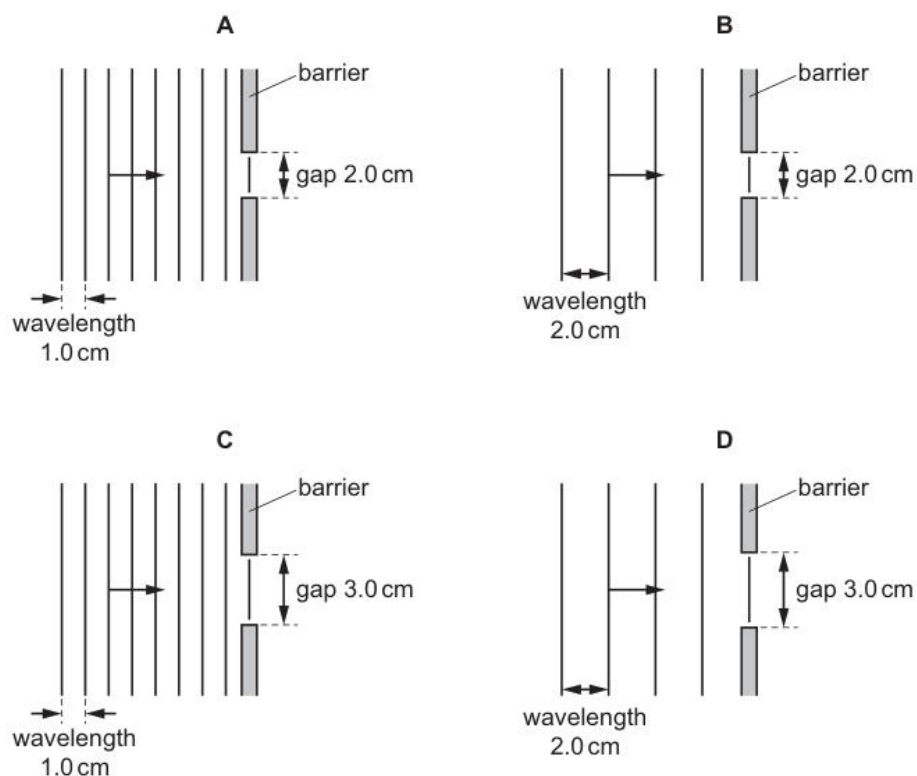
May/June 0625/21 MCQ

Question 19

19 Different waves hit barriers with different sized gaps.

The waves will diffract.

In which diagram does the greatest spreading occur?



Mark scheme — Answer: B [1]

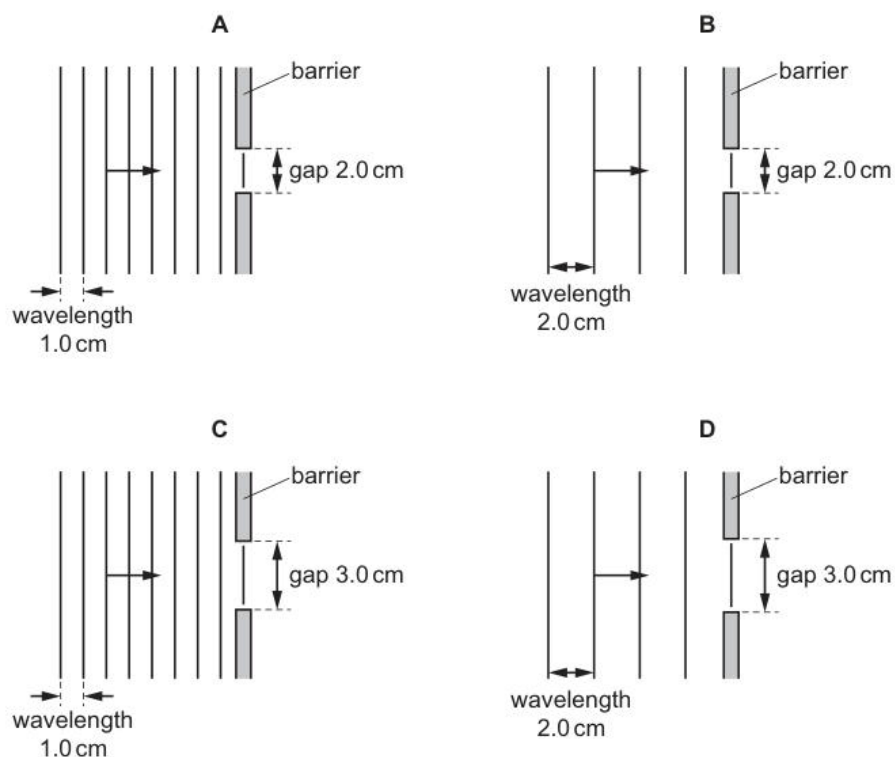
May/June 0625/22 MCQ

Question 19

19 Different waves hit barriers with different sized gaps.

The waves will diffract.

In which diagram does the greatest spreading occur?



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Mark scheme — Answer: B [1]

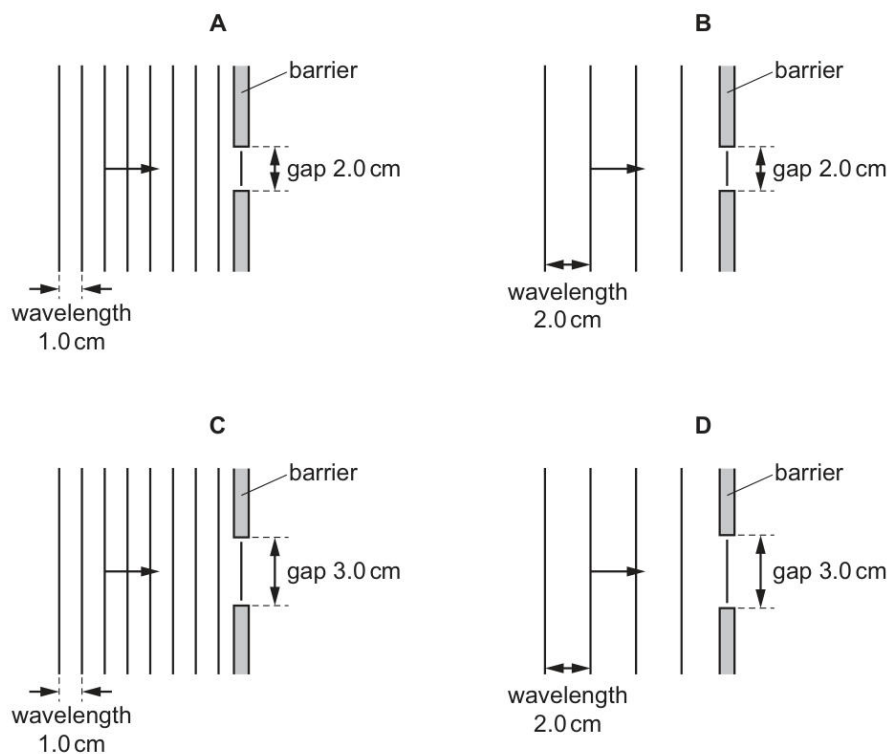
May/June 0625/23 MCQ

Question 19

19 Different waves hit barriers with different sized gaps.

The waves will diffract.

In which diagram does the greatest spreading occur?



Mark scheme — Answer: B [1]

May/June 5054/21 Theory

Question 10

14

- 10 (a) Fig. 10.1 shows two rays from an object that is placed in front of a plane mirror.

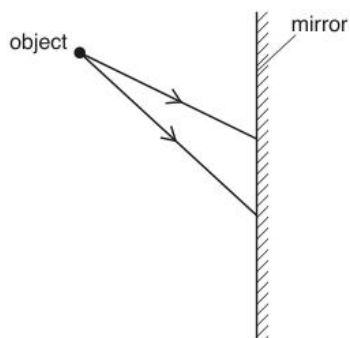


Fig. 10.1

- (i) On Fig. 10.1, draw the two reflected rays and locate the position of the image. [2]

- (ii) Apart from its position, state one characteristic of the image.

.....[1]

- (b) Fig. 10.2 shows wavefronts in a ripple tank. They move in the direction of the arrow.

The wave hits the boundary between two regions and the wave slows down as it enters the shaded region.

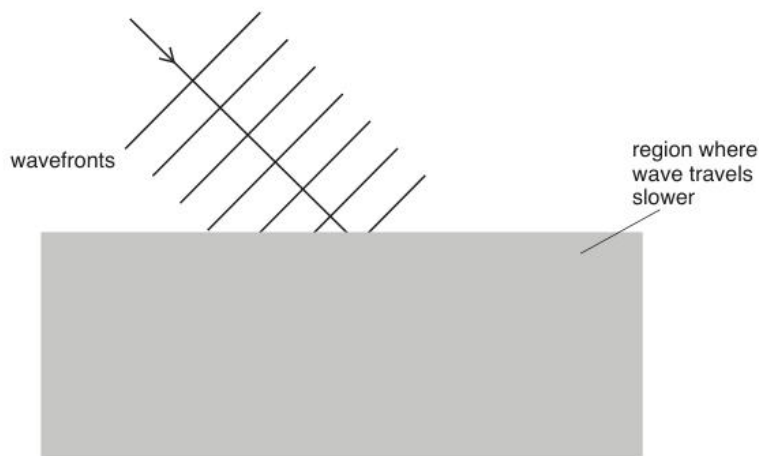


Fig. 10.2

- (i) State what is meant by *wavefront*.

.....
[2]

- (ii) Both reflection and refraction occur at the boundary.
1. On Fig. 10.2, draw the reflected wavefronts. [1]
 2. On Fig. 10.2, draw the refracted wavefronts. [2]
- (c) A loudspeaker produces a sound of frequency 2.0 kHz. The wavelength of this sound in air is 16 cm.
- (i) Calculate the speed of sound in air.

speed = [2]

- (ii) 1. State the range of frequencies that can be heard by a healthy human ear.
..... [1]
2. Calculate the smallest wavelength of sound that can be heard by a healthy human ear.

wavelength = [1]

- (iii) Describe a simple experiment to show that sound waves obey the law of reflection.
You may draw a diagram if you wish.

.....
.....
.....
..... [3]

Mark scheme — 10(a)(i) both reflected rays correct by eye B1 image in correct position shown by continuation of rays behind mirror B1
10(a)(ii) virtual or upright or laterally inverted or same size (as object) B1 10(b)(i) line (joining points) C1 line joining points / particles on wave with same phase or line joining points along a crest etc. A1 10(b)(ii)1 correct angle to surface $\pm 7^\circ$ with correct orientation and similar wavelength (by eye) B1 10(b)(ii)2 at least two lines at smaller angle to surface with correct orientation in slower medium B1 showing smaller and constant wavelength with wavefronts deviated in correct direction B1 10(c)(i) ($v =$) $f\lambda$ or $2(000) \times 16$ C1 $32\,000\text{ cm/s}$ or 320 m/s A1
10(c)(ii)1 a range with 15–25 Hz as the lowest frequency and 15–30 kHz as the highest B1 10(c)(ii)2 1.6 cm (if highest frequency is 20 000 Hz)
B1 10(c)(iii) tube or other method to produce (narrow) beam of sound on source and/or on detector B1 stated detector or stated reflector B1
detector moved to find maximum loudness or angles i and r measured with suitable experiment B1 5054/21 Question Answer Marks

Question 6

- 6 (a) Explain the difference between a longitudinal wave and a transverse wave.

You may draw a diagram, if you wish, to help your explanation.

.....

.....

.....

.....

.....[2]

- (b) A sound wave has a frequency of 3.8 kHz and a speed of 330 m/s.

- (i) Calculate the wavelength of the sound.

wavelength =[2]

- (ii) A different sound has a frequency of 3.8 Hz.

State and explain whether a human with normal hearing is able to hear this sound.

.....

.....

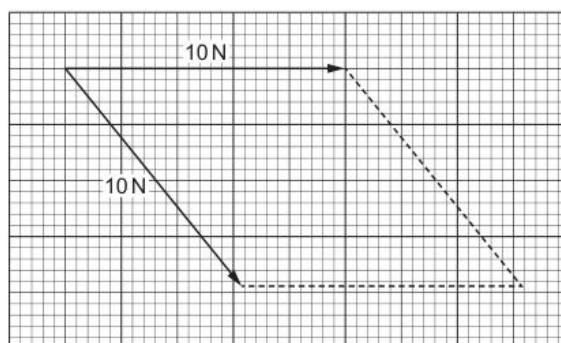
.....[1]

Mark scheme — 6(a) longitudinal - vibration / oscillation / movement to and fro and in direction of wave or has compressions and rarefactions B1 transverse – vibration / oscillation / movement up and down and at right angles to wave or has crests and troughs B1 6(b)(i) ($\lambda =$) v / f or 330 / 3800 B1 0.087 m or 8.7 cm B1 6(b)(ii) not heard and because below the range of audible frequencies or audible range is 20 – 20 000 Hz or too low a pitch / frequency B1 Question Answer Marks

Question 20

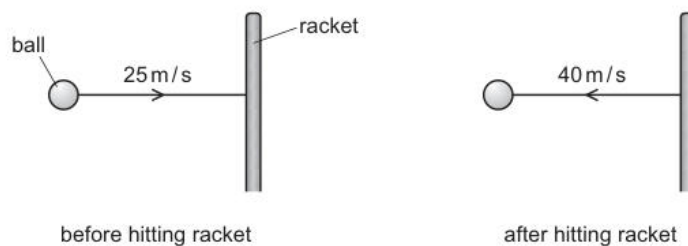
5

- 8 The diagram shows an incomplete scale drawing to find the resultant of two 10 N forces acting at a point in the directions shown.



What is the magnitude of the resultant force?

- A 7.5 N B 8.6 N C 18 N D 20 N
- 9 A tennis ball of mass 0.060 kg travels horizontally at a speed of 25 m/s. The ball hits a tennis racket and rebounds horizontally at a speed of 40 m/s.

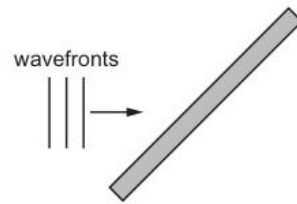


The ball is in contact with the racket for 50 ms.

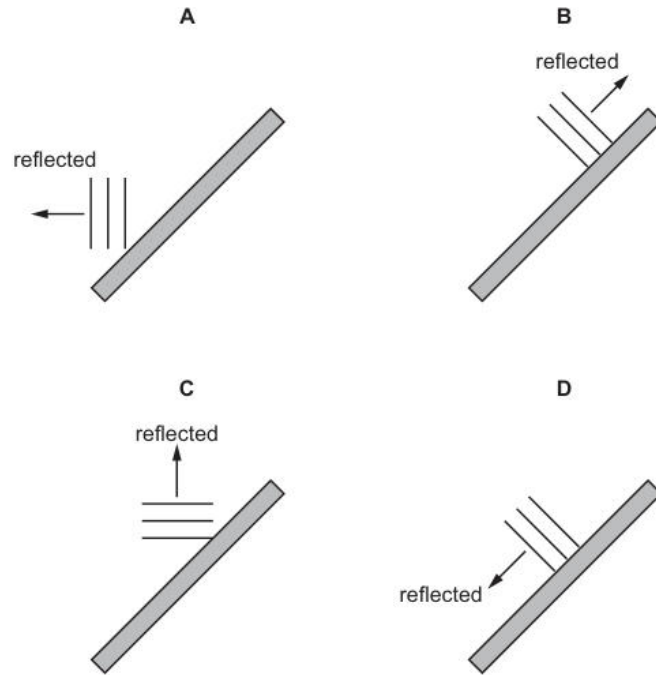
What force does the racket exert on the ball?

- A 0.018 N B 0.078 N C 18 N D 78 N

- 20 The diagram represents plane wavefronts of a water wave about to strike a solid barrier.

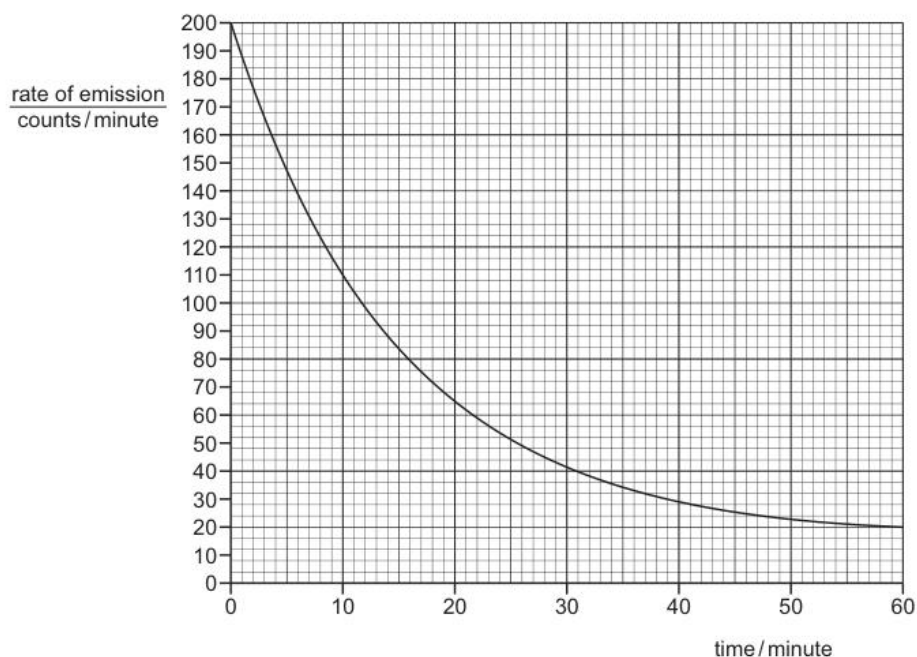


Which diagram shows the position of the wavefronts after reflection at the barrier?



- 40 The rate of emission of a radioactive source is measured until the reading reaches the background rate of 20 counts per minute.

The results are shown.



What is the best estimate of the half-life of the source?

- A 10 minutes
- B 12 minutes
- C 14 minutes
- D 30 minutes

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Mark scheme — Answer: C [1]

Question 25

25 Which wavefront is travelling at a speed closest to that of a sound wave through a solid?

- A** one that moves 10 m in 0.01 s
- B** one that moves 50 m in 0.5 s
- C** one that moves 1000 m in 100 s
- D** one that moves 2000 m in 2000 s

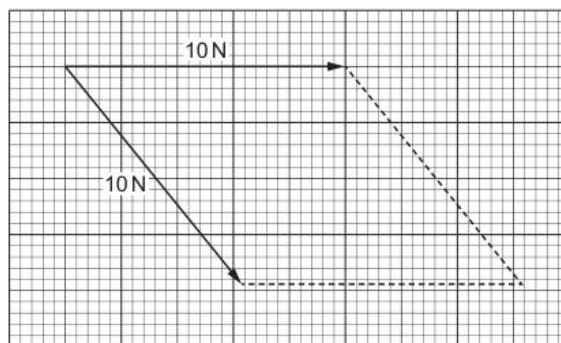
Mark scheme — Answer: A [1]

Oct/Nov 0625/22 MCQ

Question 20

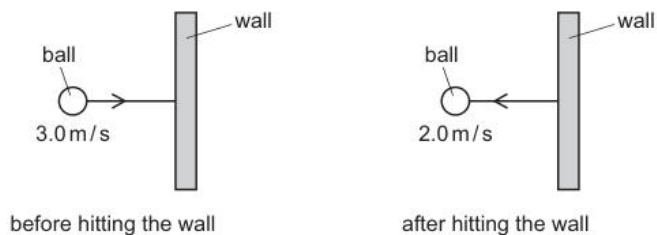
5

- 8 The diagram shows an incomplete scale drawing to find the resultant of two 10 N forces acting at a point in the directions shown.



What is the magnitude of the resultant force?

- A 7.5 N B 8.6 N C 18 N D 20 N
- 9 A ball has a mass of 0.30 kg. It moves horizontally with a speed of 3.0 m/s in the direction shown.
- The ball hits a wall.

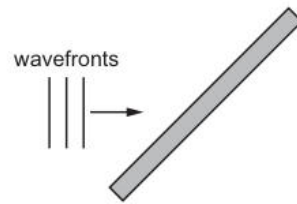


The ball rebounds from the wall with a horizontal speed of 2.0 m/s.

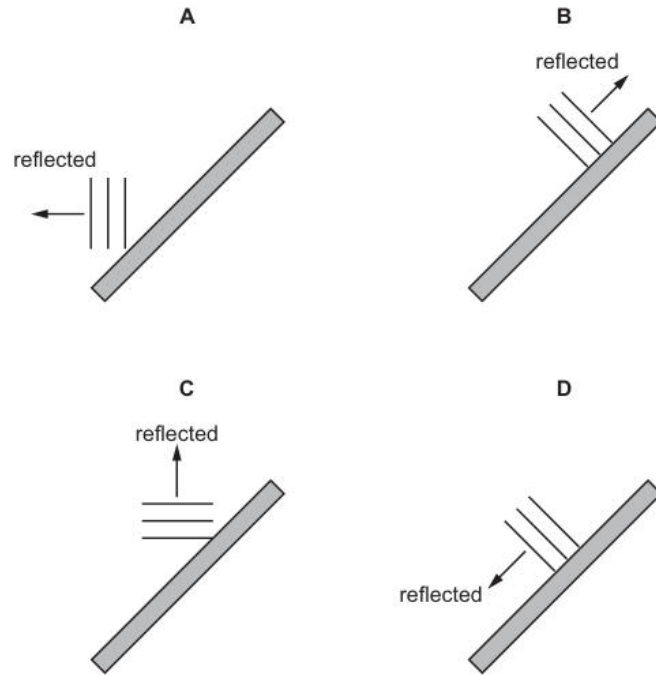
What is the change in momentum of the ball?

- A 0.30 kg m/s B 1.0 kg m/s C 1.5 kg m/s D 5.0 kg m/s
- 10 An object has a mass of 500 kg.
- It moves with a speed of 30 m/s.
- What is its kinetic energy?
- A 7.5 kJ B 15 kJ C 225 kJ D 450 kJ

- 20 The diagram represents plane wavefronts of a water wave about to strike a solid barrier.

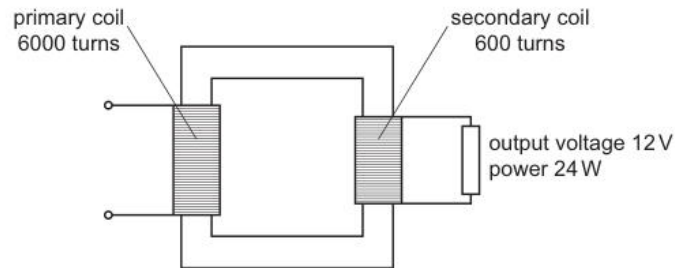


Which diagram shows the position of the wavefronts after reflection at the barrier?



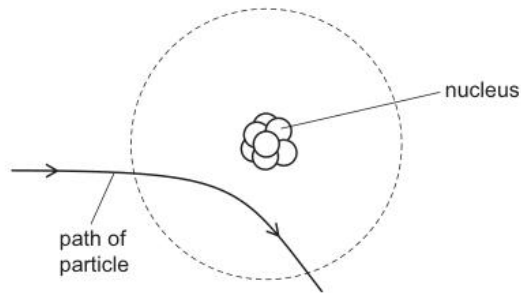
- 37 A 100% efficient transformer has 6000 turns on its primary coil and 600 turns on its secondary coil. The output voltage of the transformer is 12 V.

A resistor is connected across the secondary coil and dissipates 24 W of power.



What is the current in the primary coil of the transformer?

- A 0.050 A B 0.20 A C 5.0 A D 20 A
- 38 In the diagram, the circle represents an atom (not to scale) with the nucleus at its centre.
- A particle is emitted by a radioactive source and approaches the nucleus of the atom. The curved arrow shows the path of the particle.



What is the nature and charge of the particle?

	nature of particle	charge of particle
A	α -particle	negative
B	α -particle	positive
C	β -particle	negative
D	β -particle	positive

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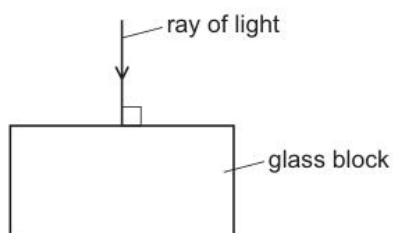
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Mark scheme — Answer: C [1]

Question 22

- 22** The diagram shows a ray of light incident on the surface of a rectangular glass block at 90° to the surface.



Which quantities remain unchanged as the light enters the glass block?

- A** direction and frequency
- B** direction and speed
- C** frequency and speed
- D** speed and wavelength

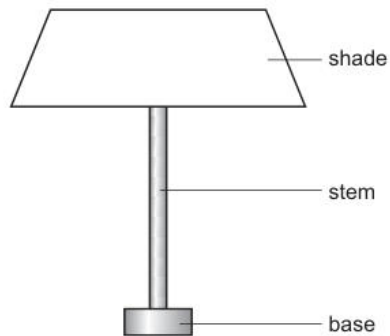
Mark scheme — Answer: A [1]

Oct/Nov 0625/23 MCQ

Question 20

5

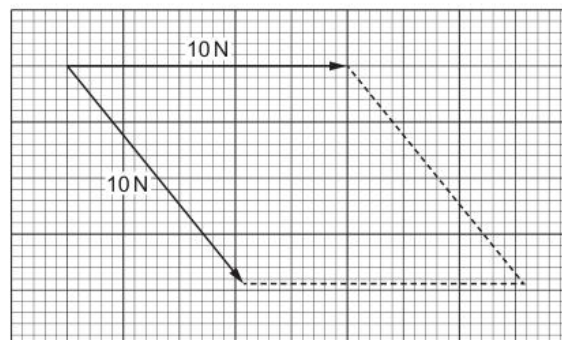
- 7 The lamp in the diagram is not very stable and falls over easily.



Which row shows changes that would definitely make the lamp more stable?

	base	centre of gravity
A	narrower	higher
B	narrower	lower
C	wider	higher
D	wider	lower

- 8 The diagram shows an incomplete scale drawing to find the resultant of two 10 N forces acting at a point in the directions shown.



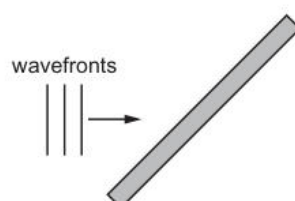
What is the magnitude of the resultant force?

- A** 7.5 N **B** 8.6 N **C** 18 N **D** 20 N

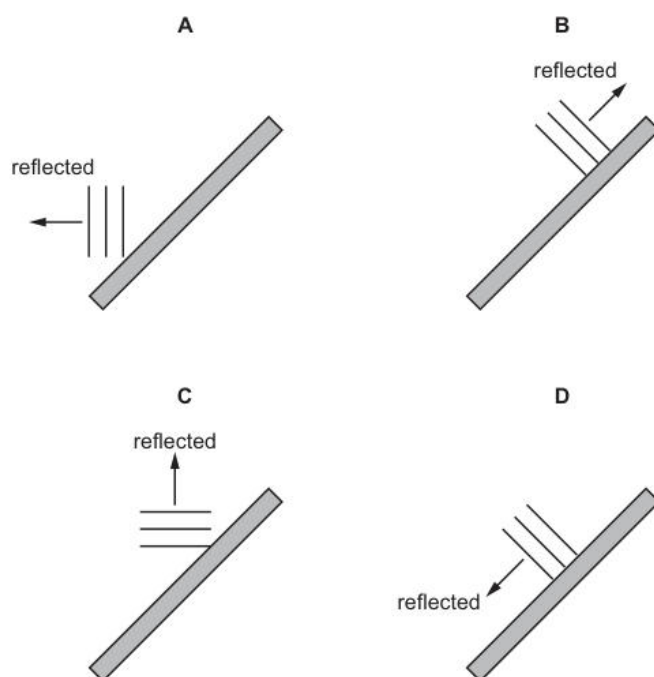
19 Why is the heating coil of a domestic immersion heater placed at the bottom of the tank?

- A Cold water is less dense than hot water and therefore sinks.
- B Cold water is more dense than hot water and therefore rises.
- C Hot water is less dense than cold water and therefore rises.
- D Hot water is more dense than cold water and therefore sinks.

20 The diagram represents plane wavefronts of a water wave about to strike a solid barrier.



Which diagram shows the position of the wavefronts after reflection at the barrier?



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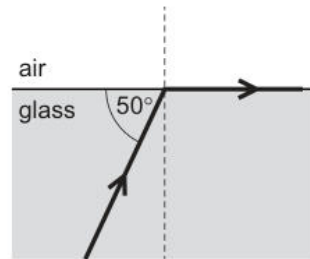
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Mark scheme — Answer: C [1]

Question 21

- 21 The diagram shows a ray of light in glass. The ray reaches a boundary with air. One weak ray of light is missing from the diagram.



Which statement is correct?

- A At the boundary, the speed of the light will become less.
- B The critical angle for light at this boundary is 50° .
- C The diagram shows an example of diffraction of light.
- D The missing ray is a weak reflected ray.

Mark scheme — Answer: D [1]

Question 25

- 25 Which statement about radio waves is correct?
- A They are used in television remote controllers.
 - B They can be detected by the human eye.
 - C They travel as longitudinal waves.
 - D They have the same speed in a vacuum as ultraviolet waves.

Mark scheme — Answer: D [1]

2018

Feb/March 0625/22 MCQ

Question 22

22 Which row shows an example of a transverse wave and an example of a longitudinal wave?

	transverse	longitudinal
A	light	radio
B	radio	sound
C	sound	water
D	water	light

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[Turn over

Mark scheme — Answer: B [1]

Question 23

23 A wave passes through a gap and diffraction causes the wave to spread out.

Which wave spreads out the most?

- A** large wavelength through a gap slightly larger than the wavelength
- B** large wavelength through a gap much smaller than the wavelength
- C** small wavelength through a gap much larger than the wavelength
- D** small wavelength through a gap much smaller than the wavelength

Mark scheme — Answer: A [1]

Question 26

- 26 Which statement describes monochromatic light?
- A light that does not diffract
 - B light that has a single frequency
 - C light that spreads out when shone through a glass prism
 - D light that travels at the same speed in all materials

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Mark scheme — Answer: B [1]

May/June 0625/21 MCQ

Question 19

- 19 Light travels at a speed of 2.0×10^8 m/s in a glass block.

In the glass, the wavelength of the light is 4.0×10^{-7} m.

What is the frequency of the light?

- A 2.0×10^{-15} Hz
- B 1.3×10^{-2} Hz
- C 80 Hz
- D 5.0×10^{14} Hz

Mark scheme — Answer: D [1]

Question 23

23 The Sun emits infra-red radiation and light.

Light from the Sun reaches the Earth in 8 minutes.

Which row gives correct information about the infra-red radiation?

	wavelength of infra-red radiation	time taken for infra-red radiation to reach Earth
A	longer than wavelength of light	8 minutes
B	longer than wavelength of light	much less than 8 minutes
C	shorter than wavelength of light	8 minutes
D	shorter than wavelength of light	much more than 8 minutes

Mark scheme — Answer: A [1]

May/June 0625/22 MCQ

Question 19

19 A tank contains water. Ripples are produced on the surface of the water.

What causes the ripples to refract?

- A** The cold water in the tank is replaced by warm water.
- B** The ripples change speed as they move from deep to shallow water.
- C** The ripples hit the wall of the tank.
- D** The ripples pass through a narrow gap.

Mark scheme — Answer: B [1]

Question 20

20 Light travels at a speed of 2.0×10^8 m/s in a glass block.

In the glass, the wavelength of the light is 4.0×10^{-7} m.

What is the frequency of the light?

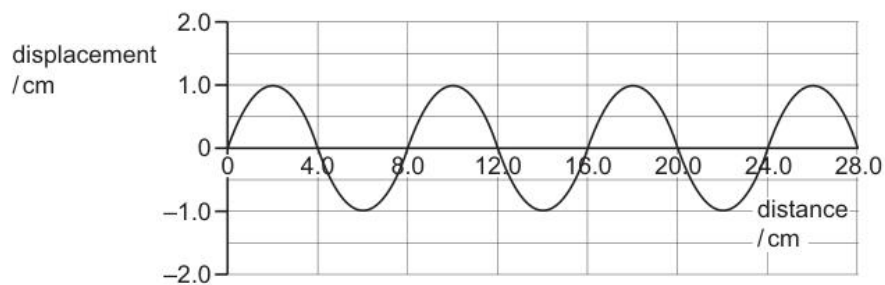
- A** 2.0×10^{-15} Hz
- B** 1.3×10^{-2} Hz
- C** 80 Hz
- D** 5.0×10^{14} Hz

Mark scheme — Answer: D [1]

May/June 0625/23 MCQ

Question 19

19 The diagram shows a wave.



Which row is correct?

	amplitude of the wave / cm	wavelength of the wave / cm
A	1.0	4.0
B	1.0	8.0
C	2.0	4.0
D	2.0	8.0

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[Turn over

Mark scheme — Answer: B [1]

Question 20

20 Light travels at a speed of 2.0×10^8 m/s in a glass block.

In the glass, the wavelength of the light is 4.0×10^{-7} m.

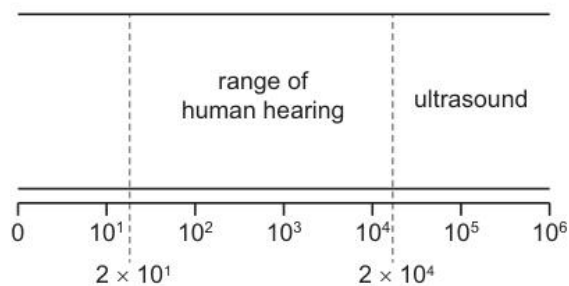
What is the frequency of the light?

- A** 2.0×10^{-15} Hz
- B** 1.3×10^{-2} Hz
- C** 80 Hz
- D** 5.0×10^{14} Hz

Mark scheme — Answer: D [1]

Question 24

24 The diagram shows the ranges of human hearing and of ultrasound waves.



To which characteristic of sound waves do the numbers on the diagram refer?

- A** amplitude in cm
- B** frequency in Hz
- C** speed in metres / second
- D** wavelength in metres

Mark scheme — Answer: B [1]

May/June 5054/21 Theory

Question 11

11 A water wave is produced in a ripple tank.

- (a) (i) Draw a diagram of a ripple tank and describe how it is used to produce and observe the wave.

.....

.....

.....

.....

.....

.....

.....

.....[4]

- (ii) The water wave in the ripple tank transfers energy.

Describe how to show that there is a transfer of energy.

.....

.....

.....

.....[2]

- (iii) State what is meant by a *wavefront*.

.....

.....

.....[2]

- (b) Fig. 11.1 shows the variation with time t of the height of water at the same place in the tank for two waves P and Q. The height is sometimes above and sometimes below the mean level which is marked as 0.

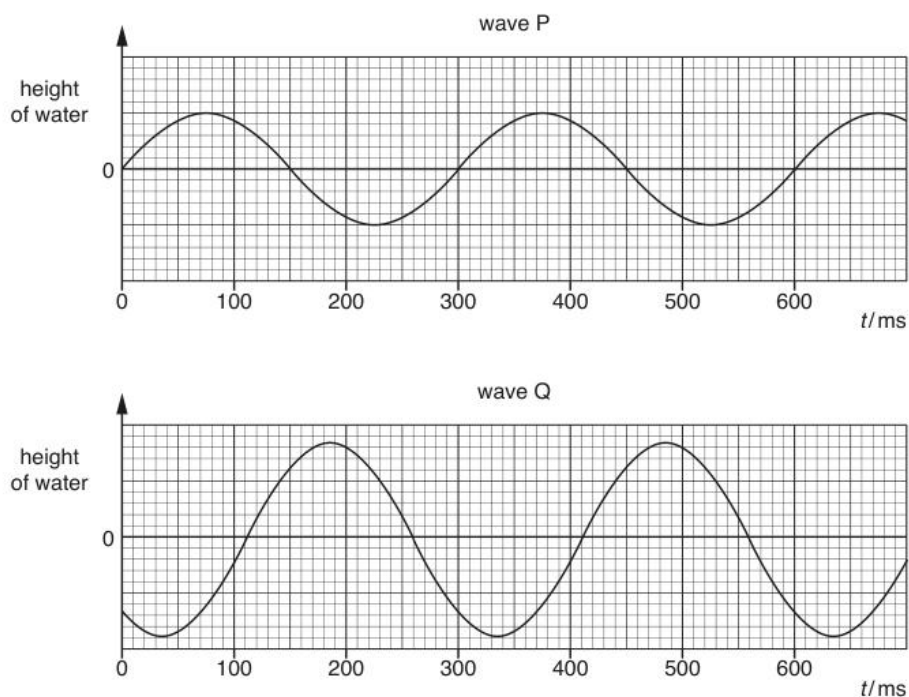


Fig. 11.1

- (i) State one difference and one similarity of the two waves.

difference

.....

similarity

.....

[2]

- (ii) The speed of the water wave is 0.20 m/s.

Determine

1. the frequency of wave P,

frequency =[3]

2. the wavelength of wave P.

wavelength =[2]

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5054/21/M/J/18

Mark scheme — 11(a)(i) tank with water B1 dipper (bar or point) touches or in surface of water B1 light source / stroboscope / video B1 dipper moves up and down (to create wave) B1 11(a)(ii) observe (surface of) water OR place particle / cork on surface OR equivalent movable / rotating object B1 moves up and down B1 11(a)(iii) line joining points M1 that are at the same point in the motion A1 11(b)(i) difference: Q has a greater / different amplitude or phase B1 similarity: same frequency; same time for one wave (period); both transverse; oscillation up and down B1 11(b)(ii) time 300 (ms) or 0.3 (s) C1 (f=) 1 / t numerical or algebraic in any form C1 3.3 Hz A1 11(b)(ii) 2 ($\lambda =$) v / f or $v = \lambda t$ – numerical or algebraic in any form C1 0.060 m A1

Question 5

2

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

- 1 A student drops a small metal object into a cylinder of oil. The object falls alongside a vertical ruler and a camera records its position at 1.0 s intervals, as shown in Fig. 1.1.

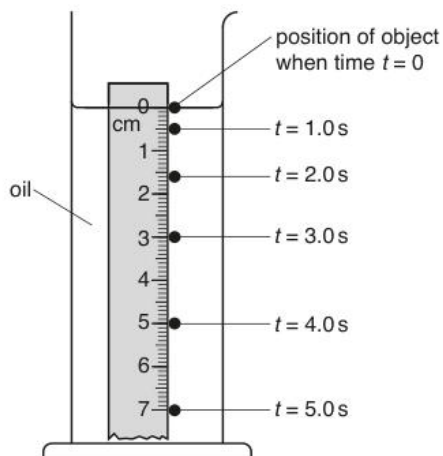


Fig. 1.1

- (a) On the grid in Fig. 1.2, plot a distance-time graph for the object.

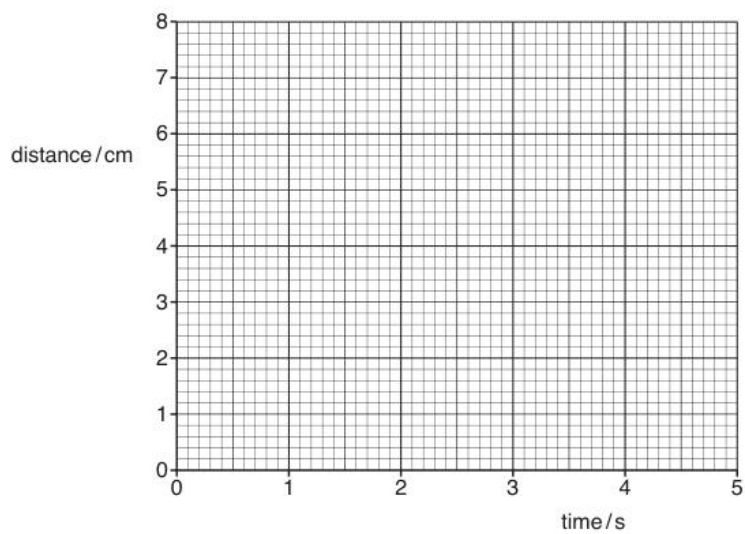


Fig. 1.2

[2]

(b) The object has a constant speed for some of the time.

(i) State and explain how a distance-time graph shows that speed is constant.

.....
.....
.....
.....[2]

(ii) Explain, in terms of the forces acting on the object, how it is able to fall at a constant speed.

.....
.....
.....
.....[2]

- 2 In an experiment, a student takes measurements and determines the extension of a spring for different loads. The apparatus is shown in Fig. 2.1.

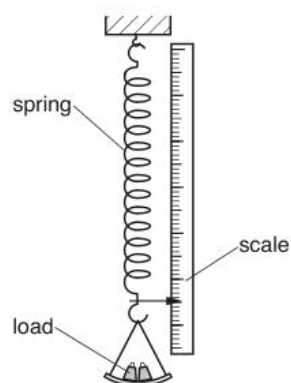


Fig. 2.1

The table in Fig. 2.2 shows the results.

load / N	0	1.0	2.0	3.0	4.0	5.0
length of spring / mm	200	235	270	305	350	420
extension / mm	0	35	70	105		

Fig. 2.2

- (a) Only some of the extensions are shown in the table.

Complete the table to show all of the extensions.

[1]

- (b) Calculate the load that produces an extension of 49 mm.

load =[2]

- (c) The student pulls a load downwards from position A to position B and holds it fixed at position B, as shown in Fig. 2.3.

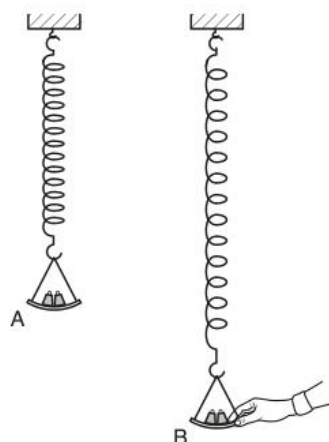


Fig. 2.3

The load is stationary at A and at B. The load has no kinetic energy at either point.

- (i) Place ticks (✓) in the boxes to show how the value of each of the forms of energy compares at A and B.

	equal at A and B	larger at A	larger at B
energy stored in spring	☐	☐	☐
gravitational potential energy of load	☐	☐	☐

[1]

- (ii) Work is done by the student to pull the load down. The law of conservation of energy states that energy cannot be created and cannot be destroyed.

Explain how this principle applies in this case.

.....

 [2]

3 Equal volumes of metal, water and air are heated from 20 °C to 80 °C.

(a) State whether the metal, the water or the air expands the most.

.....[1]

(b) Fig. 3.1 is a diagram of the metal at 20 °C. The metal is a large, square sheet with a square hole at its centre. The sheet lies flat on a table.

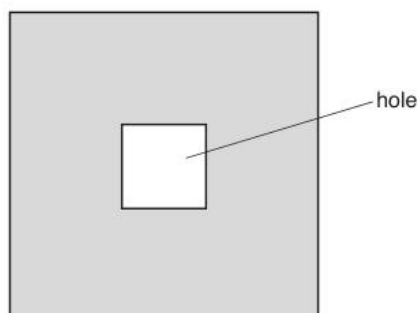


Fig. 3.1

On Fig. 3.1, draw the outline of the sheet and of the hole after the metal is heated. Make sure that the expansion is noticeable. [2]

(c) The metal has a mass of 5.0 kg and a density of $7.5 \times 10^3 \text{ kg/m}^3$.

(i) Define *density*.

.....
[1]

(ii) Calculate the volume of the metal.

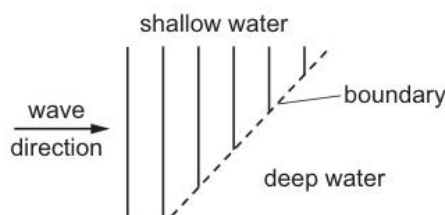
Give your answer to a suitable number of significant figures.

volume = [2]

Mark scheme — 5(a) (d=) $s \times t$ in any form algebraic or numerical, C1 $3.6 \times 10^7 \text{ m}$ A1 5(b)(i) 1st two columns correct (sound longitudinal and microwaves transverse) B1 3rd column correct microwaves electromagnetic B1 5(b)(ii) Layers / molecules / particles / spring coils close together or high(er) pressure (than atmospheric) B1 Question Answer Marks

Question 21

- 21 Plane water waves travel from a shallow region into a deeper region. They travel more quickly in the deeper water.



Which diagram shows the wave pattern in the deeper water?

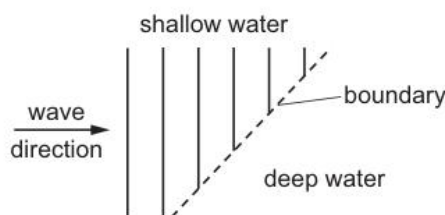


Mark scheme — Answer: C [1]

Oct/Nov 0625/22 MCQ

Question 21

- 21 Plane water waves travel from a shallow region into a deeper region. They travel more quickly in the deeper water.



Which diagram shows the wave pattern in the deeper water?



Mark scheme — Answer: C [1]

Question 23

- 23 A transmitter produces radio waves of wavelength 1500 m. It takes the waves 0.025 s to travel from the transmitter to a radio receiver.

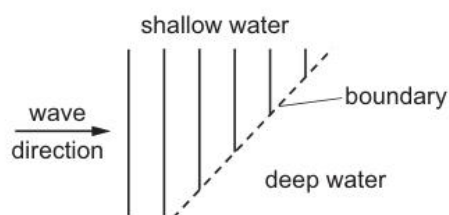
What is the distance between the radio transmitter and the receiver?

- A 5.0×10^3 m B 2.0×10^5 m C 7.5×10^6 m D 1.1×10^{10} m

Mark scheme — Answer: C [1]

Question 21

- 21** Plane water waves travel from a shallow region into a deeper region. They travel more quickly in the deeper water.



Which diagram shows the wave pattern in the deeper water?



Mark scheme — Answer: C [1]

Question 23

- 23** Light travels through air and then enters and travels through a parallel-sided glass block.

Which statement is correct?

- A** The angle of incidence is greater than the angle of refraction as the light leaves the block.
- B** The light emerging from the block is parallel to the light entering the block.
- C** The speed of the light decreases as it leaves the block.
- D** The wavelength of the light does not change as it enters the block.

Mark scheme — Answer: B [1]

Question 9

9 Some waves are longitudinal and some waves are transverse.

(a) State how a longitudinal wave differs from a transverse wave.

.....
 [1]

(b) A vibrating rod produces a water wave in a ripple tank. Fig. 9.1 shows the crests of the wave passing into the right-hand section of the tank where the depth of the water is different from the depth in the rest of the tank.

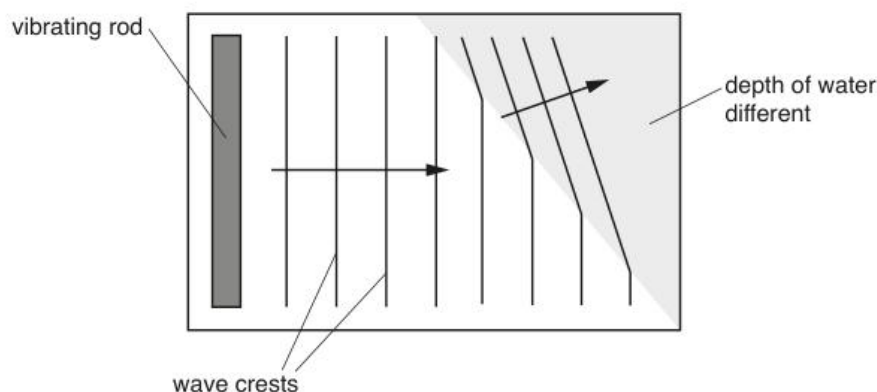


Fig. 9.1

The arrows on Fig. 9.1 show the direction of travel of the wave in the two sections of the ripple tank.

(i) In the left-hand section of the tank, the wavelength of the wave is 0.019 m and it travels at 0.17 m/s.

Calculate the frequency of the wave.

frequency = [2]

(ii) State what happens to the frequency of the wave, as it passes into the right-hand section of the tank.

..... [1]

(iii) Using Fig. 9.1, state and explain what happens to the speed of the wave as it passes into the right-hand section of the tank.

.....

 [2]

- (c) Fig. 9.2 shows light, in air, striking the vertical side of a rectangular glass block at an angle of incidence of 60° .

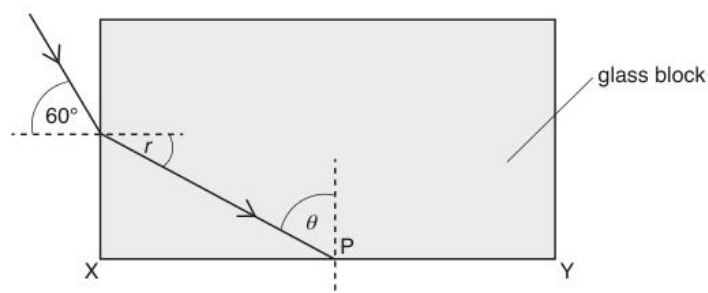


Fig. 9.2

The refractive index of the glass is 1.6. The light travels in the glass and strikes side XY at P.

- (i) Underline all the terms that describe a light wave.

electromagnetic

longitudinal

transverse

[1]

- (ii) At the point where the light enters the glass, the angle of refraction is r .

Calculate angle r .

$$r = \dots\dots\dots [2]$$

- (iii) 1. Calculate the critical angle c for light travelling in the block.

$$c = \dots\dots\dots [2]$$

2. At P, the angle θ between the ray and the normal is given by $\theta = 90^\circ - r$.

State and explain what happens to the light when it strikes side XY.

.....

 [2]

3. On Fig. 9.2, draw the path of the light after it strikes side XY at P and the path of the light when it is again travelling in the air. [2]

Mark scheme — 9(a) the vibration direction is parallel to the wave / energy travel direction or longitudinal waves have compressions and rarefactions or transverse waves have crests and troughs or longitudinal waves cannot be polarised B1 9(b)(i) ($f =$) v / λ or $0.17 / 0.019$ C1 8.9 Hz A1 9(b)(ii) (frequency) stays constant or does not change B1 9(b)(iii) (speed) decreases B1 (Fig. 9.1 shows that) wavelength decreases or refraction is towards normal or $i > r$ or top of wave (in shaded region) lags behind bottom of wave B1 9(c)(i) electromagnetic and transverse underlined B1 9(c)(ii) $n = \sin(i) / \sin(r)$ or ($r =$) $\sin^{-1}(\sin(i) / n)$ or ($r =$) $\sin^{-1}(\sin(60^\circ) / 1.6)$ C1 33° A1 5054/21 Question Answer Marks 9(c)(iii) $1/n = 1 / \sin(c)$ or ($c =$) $\sin^{-1}(1 / n)$ or $\sin^{-1}(1 / 1.6)$ C1 39° A1 2 total internal reflection B1 $\theta > c$ or $57^\circ > 39^\circ$ or angle of incidence greater than the critical angle B1 3 reflected ray at P (correct angle by eye) B1 light in air at 30° (by eye) to vertical surface B1 Question Answer Marks

Question 9

9 Both sound and ultrasound are waves that travel in solids, liquids and gases.

(a) State how sound differs from ultrasound.

.....
 [1]

(b) Sound and ultrasound are longitudinal waves that consist of compressions and rarefactions.

(i) Explain what is meant by a wave and describe how a longitudinal wave differs from a transverse wave.

.....

 [3]

(ii) Fig. 9.1 represents a longitudinal wave of frequency 25 000 Hz travelling in oil.



Fig. 9.1 (full scale)

1. On Fig. 9.1, mark two points at the centre of two different rarefactions and label each one R. [1]
2. On Fig. 9.1, draw a double-headed arrow to indicate a distance that is equal to one wavelength of the wave. [1]
3. Measure the length of the arrow drawn on Fig. 9.1 and use it to determine the speed of the wave in oil.

speed = [2]

(c) A sound wave travelling in a liquid, passes into air.

(i) State what happens to the speed of the sound wave as it enters the air.

..... [1]

- (ii) The wave in the liquid travels towards the surface at an angle. Fig. 9.2 shows the centres of the compressions of the sound wave in the liquid.

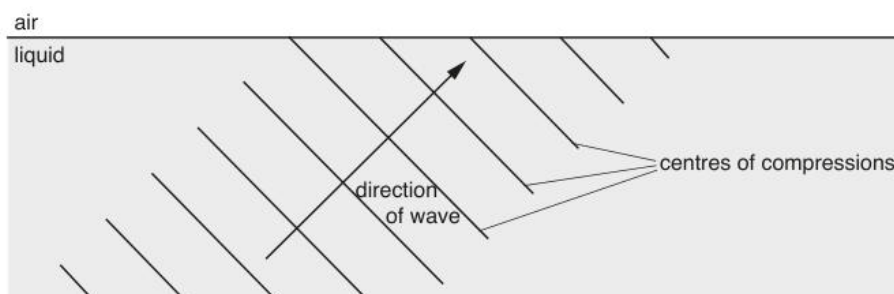


Fig. 9.2

Some compressions shown have reached the liquid-air boundary. The parts of these compressions in the air are not shown on Fig. 9.2.

On Fig. 9.2, draw the parts of these compressions that are in the air. [3]

- (d) Describe, in outline, how ultrasound is used in cleaning.

.....

.....

.....

.....

..... [3]

Mark scheme — 9(a) frequency of sound wave small(er) or its frequency is less than 20 000 Hz B1 9(b)(i) transmission of energy B1 (through a medium) with no net movement of medium or by vibrating particles B1 vibrations parallel (and antiparallel) to wave / energy travel direction or cannot be polarised B1 9(b)(ii) 1 two centres of rarefactions labelled R B1 2 distance from one point to adjacent identical point indicated (with double-headed arrow) B1 3. ($v = f\lambda$ or $25\,000 \times 0.047 / 0.048 / 0.049$ or $25\,000 \times 4.7 / 4.8 / 4.9$ or $25\,000 \times 47 / 48 / 49$ C1 1200 m / s A1 9(c)(i) decreases B1 9(c)(ii) four / five straight lines in air that touch the compressions still in the liquid and no intermediate / extra lines between the correct lines B1 at least four compressions in air parallel to each other B1 at least four straight lines at shallower angle from horizontal and slope correct B1 5054/22 Question Answer Marks 9(d) object (to be cleaned) immersed in liquid / solvent B1 object / liquid agitated / vibrated by ultrasound B1 dirt (particles) shaken off or dislodges / removes dirt B1 Question Answer Marks

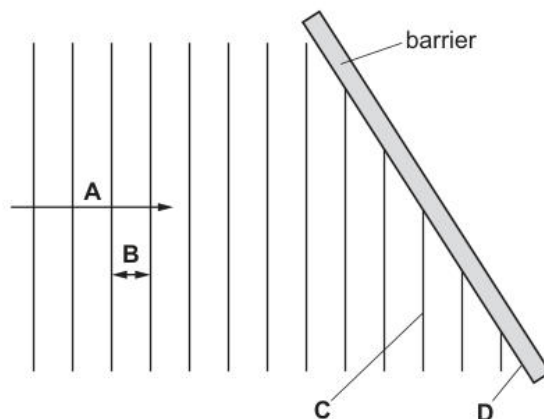
2019

Feb/March 0625/22 MCQ

Question 21

21 The diagram shows a wave before it reflects from a barrier.

Which labelled section of the diagram represents a wavefront?



Mark scheme — Answer: C [1]

Question 26

26 A sound wave passes a point. The air pressure at that point increases and then decreases 300 times every second.

Which descriptions apply to this sound wave?

	the type of wave motion	the frequency of the sound
A	longitudinal	outside human hearing range
B	longitudinal	within human hearing range
C	transverse	outside human hearing range
D	transverse	within human hearing range

Mark scheme — Answer: B [1]

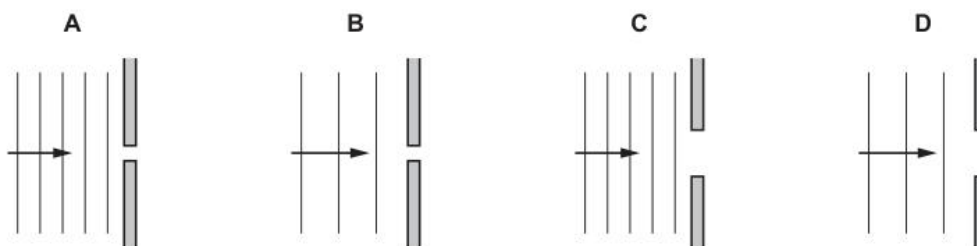
May/June 0625/21 MCQ

Question 20

20 When water waves pass through a gap they diffract.

The diagrams show wavefronts approaching a narrow gap.

In which diagram will the diffraction be least?



Mark scheme — Answer: C [1]

May/June 0625/22 MCQ

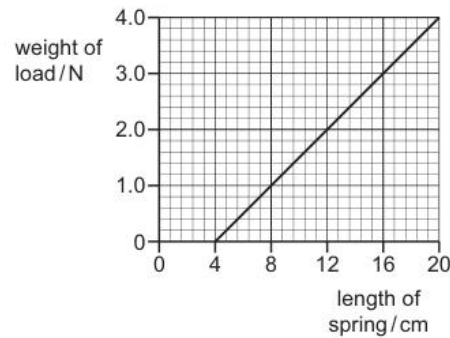
Question 20

4

- 6 An experiment is carried out to determine the spring constant for a spring.

Different loads are hung from the spring and its length is measured for each different load.

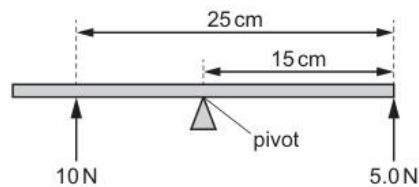
The graph shows how its length varies with load.



What is the value of the spring constant?

- A 0.20 N/cm B 0.25 N/cm C 4.0 N/cm D 5.0 N/cm
- 7 A beam is pivoted at its centre of mass.

It is acted upon by two forces, 10 N and 5.0 N, as shown.



What is the resultant moment about the pivot?

- A 25 N cm anticlockwise
 B 25 N cm clockwise
 C 175 N cm anticlockwise
 D 175 N cm clockwise

- 18 The metal surface of a kettle is hot.

What happens to the cool air outside the kettle when it comes into contact with the hot kettle?

- A The density of the air decreases and the air falls.
- B The density of the air decreases and the air rises.
- C The density of the air increases and the air falls.
- D The density of the air increases and the air rises.

- 19 Vacuum flasks usually have silvered walls that help to keep the contents of the flask hot.

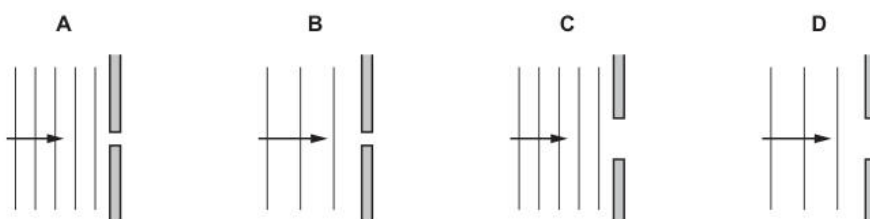
Why are the walls silvered?

- A to absorb thermal energy from the air around the flask
- B to increase the rate of convection inside the flask
- C to reduce energy loss to the surroundings by conduction
- D to reflect thermal radiation back into the flask

- 20 When water waves pass through a gap they diffract.

The diagrams show wavefronts approaching a narrow gap.

In which diagram will the diffraction be least?



- 21 Which conditions are necessary for light to be totally internally reflected?

	the incident light is in	angle of incidence
A	the less dense medium	less than the critical angle
B	the less dense medium	greater than the critical angle
C	the more dense medium	less than the critical angle
D	the more dense medium	greater than the critical angle

Mark scheme — Answer: C [1]

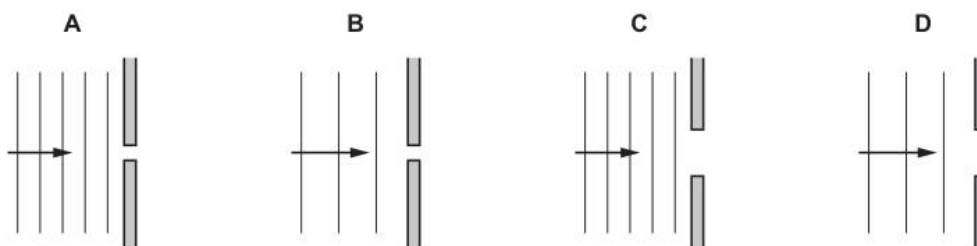
May/June 0625/23 MCQ

Question 19

19 When water waves pass through a gap they diffract.

The diagrams show wavefronts approaching a narrow gap.

In which diagram will the diffraction be least?



Mark scheme — Answer: C [1]

Question 20

20 Sound travels through air at a speed of 340 m/s. A source generates sound waves at a frequency of 1.2 kHz.

What is the wavelength of the sound waves?

- A 0.28 m B 3.5 m C 280 m D 410 m

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Mark scheme — Answer: A [1]

Question 25

25 Which statement about ultrasound is correct?

- A It has a higher frequency than audible sound, and it is a longitudinal wave.
B It has a higher frequency than audible sound, and it is a transverse wave.
C It has a lower frequency than audible sound, and it is a longitudinal wave.
D It has a lower frequency than audible sound, and it is a transverse wave.

Mark scheme — Answer: A [1]

May/June 5054/21 Theory

Question 10

10 In Fig. 10.1 the vertical lines represent the crests of a wave in a ripple tank.

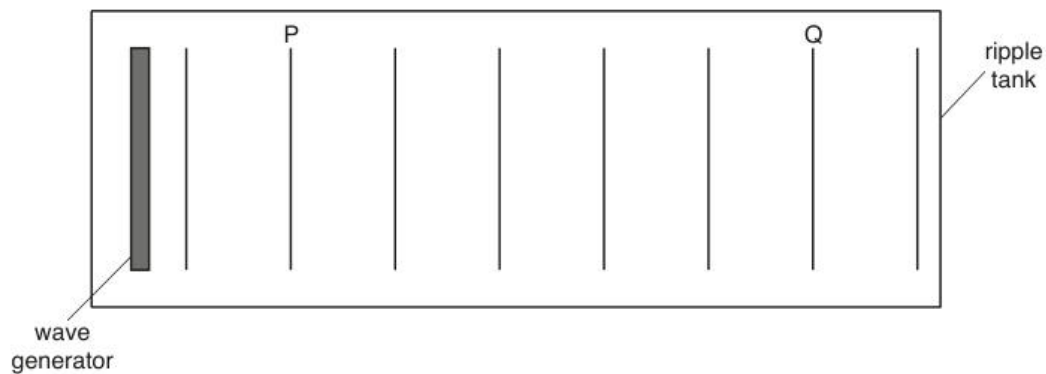


Fig. 10.1

Fig. 10.1 is drawn to scale, with 1.0 cm on the diagram representing 4.0 cm in the tank.

A crest takes 2.0 s to travel from P to Q.

(a) (i) Determine the wavelength of the wave in the ripple tank.

wavelength = [2]

(ii) Define the term *frequency*.

.....
 [2]

(iii) Determine the frequency of the wave.

frequency = [1]

- (iv) The wave generator is a bar of wood.

Describe how it is used to produce the wave shown.

.....

 [2]

- (v) Some water in the ripple tank is removed. The water becomes less deep. No other change is made.

State how the change in depth affects the wavelength and the speed of the wave.

speed
 wavelength [1]

- (b) A student reads in a textbook that the speed of sound in air is about 300 m/s.

He plans an experiment to check this value.

He asks another student to stand a long distance away in a large open space.

One student has a starting pistol. The pistol, when fired, produces a loud sound and a puff of smoke at the same instant.

- (i) Other pieces of apparatus are needed for this experiment in order to measure the speed of sound.

Name two other pieces of apparatus that are needed.

1.
 2. [2]

- (ii) Describe how all the measurements are made.

.....

 [3]

(iii) State typical values for the speed of sound in:

1. a liquid

.....

2. a solid.

.....

[2]

[Total: 15]

Mark scheme — 10(a)(i) 1.7 (cm) seen or any multiple of wavelength measured on diagram C1 6.4–7.2 cm A1 10(a)(ii) number of waves/cycles/oscillations in one second / unit time C1 number of wavelengths/crests/troughs/ wave cycles/wave oscillations generated/made/pass a point in one second/unit time or number of oscillations of a source/particle in one second / unit time A1 10(a)(iii) 2.5 Hz B1 10(a)(iv) moved up and down B1 regularly or at constant frequency / 2.5 times a second B1 10(a)(v) speed smaller / decreases and wavelength smaller / decreases B1 10(b)(i) tape measure or trundle wheel or microphones B1 stopwatch or timer B1 10(b)(ii) start timer on seeing the smoke from gun / sound picked up by microphone B1 stop timer on hearing sound B1 measure (large) distance between students B1 10(b)(iii) both between and including 1000 and 10 000 m/s B1 solid speed > liquid speed B1 5054/21 Question Answer Marks

Question 5

5 Both sound and ultrasound are longitudinal waves.

(a) Describe what is meant by *longitudinal*.

.....

.....

..... [2]

(b) Fig. 5.1 shows an ultrasound cleaner being used to clean jewellery.

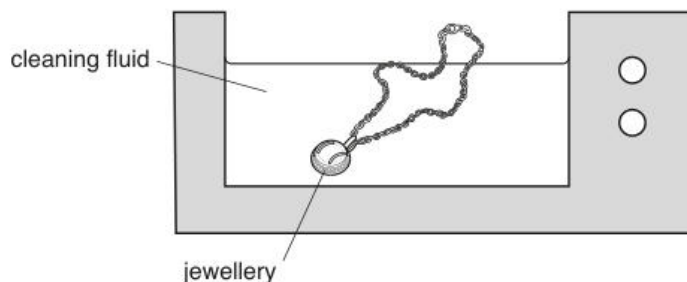


Fig. 5.1

The jewellery is lowered into the cleaning fluid and ultrasound waves of frequency 42 000 Hz are produced in the fluid.

(i) The speed of ultrasound in the fluid is 1500 m/s.

Calculate the wavelength of the ultrasound in the fluid.

wavelength = [2]

(ii) Describe how the ultrasound cleans the jewellery.

.....

.....

..... [2]

[Total: 6]

Mark scheme — 5(a) vibration / oscillation of particles or energy transfer or compressions and rarefactions B1 vibration / oscillation (of particles) parallel to energy travel direction / propagation direction B1 5054/22 Question Answer Marks 5(b)(i) ($\lambda =$) v / f or $1500 / 42\,000$ C1 0.036 m or 3.6 cm A1 5(b)(ii) (ultrasound) vibrates (the cleaning fluid / jewellery) B1 dirt shaken off B1 Question Answer Marks

Question 23

23 Four students **A**, **B**, **C** and **D**, investigate the diffraction of water waves through a gap.



Each student uses a different gap size and a different wavelength for the water waves.

Which student produces the waves which have the most diffraction?

	gap size / cm	wavelength / cm
A	2.0	1.8
B	3.0	2.1
C	4.0	2.0
D	5.0	0.9

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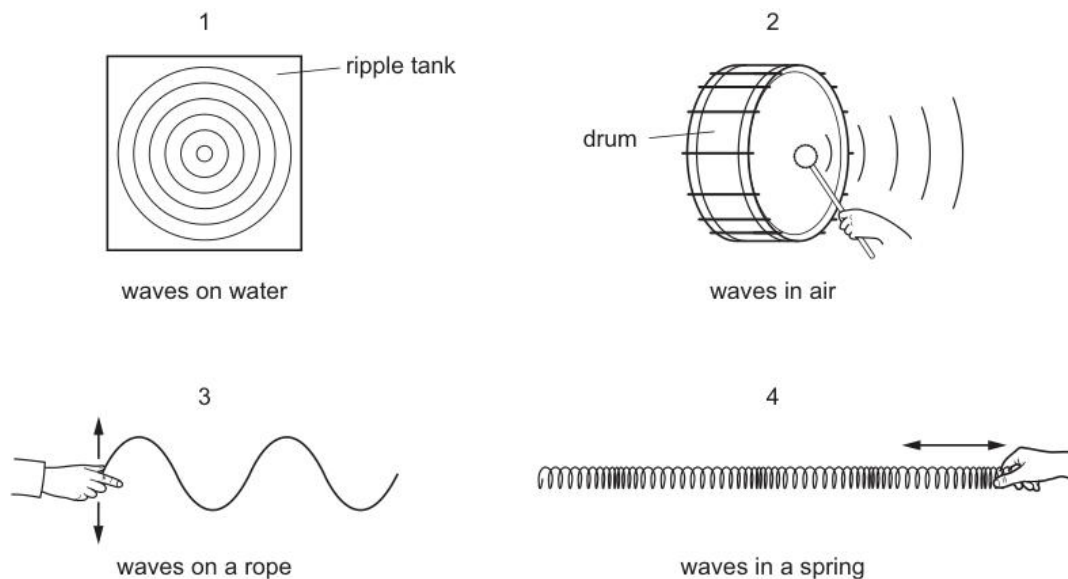
0625/22/F/M/20

[Turn over

Mark scheme — Answer: A [1]

Question 24

24 The diagrams show examples of wave motion.



Which waves are longitudinal?

- A** 1 only **B** 2 and 3 only **C** 2, 3 and 4 **D** 2 and 4 only

Mark scheme — Answer: D [1]

Question 27

- 27** The wavelength of blue light changes from $4.7 \times 10^{-7} \text{ m}$ to $3.5 \times 10^{-7} \text{ m}$ as it passes from air to water.

What is the speed of this light in water?

- A** $7.4 \times 10^7 \text{ m/s}$
- B** $1.3 \times 10^8 \text{ m/s}$
- C** $2.2 \times 10^8 \text{ m/s}$
- D** $3.0 \times 10^8 \text{ m/s}$

Mark scheme — Answer: C [1]

May/June 0625/21 MCQ

Question 20

- 20** A wave of frequency 6600 Hz travels 1320 m in 4.0 s.

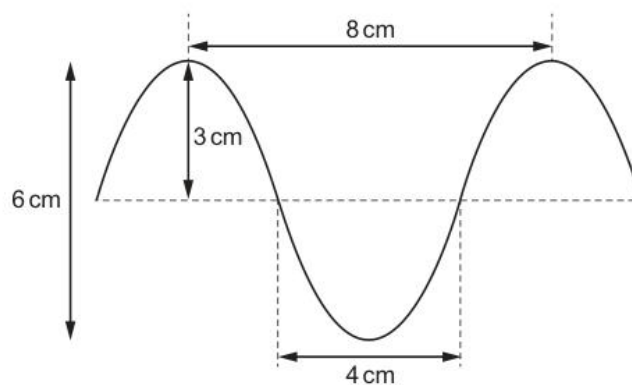
What is the wavelength?

- A** 0.050 m **B** 0.80 m **C** 1.3 m **D** 20 m

Mark scheme — Answer: A [1]

Question 21

21 The diagram shows a wave.



What are the amplitude and the wavelength of this wave?

	amplitude / cm	wavelength / cm
A	3	4
B	3	8
C	6	4
D	6	8

Mark scheme — Answer: B [1]

Question 25

25 Sound travels through air as a series of compressions and rarefactions.

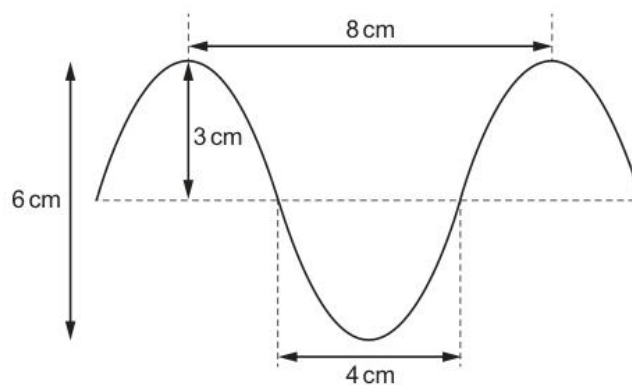
Which statement correctly compares a compression with a rarefaction?

- A** In a compression the wavelength is longer than in a rarefaction.
- B** In a compression the wavelength is shorter than in a rarefaction.
- C** In a compression the density of the air is greater than in a rarefaction.
- D** In a compression the density of the air is lower than in a rarefaction.

Mark scheme — Answer: C [1]

Question 20

20 The diagram shows a wave.



What are the amplitude and the wavelength of this wave?

	amplitude / cm	wavelength / cm
A	3	4
B	3	8
C	6	4
D	6	8

Mark scheme — Answer: B [1]

Question 21

21 The frequency of the microwaves used in a microwave oven is 2400 MHz.

What is the wavelength of these microwaves?

- A** 0.125 m **B** 8.00 m **C** 125 m **D** 7200 m

Mark scheme — Answer: A [1]

Question 20

D	20	2
----------	----	---

- 6 A rectangular gymnasium is 50 m long, 25 m wide and 8.0 m high.
The density of air is 1.2 kg/m^3 .
What is the best estimate of the mass of air in the gymnasium?
A 0.00012 kg **B** 100 kg **C** 8300 kg **D** 12 000 kg
- 7 Which moving object has a resultant force acting on it?
A a diver rising vertically through water at constant speed
B an aircraft circling an airport at constant speed
C a parachutist descending vertically at terminal velocity
D a train going up a straight slope at constant speed
- 8 Which quantity is **not** a vector?
A acceleration
B temperature
C velocity
D weight

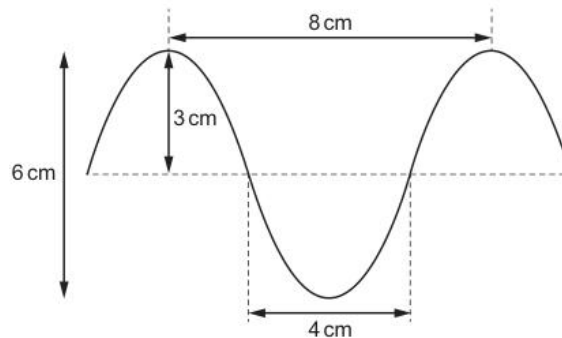
- 20 An earthquake-monitoring station records the arrival of 16 complete waves of an earthquake wave in 20 s.

The speed of the earthquake wave is 6.0 km/s.

What is the wavelength of the earthquake wave?

- A 1.3×10^{-4} m
 B 2.1×10^{-4} m
 C 4.8×10^3 m
 D 7.5×10^3 m

- 21 The diagram shows a wave.



What are the amplitude and the wavelength of this wave?

	amplitude / cm	wavelength / cm
A	3	4
B	3	8
C	6	4
D	6	8

- 22 Which statement is correct?

- A The speed of light in glass is equal to the speed of light in a vacuum multiplied by the refractive index of glass.
 B The incident angle of a light ray at an air-glass surface is the angle between the ray and the glass surface.
 C The sine of the critical angle at an air-glass surface is equal to $\frac{1}{\text{refractive index of glass}}$.
 D The angle of refraction for light passing through an air-glass surface is proportional to the angle of incidence at that surface.

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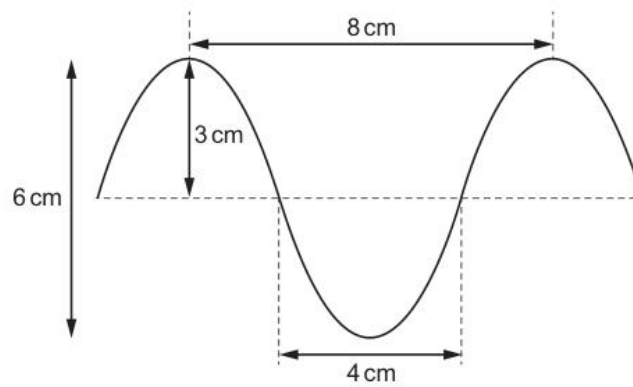
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Mark scheme — Answer: D [1]

Question 21

21 The diagram shows a wave.



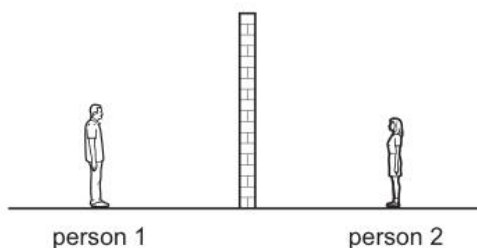
What are the amplitude and the wavelength of this wave?

	amplitude / cm	wavelength / cm
A	3	4
B	3	8
C	6	4
D	6	8

Mark scheme — Answer: B [1]

Question 25

25 Two people are standing outdoors on either side of a high wall.



Person 1 can hear person 2 talking although he cannot see her.

Which statement explains this?

- A The sound waves have diffracted around the wall.
- B The sound waves have passed unaffected through the wall.
- C The sound waves have reflected around the wall.
- D The sound waves have refracted around the wall.

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Mark scheme — Answer: A [1]

Oct/Nov 0625/21 MCQ

Question 21

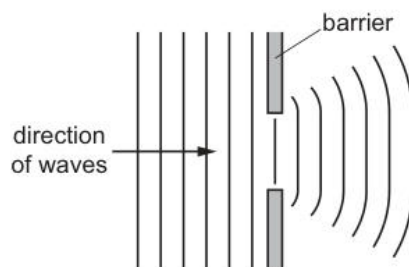
21 Which row correctly describes light waves?

	wave type	direction of vibrations
A	longitudinal	parallel to direction of wave travel
B	longitudinal	perpendicular to direction of wave travel
C	transverse	parallel to direction of wave travel
D	transverse	perpendicular to direction of wave travel

Mark scheme — Answer: D [1]

Question 22

22 The diagram shows part of a diffracted wave pattern.



Changes are made to the wavelength and to the gap size to produce a semicircular diffracted wave pattern.

Which row produces the required semicircular diffracted wave pattern?

	gap in barrier	wavelength
A	larger	same
B	larger	smaller
C	same	larger
D	same	smaller

Mark scheme — Answer: C [1]

Oct/Nov 0625/22 MCQ

Question 21

21 Which row correctly describes light waves?

	wave type	direction of vibrations
A	longitudinal	parallel to direction of wave travel
B	longitudinal	perpendicular to direction of wave travel
C	transverse	parallel to direction of wave travel
D	transverse	perpendicular to direction of wave travel

Mark scheme — Answer: D [1]

Question 22

22 A radio transmitter broadcasts at a frequency of 200 kHz.

What is the wavelength of these radio waves?

- A** $6.7 \times 10^{-4} \text{ m}$ **B** 1.5 m **C** $1.5 \times 10^3 \text{ m}$ **D** $1.5 \times 10^6 \text{ m}$

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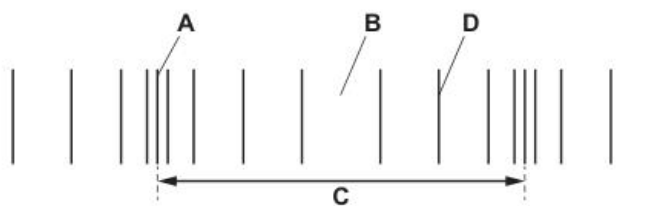
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Mark scheme — Answer: C [1]

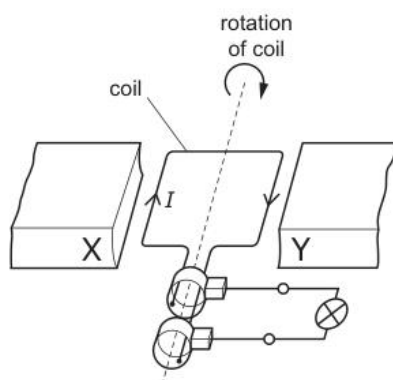
Question 24

24 A student draws a diagram to illustrate the different sections of a longitudinal wave.

Which labelled section is a rarefaction?



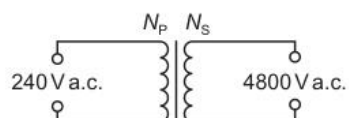
- 34 The diagram shows an a.c. generator used to power a lamp. The coil rotates in a clockwise direction.



Which magnetic poles are X and Y?

	X	Y
A	N pole	N pole
B	N pole	S pole
C	S pole	N pole
D	S pole	S pole

- 35 A transformer is needed to convert a supply of 240 V a.c. into 4800 V a.c..



Which pair of coils would be suitable for this transformer?

	number of turns on primary coil N_p	number of turns on secondary coil N_s
A	50	1 000
B	240	48 000
C	480	24
D	2000	100

Mark scheme — Answer: B [1]

Oct/Nov 0625/23 MCQ

Question 21

21 Which row correctly describes light waves?

	wave type	direction of vibrations
A	longitudinal	parallel to direction of wave travel
B	longitudinal	perpendicular to direction of wave travel
C	transverse	parallel to direction of wave travel
D	transverse	perpendicular to direction of wave travel

Mark scheme — Answer: D [1]

Question 22

22 A water wave has a speed of 2.0 m/s.

4.0 complete waves pass a point every 10 seconds.

What is the wavelength of the wave?

A 0.50 m **B** 0.80 m **C** 5.0 m **D** 8.0 m

Mark scheme — Answer: C [1]

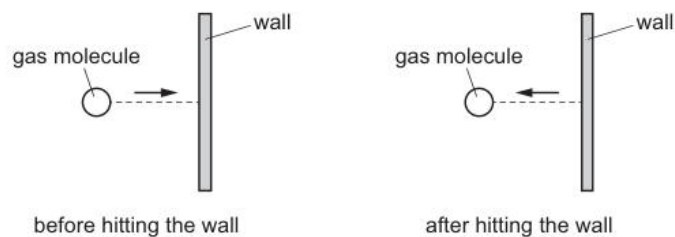
2021

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Question 20

4

- 9 A gas molecule strikes the wall of a container. The molecule rebounds with the same speed.



What happens to the kinetic energy and what happens to the momentum of the molecule?

	kinetic energy	momentum
A	changes	changes
B	changes	stays the same
C	stays the same	changes
D	stays the same	stays the same

- 10 A horizontal force pulls a box along a horizontal surface.

The box gains 30 J of kinetic energy and 10 J of thermal energy is produced by the friction between the box and the surface.

How much work is done by the force?

- A** 10 J **B** 20 J **C** 30 J **D** 40 J

- 11 A crane is used to lift loads vertically.

The output power of the crane to lift a car is P .

The crane then lifts a lorry, which has 3.0 times the weight of the car, through 0.25 of the distance in 0.50 of the time.

What is the output power of the crane now?

- A** $\frac{3P}{8}$ **B** $\frac{3P}{2}$ **C** $\frac{8P}{3}$ **D** $6P$

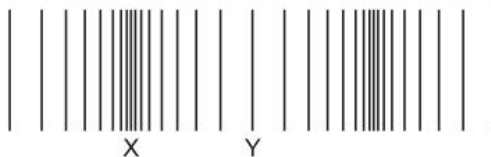
- 18 One end of a copper rod is heated.

What is one method by which thermal energy is transferred in the copper rod?

- A Free electrons transfer energy from the cooler end to the hotter end.
 - B Free electrons transfer energy from the hotter end to the cooler end.
 - C Molecules of copper move from the cooler end to the hotter end.
 - D Molecules of copper move from the hotter end to the cooler end.
- 19 Which change will cause a decrease in the rate of radiation emitted by an object?
- A changing the surface colour from white to black
 - B changing the surface texture from dull to shiny
 - C increasing the surface temperature
 - D increasing the surface area
- 20 What is the approximate wavelength in air of the highest frequency sound that can be heard by a normal healthy person?
- A 0.02 m B 60 m C 20 000 m D 7 000 000 m
- 21 What causes the change in direction when light travels from air into glass?
- A The amplitude of the light changes.
 - B The colour of the light changes.
 - C The frequency of the light changes.
 - D The speed of the light changes.
- 22 Light from a torch is incident on a plane mirror. The angle of incidence is 38° .
- What is the angle of reflection?
- A 38° B 52° C 76° D 142°

Question 26

26 The diagram represents a sound wave.



What are the names of the parts of the sound wave labelled X and Y?

	X	Y
A	amplitude	wavelength
B	compression	rarefaction
C	rarefaction	amplitude
D	wavelength	compression

Mark scheme — Answer: B [1]

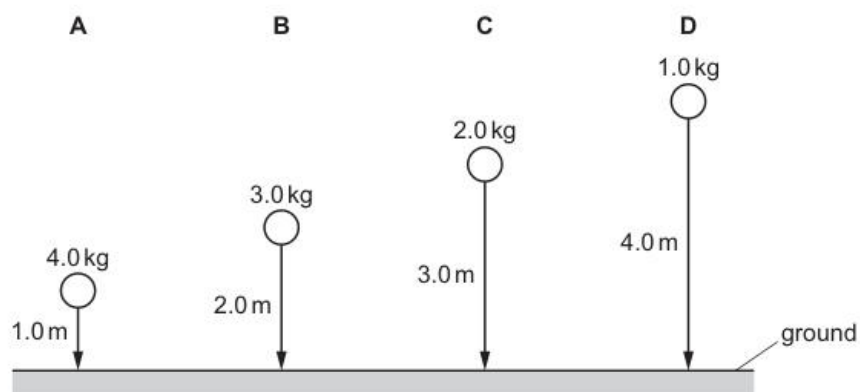
May/June 0625/21 MCQ

Question 3

- 3 Four balls with different masses are dropped from the heights shown.

Air resistance may be ignored.

Which ball has the smallest average speed?

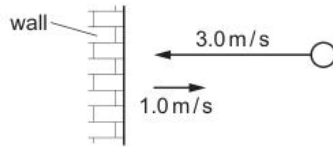


- 7 A cart has a mass of 10 kg. A boy pushes on the cart horizontally with a force of 50 N. The cart accelerates at 0.50 m/s^2 .

What is the frictional force acting on the cart?

- A 5.0 N B 20 N C 30 N D 45 N

- 8 A ball has a mass of 2.0 kg. The ball approaches a wall at a speed of 3.0 m/s and rebounds at a speed of 1.0 m/s .



What is the impulse on the wall?

- A 4.0 N B 4.0 Ns C 8.0 N D 8.0 Ns

- 9 Which situation involves no work being done and no energy being transferred?

- A a car skidding to a stop on a road
 B a crane lifting a load
 C a heavy load hanging from a strong bar
 D a student dragging a big box over a rough floor

- 10 A student suggests that there are several ways of transferring energy to a small, stationary block of iron on a smooth table. He makes the following suggestions.

- 1 Heat it.
- 2 Shine light on it.
- 3 Pass a current through it.

Which suggestions are correct?

- A 1 and 2 only B 1 and 3 only C 2 and 3 only D 1, 2 and 3

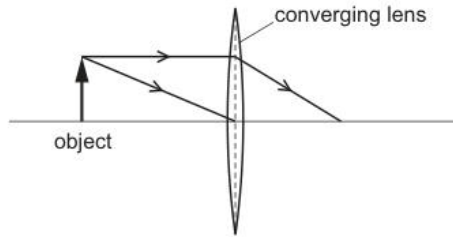
- 11 An engine produces 240 kJ of energy in 2.0 minutes.

What is the power output of the engine?

- A 2.0 kW B 120 kW C 480 kW D 28 800 kW

- 24 An object is placed in front of a thin converging lens.

The diagram shows the paths of two rays from the top of the object.



An image of the object is formed on a screen to the right of the lens.

How does this image compare with the object?

- A It is larger and inverted.
 - B It is larger and the same way up.
 - C It is smaller and inverted.
 - D It is smaller and the same way up.
- 25 A remote-controlled vehicle is travelling on the surface of a planet. The vehicle senses an obstacle ahead. It sends a radio message to the control room from where it is being controlled. The control room is 2.4×10^6 km away from the vehicle. The control room sends a message back to the vehicle telling it to stop.

What is the minimum time that elapses between the vehicle sensing the obstacle and receiving the message back from the control room?

- A 8.0 ms B 16 ms C 8.0 s D 16 s
- 26 A sound wave is travelling outwards from a loudspeaker into the surrounding air.

Here are three statements.

- 1 The air pressure is lower at a rarefaction compared with undisturbed air.
- 2 The density of the air is less at a compression compared with undisturbed air.
- 3 The distance from a compression to a rarefaction equals half a wavelength.

Which statements about the sound wave are correct?

- A 1 and 2 only B 1 and 3 only C 2 and 3 only D 1, 2 and 3

Question 21

- 21 A water wave passes into a region where the wave travels more slowly.

As it passes into the slow region, what happens to the frequency and what happens to the wavelength of the wave?

	frequency	wavelength
A	decreases	remains the same
B	increases	remains the same
C	remains the same	decreases
D	remains the same	increases

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Mark scheme — Answer: C [1]

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Question 21

- 21 A water wave passes into a region where the wave travels more slowly.

As it passes into the slow region, what happens to the frequency and what happens to the wavelength of the wave?

	frequency	wavelength
A	decreases	remains the same
B	increases	remains the same
C	remains the same	decreases
D	remains the same	increases

Mark scheme — Answer: C [1]

May/June 0625/23 MCQ

Question 21

21 A water wave passes into a region where the wave travels more slowly.

As it passes into the slow region, what happens to the frequency and what happens to the wavelength of the wave?

	frequency	wavelength
A	decreases	remains the same
B	increases	remains the same
C	remains the same	decreases
D	remains the same	increases

Mark scheme — Answer: C [1]

May/June 5054/22 Theory

Question 10

16

- 10 (a) Fig. 10.1 shows a water wave moving from shallow into deep water. The wavefronts shown represent the crests of the wave.

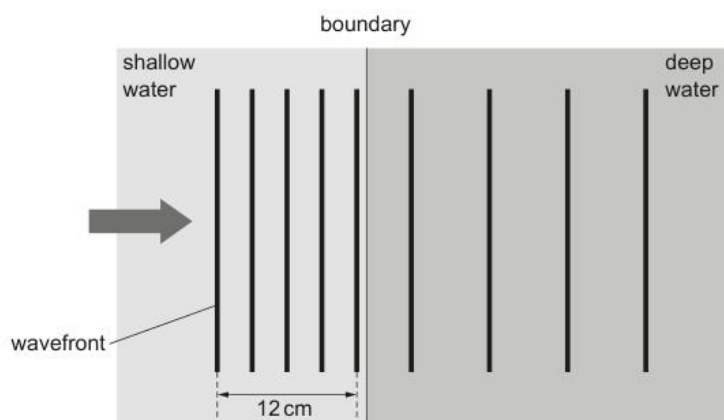


Fig. 10.1 (not to scale)

The water wave is made by dipping a wooden bar up and down in the water. The bar makes 10 complete up and down movements in 5.0 s.

- (i) Determine the wavelength of the wave in the shallow water.

wavelength = [1]

- (ii) Calculate the frequency of the wave produced.

frequency = [2]

- (iii) Determine the speed of the water wave in the shallow water.

speed = [2]

- (iv) On Fig. 10.2, draw lines to show what happens to the wavelength, the frequency and the speed of the water wave as it enters the deep water.

One line has been drawn for you.

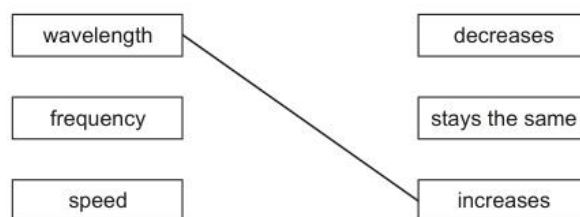


Fig. 10.2

[1]

- (v) The direction of the boundary is now altered. The water wave now enters the deep water from the shallow water at an angle, as shown in Fig. 10.3.

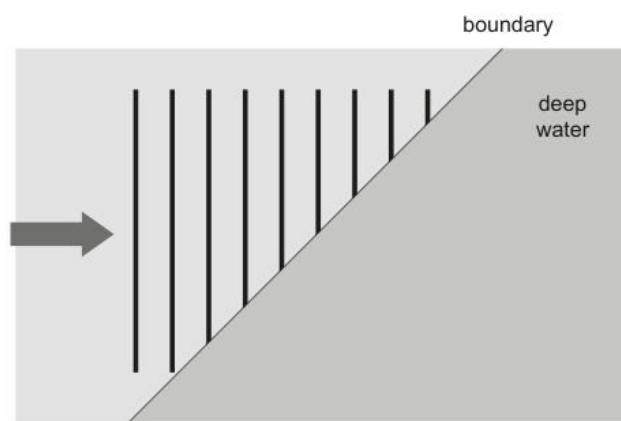


Fig. 10.3

On Fig. 10.3, draw the wavefronts in the deep water.

[3]

(b) White light is made up of different colours.

- (i) State the name of four of the colours in the visible spectrum and place them in order from the smallest wavelength to the largest wavelength.

smallest wavelength

.....

.....

largest wavelength

[2]

- (ii) A narrow beam of white light can be split into different colours.

Fig. 10.4 shows rays of white light emitted from a lamp.

Complete Fig. 10.4 to show how a narrow beam is produced from these rays and how a spectrum is shown on the screen. Label your diagram.

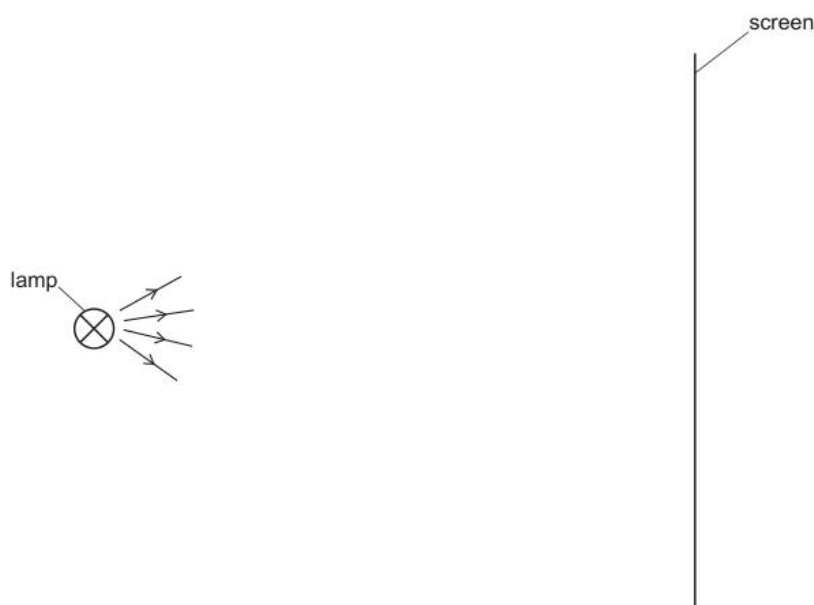


Fig. 10.4

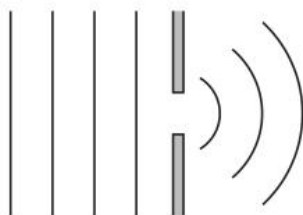
[4]

[Total: 15]

Mark scheme — 10(a)(i) 3 (.0) cm B1 10(a)(ii) number of waves per second or 5 / 10 or $f = 1 / T$ C1 2(.0) Hz A1 10(a)(iii) ($v = f \lambda$ or $s = d / t$ numerical or algebraic in any form C1 6(.0) cm / s A1 10(a)(iv) frequency stays the same and speed increases B1 5054/22 Question Answer Marks 10(a)(v) ■ three consecutive incident wavefronts meeting straight refracted wavefronts on the boundary without any wavefronts in-between B1 constant wavelength shown between all refracted wavefronts B1 all wavefronts refracted in correct direction, i.e. lines between the vertical and the normal B1 10(b)(i) four colours from violet, indigo, blue, green, yellow, orange, red B1 correct order of four of the above colours B1 10(b)(ii) prism B1 slit (to produce narrow beam) B1 dispersion shown at first or second face, i.e. a ray becoming at least two rays at least one face B1 correct refraction on first and second faces for all rays shown B1 Question Answer Marks

Question 17

- 17 The diagram shows waves in a ripple tank containing water.



The waves approach a barrier and pass through the gap in the barrier.

The size of the gap is about the same size as the wavelength of the ripples.

The gap size is increased.

What happens to the ripple pattern to the right of the barrier?

- A The ripples are closer together.
- B The ripples are further apart.
- C The ripples are more curved.
- D The ripples are less curved.

Mark scheme — Answer: D [1]

Question 19

- 19 Two beams of light are both the same colour of red. One beam is travelling through air. The other beam is travelling through water. Each beam has a different brightness.

Which quantity is the same for both sets of waves?

- A amplitude
- B frequency
- C speed
- D wavelength

Mark scheme — Answer: B [1]

Question 22

22 The Sun emits infrared radiation and light.

Light from the Sun reaches the Earth in 8 minutes.

Which row gives correct information about the infrared radiation?

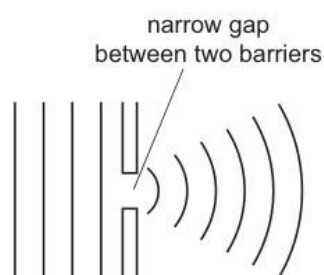
	wavelength of infrared radiation	time taken for infrared radiation to reach the Earth
A	longer than wavelength of light	8 minutes
B	longer than wavelength of light	much less than 8 minutes
C	shorter than wavelength of light	8 minutes
D	shorter than wavelength of light	much more than 8 minutes

Mark scheme — Answer: A [1]

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Question 17

17 The diagram shows the pattern of water waves as they pass through a narrow gap.



Which row names the process shown and describes the effect of using a wider gap?

	name of process	wider gap
A	refraction	waves spread out less
B	refraction	waves spread out more
C	diffraction	waves spread out less
D	diffraction	waves spread out more

Mark scheme — Answer: C [1]

Question 18

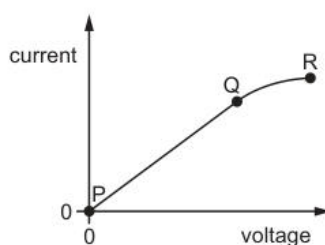
18 Which row is **not** correct for a wave on the surface of water?

	quantity	usual unit
A	amplitude	m
B	frequency	Hz
C	wavelength	λ
D	speed	m/s

- 26 Which circuit symbol represents a component used to measure electric current?

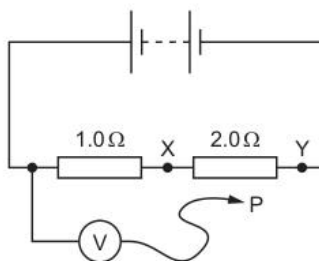


- 27 The graph shows the current–voltage characteristic for a conductor.



Where on the graph can Ohm's law be applied to the conductor?

- A** at Q only
B between P and Q
C between P and R
D between Q and R
- 28 The diagram shows a circuit containing two resistors of resistance $1.0\ \Omega$ and $2.0\ \Omega$.
- A voltmeter is connected across the $1.0\ \Omega$ resistor by connecting P to X.
- The reading on the voltmeter is $6.0\ \text{V}$.



P is moved to point Y in the circuit.

What is the new reading on the voltmeter?

- A** $3.0\ \text{V}$ **B** $6.0\ \text{V}$ **C** $12\ \text{V}$ **D** $18\ \text{V}$

Question 22

22 The Sun emits infrared radiation and light.

Light from the Sun reaches the Earth in 8 minutes.

Which row gives correct information about the infrared radiation?

	wavelength of infrared radiation	time taken for infrared radiation to reach the Earth
A	longer than wavelength of light	8 minutes
B	longer than wavelength of light	much less than 8 minutes
C	shorter than wavelength of light	8 minutes
D	shorter than wavelength of light	much more than 8 minutes

Mark scheme — Answer: A [1]

Question 23

23 Which statement about electromagnetic waves is **not** correct?

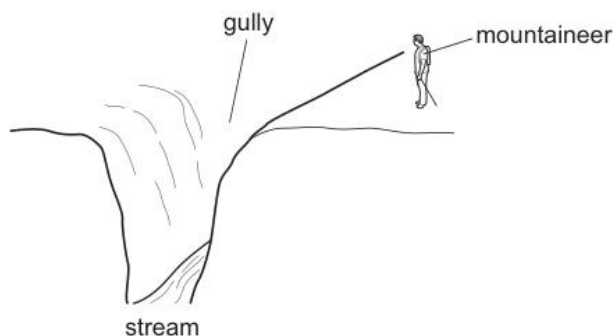
- A** They travel at 3×10^8 m/s in a vacuum.
- B** They transfer energy.
- C** They travel at 340 m/s in air.
- D** They are transverse waves.

Mark scheme — Answer: C [1]

Oct/Nov 0625/23 MCQ

Question 17

- 17 In the diagram, the mountaineer can hear the stream although he cannot see it. When he is closer to the gully, he can both hear and see the stream. When he is further from the gully, he can neither hear nor see the stream.



Which statement is **not** correct?

- A As he approaches the gully edge, he hears first the short wavelength, higher frequencies and then the long wavelength, lower frequencies.
- B As he approaches the gully edge, the sound becomes louder.
- C He hears the stream because some of the sound is diffracted.
- D He hears the stream because some of the sound is reflected from the opposite wall of the gully.

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Mark scheme — Answer: A [1]

Question 22

- 22 The Sun emits infrared radiation and light.

Light from the Sun reaches the Earth in 8 minutes.

Which row gives correct information about the infrared radiation?

	wavelength of infrared radiation	time taken for infrared radiation to reach the Earth
A	longer than wavelength of light	8 minutes
B	longer than wavelength of light	much less than 8 minutes
C	shorter than wavelength of light	8 minutes
D	shorter than wavelength of light	much more than 8 minutes

Mark scheme — Answer: A [1]

Oct/Nov 5054/21 Theory

Question 8

- 8 A loudspeaker is made from a coil of wire fixed to a cardboard tube. The tube is attached to a cardboard cone.

Fig. 8.1 shows part of the arrangement of the loudspeaker.

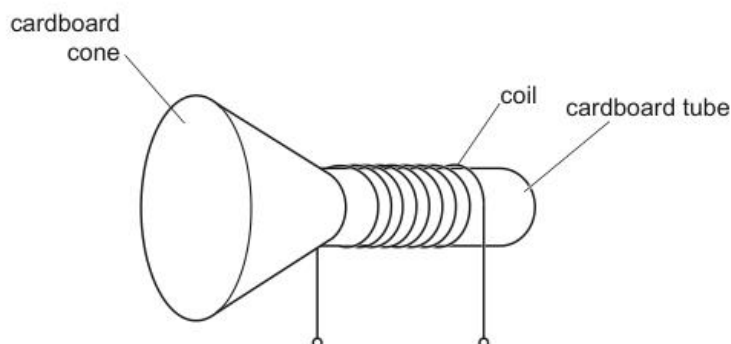


Fig. 8.1

When there is a current in the coil, the coil experiences a force.

- (a) State what else is needed in a loudspeaker to make a current-carrying wire experience a force.

..... [1]

- (b) A student connects the coil to the output of an alternating current (a.c.) generator. Fig. 8.2 shows how the electromotive force (e.m.f.) produced by the generator varies with time.

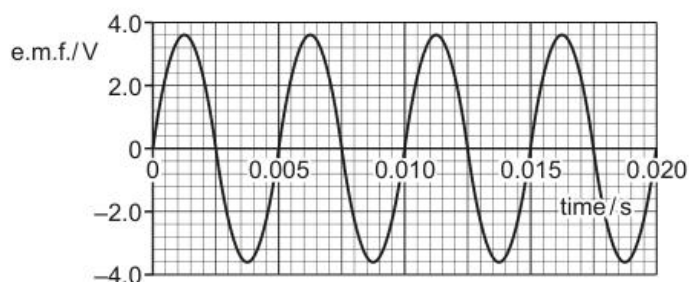


Fig. 8.2

The coil, tube and cone vibrate backwards and forwards.

- (i) Explain why the e.m.f. shown in Fig. 8.2 makes the coil vibrate.

.....

 [2]

- (ii) The vibrating cone produces sound in the surrounding air.

Explain, in terms of molecules, how the cone produces a sound wave that travels through the air.

.....

.....

.....

.....

..... [3]

- (iii) Sound is a longitudinal wave.

Describe the difference between a longitudinal wave and a transverse wave.

.....

.....

.....

..... [2]

- (iv) Using Fig. 8.2, determine the number of times that the cone reverses its direction of motion in 1.0 s.

number = [2]

- (v) The speed of sound in air is 340 m/s.

Calculate the wavelength of the sound.

wavelength = [2]

- (c) The student adjusts the generator so that the maximum voltage of the output is now 3.0 V. Everything else stays the same as the output shown on the graph in Fig. 8.2.

(i) Explain any effect on the loudness of the sound.

.....
..... [2]

(ii) Explain any effect on the pitch of the sound.

.....
..... [1]

[Total: 15]

Question 9 starts over the page

Mark scheme — 8(a)(i) magnetic field or magnet B1 8(b)(i) the current (in coil continually) reverses (direction) or the force (on the coil continually) reverses B1 the current (in the coil continually) reverses (direction) and the force (on the coil continually) reverses B1 5054/21 Question Answer Marks 8(b)(ii) cone moves forwards and air is compressed / hits (air) molecules B1 cone moves backwards and rarefaction produced / molecules sucked back B1 (air) molecules vibrate or production of compressions and rarefactions continues B1 8(b)(iii) vibration direction is parallel to energy travel / propagation direction B1 in transverse wave, vibration direction is perpendicular to propagation / energy travel direction B1 8(b)(iv) (time period =) 0.0050 (s) or 1 / 0.0050 or 200 (times per second) C1 400 (times per second) A1 8(b)(v) ($\lambda =$) c / v or 340 / 200 C1 1.7 m A1 8(c)(i) decreases / less loud / quieter B1 amplitude of sound / vibration decreases or (maximum) force on coil decreases B1 8(c)(ii) (pitch) stays the same / unchanged and no change to frequency / time period B1 Question Answer Marks

2022

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Question 21

21 A tank contains water. Ripples are produced on the surface of the water. Refraction is observed.

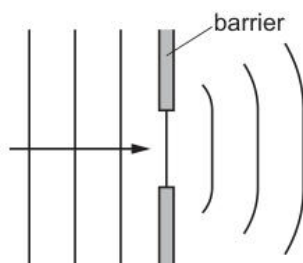
What causes the ripples to refract?

- A The cold water in the tank is replaced by warm water.
- B The ripples change speed as they move from deep to shallow water.
- C The ripples hit the wall of the tank.
- D The ripples pass through a narrow gap.

Mark scheme — Answer: B [1]

Question 22

22 The diagram shows wavefronts of a water wave passing through a gap in a barrier.



Which change will increase the diffraction of the wave as it passes through the gap?

- A Increase the amplitude of the wave.
- B Increase the width of the gap.
- C Reduce the depth of water.
- D Reduce the frequency of the wave.

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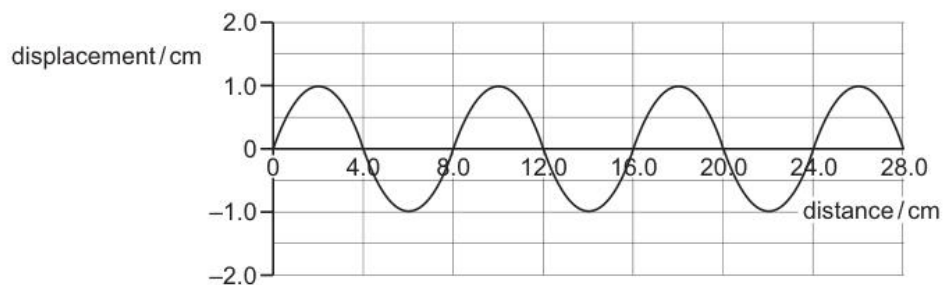
0625/22/F/M/22

Mark scheme — Answer: D [1]

May/June 0625/21 MCQ

Question 21

21 The diagram shows a wave.



Which row is correct?

	amplitude of the wave / cm	wavelength of the wave / cm
A	1.0	4.0
B	1.0	8.0
C	2.0	4.0
D	2.0	8.0

Mark scheme — Answer: B [1]

Question 22

22 A sound wave is created by a loudspeaker that vibrates backwards and forwards 96 000 times per minute.

The speed of sound is 320 m/s.

What is the wavelength of the sound wave?

- A** 0.20 m **B** 5.0 m **C** 300 m **D** 18 000 m

Mark scheme — Answer: A [1]

Question 27

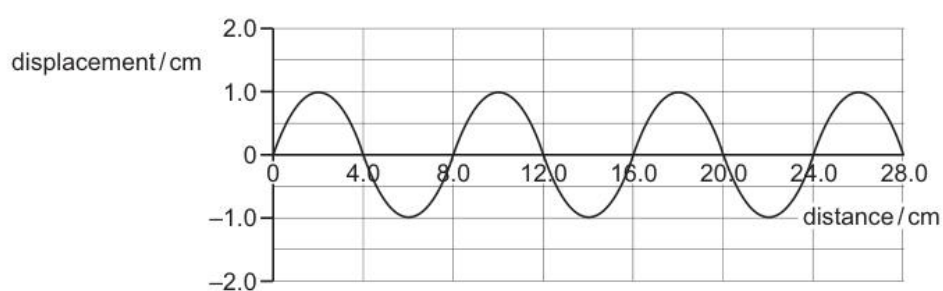
- 27 Which statement correctly compares radio waves and X-rays?
- A Radio waves have a longer wavelength and a greater speed in a vacuum.
 - B Radio waves have a longer wavelength and the same speed in a vacuum.
 - C Radio waves have a shorter wavelength and a greater speed in a vacuum.
 - D Radio waves have a shorter wavelength and the same speed in a vacuum.

Mark scheme — Answer: B [1]

May/June 0625/22 MCQ

Question 21

- 21 The diagram shows a wave.



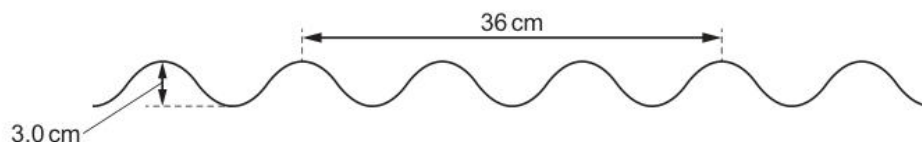
Which row is correct?

	amplitude of the wave / cm	wavelength of the wave / cm
A	1.0	4.0
B	1.0	8.0
C	2.0	4.0
D	2.0	8.0

Mark scheme — Answer: B [1]

Question 22

22 The water wave shown has a frequency of 4.0 Hz.



What is the speed of the wave?

- A** 3.0 cm/s **B** 12 cm/s **C** 48 cm/s **D** 72 cm/s

Mark scheme — Answer: C [1]

Question 27

27 Which statement correctly compares radio waves and X-rays?

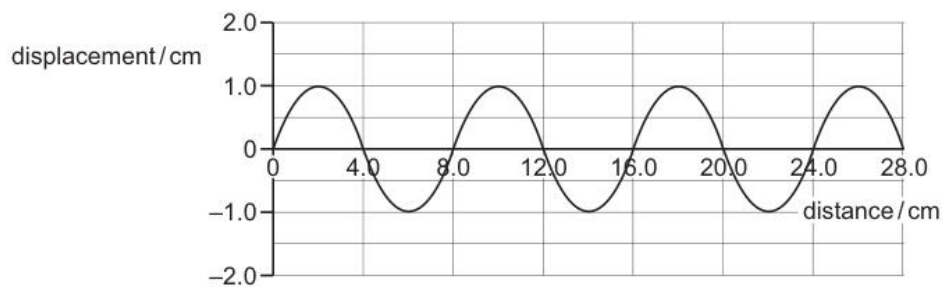
- A** Radio waves have a longer wavelength and a greater speed in a vacuum.
- B** Radio waves have a longer wavelength and the same speed in a vacuum.
- C** Radio waves have a shorter wavelength and a greater speed in a vacuum.
- D** Radio waves have a shorter wavelength and the same speed in a vacuum.

Mark scheme — Answer: B [1]

May/June 0625/23 MCQ

Question 21

21 The diagram shows a wave.



Which row is correct?

	amplitude of the wave / cm	wavelength of the wave / cm
A	1.0	4.0
B	1.0	8.0
C	2.0	4.0
D	2.0	8.0

Mark scheme — Answer: B [1]

Question 22

22 The wavelength of a beam of X-rays, travelling through air, is 5.4×10^{-10} m.

What is its frequency?

- A** 5.6×10^{-17} Hz
- B** 5.6×10^{11} Hz
- C** 5.6×10^{17} Hz
- D** 5.6×10^{18} Hz

Mark scheme — Answer: C [1]

Question 27

27 Which statement correctly compares radio waves and X-rays?

- A** Radio waves have a longer wavelength and a greater speed in a vacuum.
- B** Radio waves have a longer wavelength and the same speed in a vacuum.
- C** Radio waves have a shorter wavelength and a greater speed in a vacuum.
- D** Radio waves have a shorter wavelength and the same speed in a vacuum.

Mark scheme — Answer: B [1]

May/June 5054/22 Theory

Question 8

- 8 (a) Fig. 8.1 shows a ripple tank and the crests of the water wave that is produced in it.

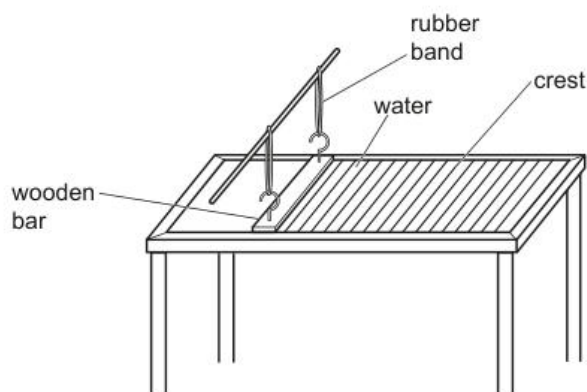


Fig. 8.1

The frequency of the water wave is 2.0 Hz and its amplitude is 3.0 mm .

- (i) Calculate the number of crests produced in 1.5 s .

number = [1]

- (ii) The height of the wave is measured from the level of the undisturbed surface of the water. The height of the wave at one point is 0 at time = 0.

On Fig. 8.2, draw a graph to show how the height of the wave at this point varies with time.

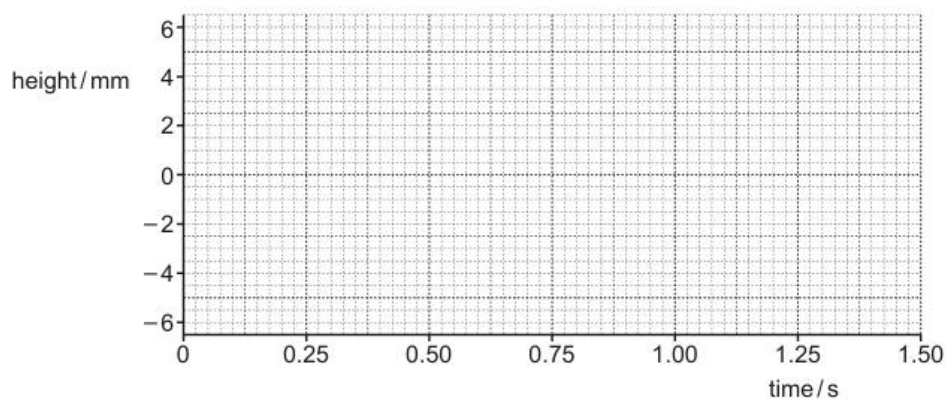


Fig. 8.2

[2]

- (b) (i) The frequency of the wave is increased.

Describe how the apparatus shown in Fig. 8.1 is adjusted so that the frequency of the wave is increased.

.....
 [1]

- (ii) State what happens to the speed and wavelength of the wave as the frequency increases.

speed
 wavelength [2]

- (c) The apparatus shown in Fig. 8.1 can be used to demonstrate refraction.

- (i) State the additional apparatus needed to demonstrate refraction.

..... [1]

- (ii) Draw on Fig. 8.3 to show the refraction of the water wave.

Label a boundary where the refraction occurs.

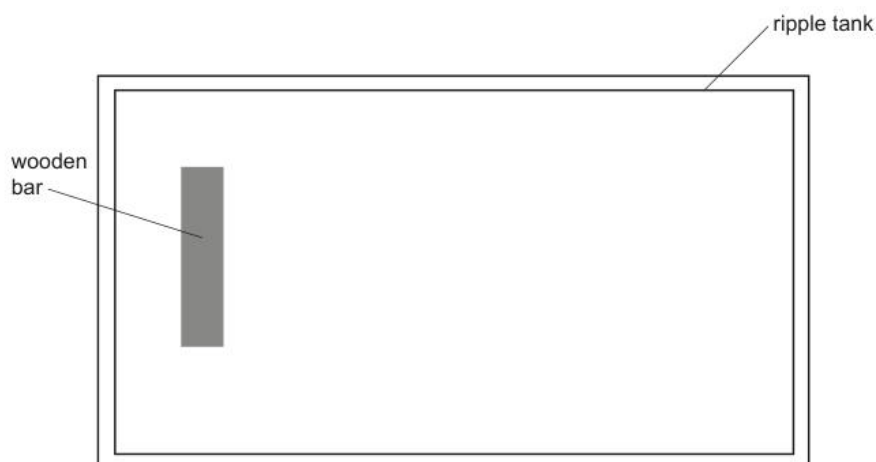


Fig. 8.3

[3]

- (d) Fig. 8.4 shows a connection to the internet made from a remote station A using a satellite above the Earth's surface.

Data is sent between stations A and B using microwaves which travel to and from the satellite.

Station B is connected directly to the internet using optical fibre.

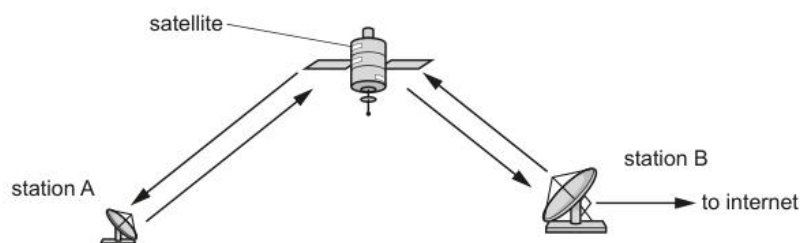


Fig. 8.4 (not to scale)

The speed of microwaves is 3.0×10^8 m/s.

Stations A and B are each 560 km from the satellite.

- (i) Calculate the time taken for data to travel from A to B using microwaves.

time = [3]

- (ii) The same data can be sent from A to B in a shorter time along the surface of the Earth using infrared waves to carry the information in optical fibres.

State **one** property that infrared waves and microwaves have in common.

.....
 [1]

- (iii) Suggest **one** other advantage of using optical fibres to connect stations A and B directly.

.....
 [1]

[Total: 15]

Question 9 begins over the page.

Mark scheme — 8(a)(i) 3 B1 8(a)(ii) shape approx. sinusoidal with amplitude 3 mm for the first complete wave (may then change) B1 1 wave taking 0.5 s for first wave shown within half a square or 1.5 s shown for the number of waves in (a)(i) B1 8(b)(i) move bar up and down more often (per second) / faster or make rubber band(s) strong(er) / short(er) / tight(er) / more in number or use bar of less mass / weight B1 8(b)(ii) speed remains the same B1 wavelength decreases B1 8(c)(i) (piece of) plastic / (piece of) glass / block / plate or a shallow(er) / deep(er) area / boundary B1 8(c)(ii) diagram showing within the ripple tank: ■ crests parallel to bar hitting boundary with a boundary labelled B1 ■ correct refraction of plane crests either towards or away from normal on the same side of the normal or no direction change if crests are initially parallel to boundary but with a wavelength change B1 ■ different wavelength shown before and after refraction and constant before and after (by eye) B1 8(d)(i) (t =) distance / speed algebraic or numerical seen in any form C1 2 ■ 560(000) seen or used or 0.0019 (s) C1 0.0037 s A1 5054/22 Question Answer Marks 8(d)(ii) One of (both): ■ reflect / refract ■ have same speed (in a vacuum) ■ travel in a vacuum / don't need a medium ■ are electromagnetic ■ are transverse ■ carry energy / cause heating ■ have larger wavelength / smaller frequency than visible light B1

8(d)(iii) less likely to be hacked / intercepted / greater security or more channels / better bandwidth possible / more data or less likely for transmission to be interrupted or cost of satellites saved / satellites expensive B1 Question Answer Marks

Oct/Nov 0625/21 MCQ

Question 1

- 1 Which measuring instrument is used to measure the diameter of a thin metal wire?
- A 30 cm rule
 - B measuring tape
 - C metre rule
 - D micrometre screw gauge

- 9 An object has kinetic energy of 200 J.

A constant resultant force of 190 N is applied in the direction of its motion through a distance of 10 m.

What is the final kinetic energy of the object?

- A 390 J B 1700 J C 2000 J D 2100 J

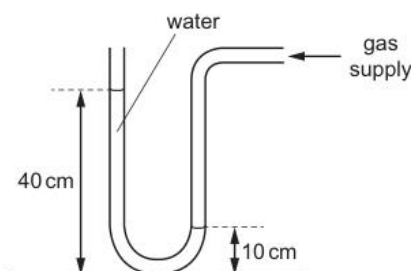
- 10 The statements describe what happens when the power of a machine is increased.

- 1 The work done in a given time decreases.
- 2 The work done in a given time increases.
- 3 The time taken to do a given quantity of work decreases.
- 4 The time taken to do a given quantity of work increases.

Which statements are correct?

- A 1 and 3 B 1 and 4 C 2 and 3 D 2 and 4

- 11 A manometer containing water is used to measure the pressure of a gas supply, as shown.

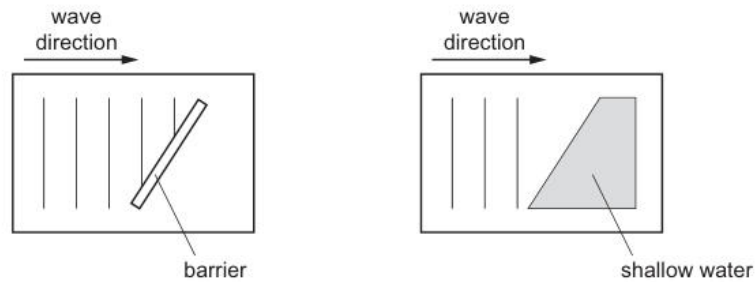


The density of water is 1000 kg/m^3 .

What is the pressure of the gas supply?

- A 300 Pa above atmospheric pressure
 B 400 Pa above atmospheric pressure
 C 3000 Pa above atmospheric pressure
 D 4000 Pa above atmospheric pressure

18 The diagrams show two sets of wavefronts in a ripple tank.



A student makes two statements about the waves.

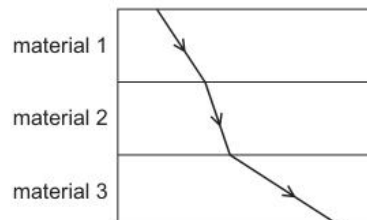
- 1 When the waves reflect from the barrier the direction changes but the wavelength remains the same.
- 2 When the waves refract as they enter the shallow water the direction remains the same, but the wavelength changes.

Which statements are correct?

- A** statement 1 and statement 2
B statement 1 only
C statement 2 only
D neither statement 1 nor statement 2

19 A composite block is made by joining together three transparent materials.

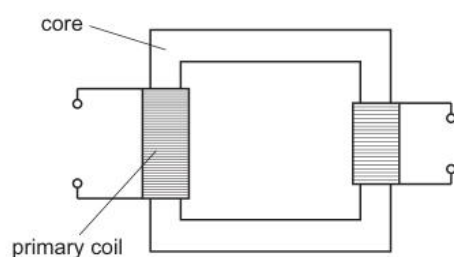
The diagram shows a ray of light passing through the composite block.



Which list gives the three materials in order of the speeds of light in the materials, from slowest to fastest?

- A** 1 → 2 → 3 **B** 1 → 3 → 2 **C** 2 → 1 → 3 **D** 2 → 3 → 1

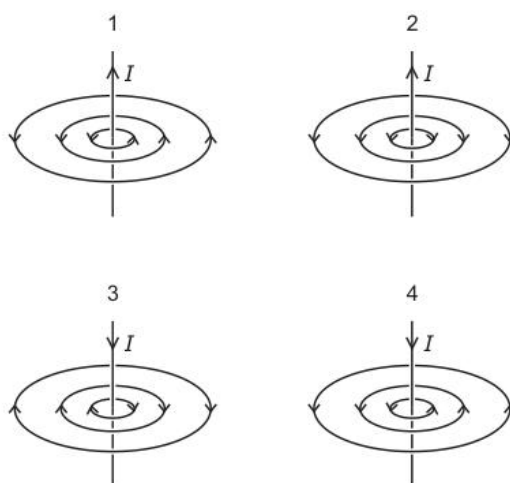
34 The diagram represents a transformer.



Which row shows materials suitable for making the core and the primary coil?

	core	primary coil
A	iron	copper
B	iron	plastic
C	steel	copper
D	steel	plastic

35 The diagrams show the magnetic field lines around a wire carrying a current, I .



Which diagrams are correct?

- A** 1 only **B** 2 and 3 **C** 4 only **D** 1 and 3

Question 2

- 2 A car starts from rest.

The table shows the readings from its speedometer every 10 s.

time / s	0	10	20	30	40	50	60
$\frac{\text{speed}}{\text{m/s}}$	0	4	8	12	12	12	12

Which row describes the car's motion in the first 30 seconds and in the last 30 seconds?

	motion during first 30 s	motion during last 30 s
A	non-zero acceleration	at rest
B	zero acceleration	constant speed
C	zero acceleration	at rest
D	non-zero acceleration	constant speed

- 9 An object has kinetic energy of 200 J.

A constant resultant force of 190 N is applied in the direction of its motion through a distance of 10 m.

What is the final kinetic energy of the object?

- A 390 J B 1700 J C 2000 J D 2100 J

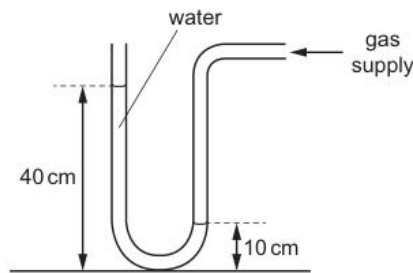
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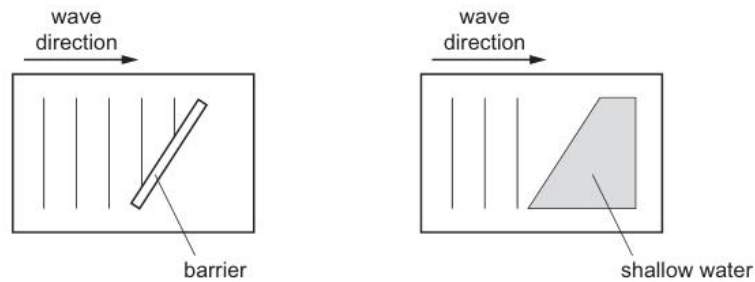


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What is the pressure of the gas supply?

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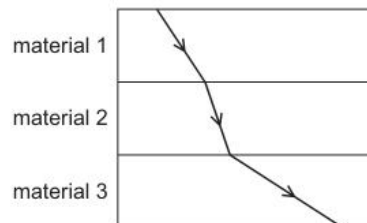
- 1 When the waves reflect from the barrier the direction changes but the wavelength remains the same.
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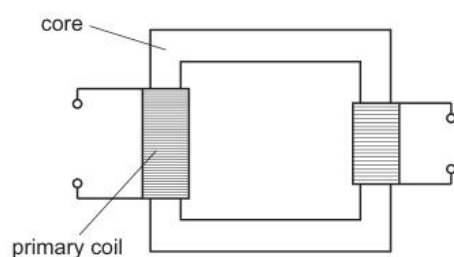
The diagram shows a ray of light passing through the composite block.



Which list gives the three materials in order of the speeds of light in the materials, from slowest to fastest?

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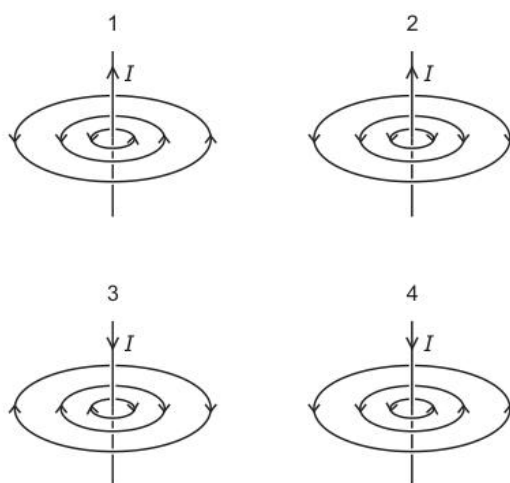
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Which diagrams are correct?

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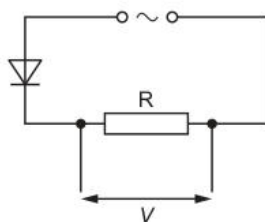
Question 17

17 A wave source produces 3000 crests every minute. The wave has a speed of 300 m/s.

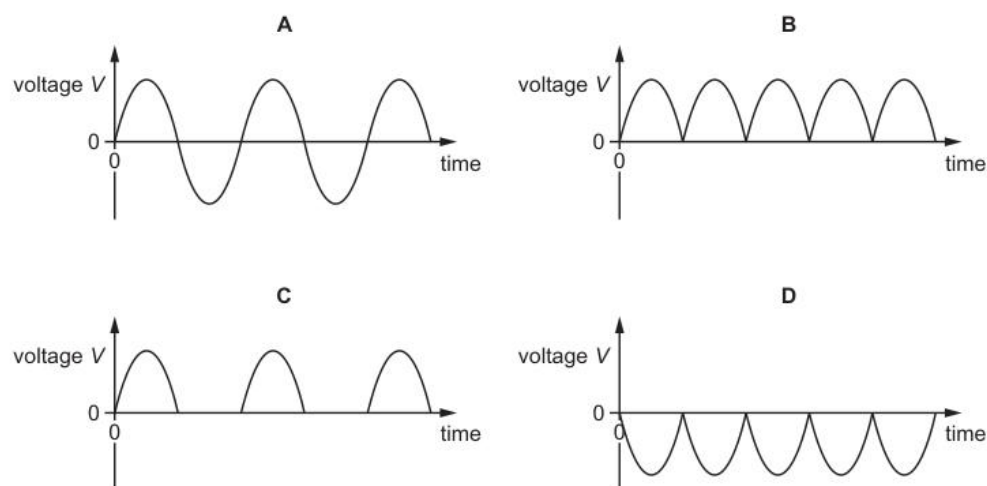
What is the wavelength of the wave?

- A** 0.10 m **B** 0.17 m **C** 6.0 m **D** 10.0 m

- 30 An alternating current (a.c.) power supply is connected in series with a resistor R and a diode.



Which graph shows how the voltage V across the resistor R varies with time?



- 31 A student makes four resistors using different pieces of wire. The wires have different diameters and lengths. All the pieces of wire are made of the same material.

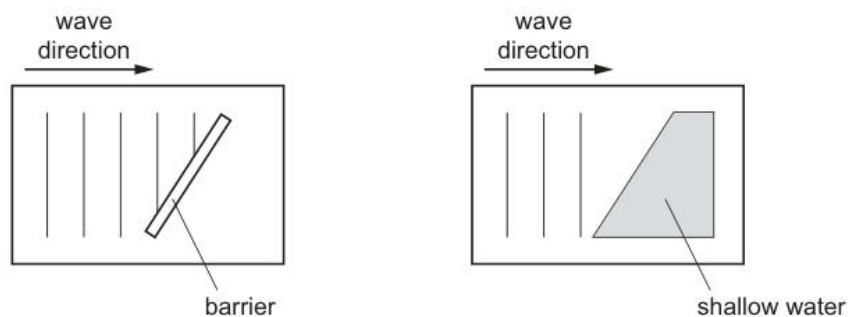
Which piece of wire will make the resistor with the largest resistance?

	diameter / mm	length / cm
A	0.8	10
B	0.8	17
C	2.0	10
D	2.0	17

Mark scheme — Answer: C [1]

Question 18

18 The diagrams show two sets of wavefronts in a ripple tank.



A student makes two statements about the waves.

- 1 When the waves reflect from the barrier the direction changes but the wavelength remains the same.
- 2 When the waves refract as they enter the shallow water the direction remains the same, but the wavelength changes.

Which statements are correct?

- A statement 1 and statement 2
- B statement 1 only
- C statement 2 only
- D neither statement 1 nor statement 2

Mark scheme — Answer: B [1]

Oct/Nov 0625/22 MCQ

Question 17

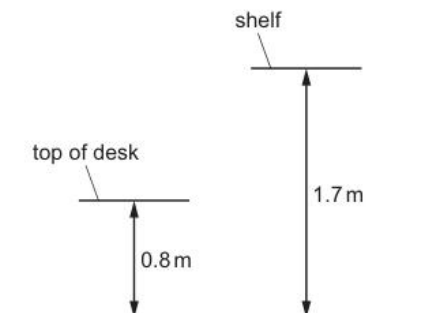
5

- 7 A trolley of mass 4.0 kg travelling with a velocity of 4.0 m/s collides with a trolley of mass 2.0 kg travelling with a velocity of 2.0 m/s in the same direction. After the collision, the velocity of the 4.0 kg trolley is reduced to 3.0 m/s.



What is the velocity v of the 2.0 kg trolley after the collision?

- A 0.25 m/s B 4.0 m/s C 5.0 m/s D 16 m/s
- 8 An object falls towards the Earth's surface.
- What happens to the gravitational potential energy and to the kinetic energy of the object?
- | | gravitational potential energy | kinetic energy |
|---|--------------------------------|----------------|
| A | decreases | decreases |
| B | decreases | increases |
| C | increases | decreases |
| D | increases | increases |
- 9 A boy takes 0.60 s to lift a book of mass 0.60 kg from the top of a desk and place it on a shelf. The top of the desk is 0.80 m above the floor, and the shelf is 1.7 m above the floor. The gravitational field strength is 10 N/kg.



Which power does the boy develop?

- A 0.9 W B 1.7 W C 9.0 W D 17 W

- 15 A scientist is determining the specific latent heat of vaporisation of a liquid.

He puts the liquid in a vacuum flask and heats it with a 100 W heater. The mass of liquid in the vacuum flask when it starts to boil is 300 g. He continues to heat the liquid for a further 12 minutes after which the mass of the remaining liquid is 100 g.

What is the specific latent heat of vaporisation of the liquid?

(Assume that all the thermal energy from the heater is used to vaporise the liquid.)

- A 6000 J/kg
 B 240 000 J/kg
 C 360 000 J/g
 D 360 000 J/kg
- 16 Which piece of equipment is designed to produce a type of electromagnetic wave?
- A electric fire
 B electric generator
 C electric motor
 D electromagnet
- 17 A sound wave travels from air into water.

Which row describes what happens to the frequency and the wavelength of the wave?

	frequency	wavelength
A	decreases	increases
B	decreases	stays the same
C	stays the same	decreases
D	stays the same	increases

Mark scheme — Answer: D [1]

Oct/Nov 0625/23 MCQ

Question 17

- 17 A passing boat causes a floating object on a lake to bob up and down 18 times in 12 s. The wavelength of the wave created by the boat is 48 cm.

What is the velocity of these water waves?

- A 32 cm/s B 72 cm/s C 576 cm/s D 864 cm/s

Mark scheme — Answer: B [1]

Question 18

- 18 Wavefronts are incident on a boundary.

What is needed for the wave to refract at the boundary?

- A a shiny surface at the boundary
B a small gap in the boundary
C different mediums either side of the boundary in which the frequency of the wave is different
D different mediums either side of the boundary in which the speed of the wave is different

Mark scheme — Answer: D [1]

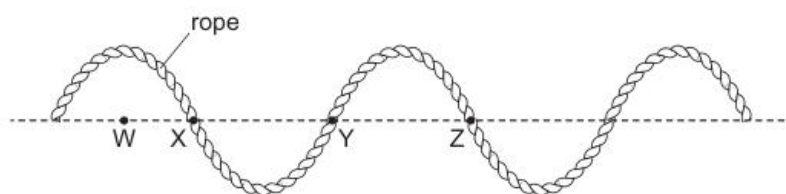
2023

Feb/March 0625/22 MCQ

Question 18

18 A transverse wave moves along a rope.

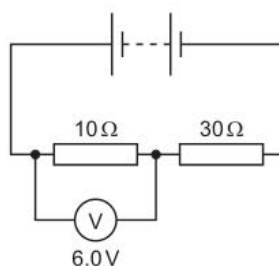
The diagram shows the position of the rope at one particular time.



Which two labelled points are one wavelength apart?

- A** W and X **B** W and Z **C** X and Z **D** Y and Z

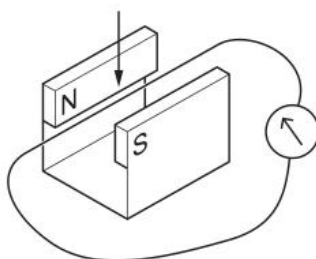
- 29 The diagram shows a circuit which includes two resistors and a battery.



The voltmeter reads 6.0 V.

What is the potential difference across the 30 Ω resistor?

- A 2.0 V B 6.0 V C 18 V D 24 V
- 30 A wire is moved down in a direction perpendicular to the magnetic field.



Three changes are suggested.

- 1 The speed of the movement of the wire is increased.
- 2 The magnetic field strength is decreased.
- 3 The direction of the magnetic field is reversed.

Which changes increase the electromotive force (e.m.f.) induced in the wire?

- A 1 and 3 B 1 only C 2 and 3 D 3 only

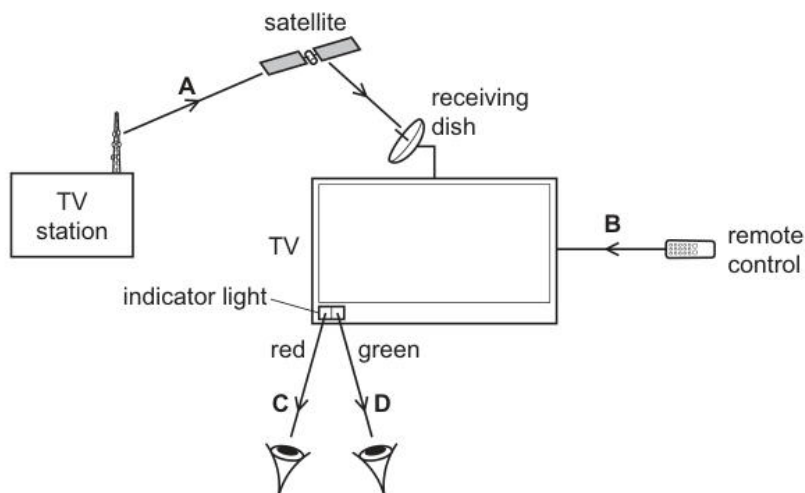
Question 22

22 A television (TV) station transmits a signal to a television receiving dish.

The television has an on/off indicator light.

The television is switched on by a remote control which changes the indicator light from red to green.

Which electromagnetic wave used in these actions has the longest wavelength?



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Mark scheme — Answer: A [1]

May/June 0625/21 MCQ

Question 21

21 Visible light has wavelengths in the range 4.0×10^{-7} m to 7.0×10^{-7} m.

What is the range of the frequencies of visible light?

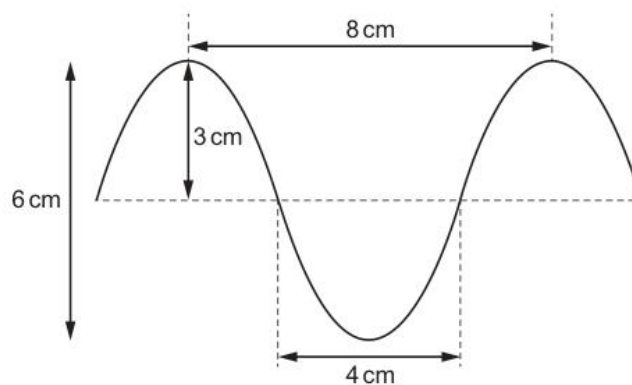
- A** 0.12 Hz to 0.21 Hz
- B** 120 Hz to 210 Hz
- C** 4.3×10^{11} Hz to 7.5×10^{11} Hz
- D** 4.3×10^{14} Hz to 7.5×10^{14} Hz

Mark scheme — Answer: D [1]

May/June 0625/22 MCQ

Question 17

17 The diagram shows a wave.



What are the amplitude and the wavelength of this wave?

	amplitude / cm	wavelength / cm
A	3	4
B	3	8
C	6	4
D	6	8

Mark scheme — Answer: B [1]

Question 21

21 A radio station broadcasts a signal with a frequency of 89 MHz.

What is the wavelength of this signal?

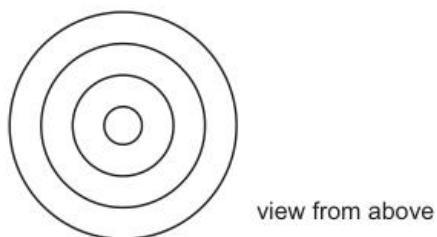
- A** 3.7 μm **B** 3.4 m **C** 3.7 km **D** 3.4 Mm

Mark scheme — Answer: B [1]

May/June 0625/23 MCQ

Question 17

- 17** A drop of water from a tap falls onto the surface of some water of constant depth.



Water waves spread out on the surface of the water.

Which statement is correct?

- A** The waves are longitudinal and travel at the same speed in all directions.
- B** The waves are longitudinal and travel more quickly in one direction than in others.
- C** The waves are transverse and travel at the same speed in all directions.
- D** The waves are transverse and travel more quickly in one direction than in others.

31 Which component forms part of a d.c. motor but **not** a simple moving coil a.c. generator?

- A the coil
- B the brushes
- C the magnet
- D the split-ring commutator

32 A transformer has 5500 turns on the primary coil and 500 turns on the secondary coil.

The output of the secondary coil is 110 V a.c. and is connected to a heater. The transformer is 100% efficient.

The heater produces a power of 132 W.

What is the current in the primary coil?

- A 0.11 A B 0.12 A C 11 A D 12 A

33 The scattering of alpha-particles from a thin gold foil produces the following observations.

- Most of the alpha-particles pass through the foil.
- Most of the alpha-particles are virtually undeflected.
- A small fraction of the alpha-particles are deflected through large angles.
- A very small fraction of the alpha-particles bounce back from the foil.

Which conclusion does **not** follow from these observations?

- A Most of the mass of the gold atom is in its nucleus.
- B Most of the atom is empty space.
- C The nucleus consists of protons and neutrons.
- D The nucleus must be charged.

34 A nuclide of chlorine has the symbol shown.



What is the nucleon number of this nuclide of chlorine?

- A 17 B 18 C 35 D 52

Mark scheme — Answer: C [1]

May/June 5054/21 Theory

Question 7

- 7 A water wave in a ripple tank diffracts as it passes through the gap in a barrier.

Fig. 7.1 shows a drawing made by a student of the crests in the pattern.

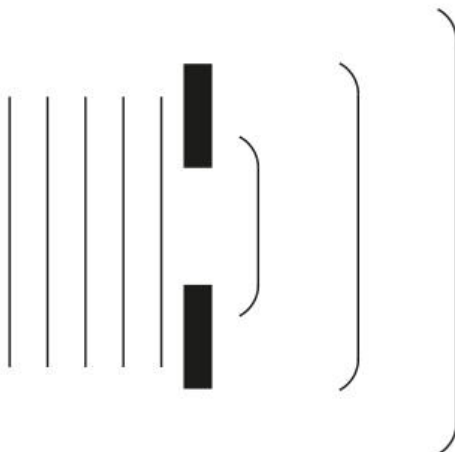


Fig. 7.1

- (a) State **one** way in which the student's drawing of the crests is wrong.

.....
 [1]

- (b) The gap in the barrier is now made smaller than the wavelength, as shown in Fig. 7.2.

Complete Fig. 7.2 with at least three crests to show the new diffraction pattern.



Fig. 7.2

[2]

- (c) In one part of the ripple tank, the water wave has a speed of 6.0 cm/s and a wavelength of 2.0 cm.

The wave then passes into a shallower region of the tank.

The speed of the wave in the shallow region is 4.0 cm/s.

- (i) Define the term 'wavelength'.

.....
 [1]

- (ii) Calculate the frequency of the wave.

frequency = Hz [2]

- (iii) Calculate the wavelength of the wave in the shallow part of the tank.

wavelength = cm [1]

[Total: 7]

Mark scheme — 7(a) wavelength has altered / too large or crests in shadow should be bent / quarter circular above or below gap or vertical lines too straight on two crests B1 7(b) at least 3 semicircles centered on the gap B1 wavelength constant and the same on right and left of gap and some bending B1 7(c)(i) distance between two (consecutive) identical points such as two (consecutive) crests B1 7(c)(ii) $v = f \lambda$ numerical or algebraic e.g. $f = 6 / 2$ (Hz) C1 3(0) Hz) A1 7(c)(iii) 1.3 (cm) B1 5054/21 Question Answer Marks

Question 5

5 Water waves are transverse waves.

(a) Underline **two** other examples of transverse waves.

seismic P-waves seismic S-waves sound X-rays [1]

(b) Fig. 5.1 shows a wooden bar and a glass block in a ripple tank. The depth of water in the tank is less than the height of the glass block.

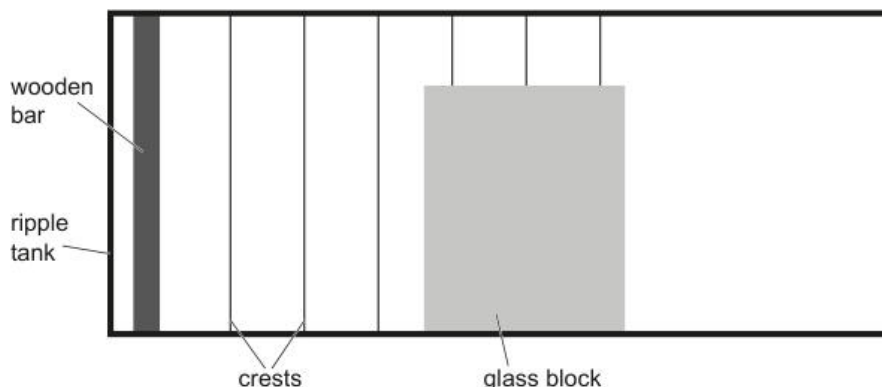


Fig. 5.1 (not to scale)

The wooden bar moves up and down once every 0.15 s to create the crests.

(i) The speed of the water wave is 27 cm/s.

Calculate the frequency and the wavelength of the wave.

frequency = Hz

wavelength = cm
[3]

(ii) The wave diffracts at the right-hand edge of the glass block.

On Fig. 5.1 draw **two** crests after they pass the glass block to show the diffraction. [2]

(iii) Describe how a wave with a smaller wavelength is made with the wooden bar.

.....
..... [1]

(iv) Describe how a decrease in wavelength affects the diffraction.

.....
..... [1]

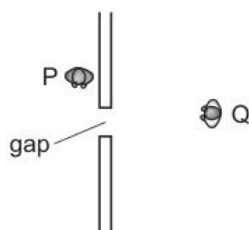
[Total: 8]

[Turn over]

Mark scheme — 5(a) seismic S-waves and X-rays B1 5(b)(i) (wavelength =) v / f in any form or (distance =) vt or $f = 1 / T$ or $1 / 0.15$ C1 frequency 6.7 (Hz) A1 wavelength 4.0 (cm) A1 5(b)(ii) straight wavefronts above block to the right with same wavelength B1 correct diffraction – wavefronts quarter circles with centre at top corner B1 5(b)(iii) increase frequency of movement of (wooden) bar B1 5054/22 Question Answer Marks 5(b)(iv) less diffraction or less bending B1 Question Answer Marks

Question 17

- 17 Two men, P and Q, stand close to a gap in a wall, as shown. Man P cannot see man Q but man P can hear man Q speaking.



Which statement explains this?

- A Light waves do not diffract at all because they are electromagnetic waves.
- B Light waves have a range of frequencies but sound has just one frequency.
- C Sound waves are of a higher frequency than light waves.
- D Sound waves diffract a lot because their wavelength is a similar size to the width of the gap.

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[Turn over

Mark scheme — Answer: D [1]

Question 18

- 18 Which quantities relating to a wave on the surface of water can **both** be measured in metres?
- A amplitude and frequency
 - B amplitude and wavelength
 - C amplitude and wave speed
 - D frequency and wavelength

Mark scheme — Answer: B [1]

Question 22

- 22 A radio transmitter broadcasts at a frequency of 200 kHz.

What is the wavelength of these radio waves?

- A 6.7×10^{-4} m B 1.5 m C 1.5×10^3 m D 1.5×10^6 m

Mark scheme — Answer: C [1]

Oct/Nov 0625/22 MCQ

Question 17

17 A student writes down some facts about two transverse waves.

Wave 1 has a frequency f and a velocity v .

Wave 2 has four times the frequency of wave 1 and is travelling at a velocity of $2v$.

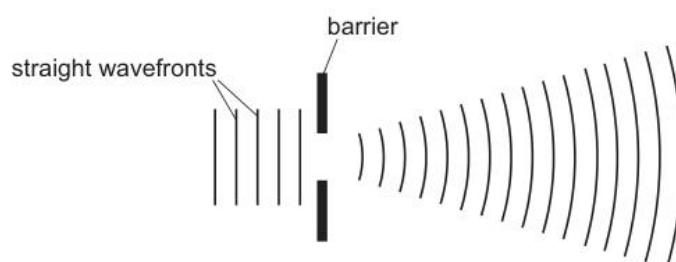
What is the wavelength of wave 2 in terms of f and v ?

- A $\frac{2f}{v}$ B $8fv$ C $\frac{2v}{f}$ D $\frac{v}{2f}$

Mark scheme — Answer: D [1]

Question 18

18 Straight wavefronts on the surface of a ripple tank approach a gap in a barrier. The diagram shows how the wavefronts change shape as they pass through the gap.



What is the name of this effect?

- A diffraction
B propagation
C reflection
D refraction

Mark scheme — Answer: A [1]

Question 22

22 The frequency of the microwaves used in a microwave oven is 2400 MHz.

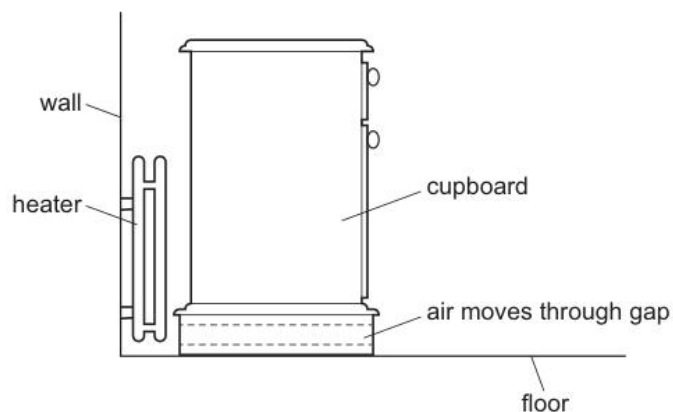
What is the wavelength of these microwaves?

- A 0.125 m B 8.00 m C 125 m D 7200 m

Mark scheme — Answer: A [1]

Question 16

- 16 A cupboard is placed in front of a heater. Air can move through a gap under the cupboard.



Which row describes the temperature and the direction of movement of the air in the gap?

	air temperature	air direction
A	cool	away from the heater
B	cool	towards the heater
C	warm	away from the heater
D	warm	towards the heater

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Mark scheme — Answer: B [1]

Question 17

- 17 Light diffracts when it enters a telescope. This causes the image to blur slightly. The amount of diffraction depends on the diameter of the hole through which the light enters the telescope and the wavelength of the light.

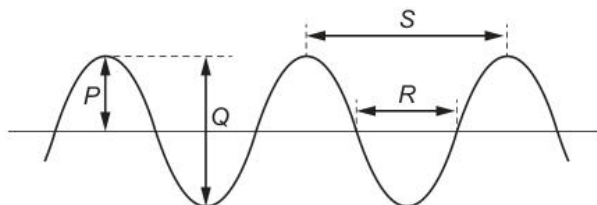
Which combination for diameter and wavelength will result in the sharpest image (least blurring)?

	diameter	wavelength
A	large	long
B	large	short
C	small	long
D	small	short

Mark scheme — Answer: B [1]

Question 18

18 The diagram shows a transverse wave.



Which row identifies the amplitude and the wavelength of the wave?

	amplitude	wavelength
A	<i>P</i>	<i>R</i>
B	<i>P</i>	<i>S</i>
C	<i>Q</i>	<i>R</i>
D	<i>Q</i>	<i>S</i>

Mark scheme — Answer: B [1]

Oct/Nov 5054/21 Theory

Question 5

5 Most of the energy emitted by the Sun is in three regions of the electromagnetic spectrum.

(a) The name of one of these three regions is the ultraviolet region.

(i) State the names of all three regions in order of increasing frequency.

smallest frequency

intermediate frequency

largest frequency [2]

(ii) Some of the ultraviolet radiation emitted by the Sun reaches the surface of the Earth.

Give one possible damaging effect of ultraviolet radiation on the human body and the property of ultraviolet radiation that causes this damage.

damaging effect

.....

property

..... [2]

(b) Electromagnetic radiation is a transverse wave.

Describe the difference between a transverse wave and a longitudinal wave.

.....

.....

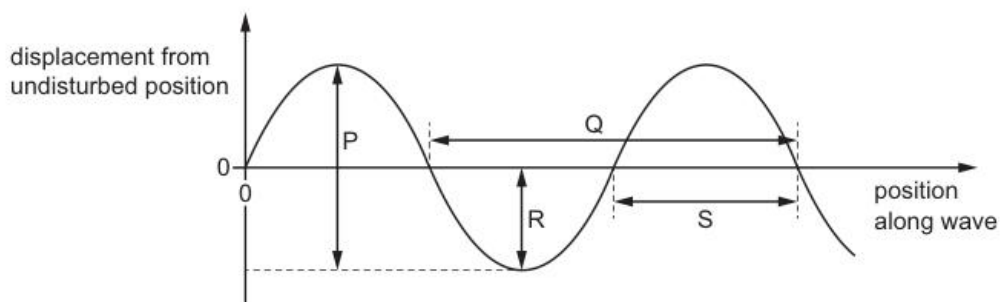
..... [2]

[Total: 6]

Mark scheme — 5(a)(i) infrared (radiation) and visible (light) and ultraviolet (radiation) B1 infrared (radiation) and visible (light) and ultraviolet (radiation) and in this order B1 5(a)(ii) damaging effect: (skin) cancer / cataracts B1 property: it / ultraviolet radiation is ionising B1 5054/21 Question Answer Marks 5(b) in a transverse wave, the vibration direction is perpendicular / at right angles to the propagation direction B1 in a longitudinal wave, the vibration direction is parallel to the propagation direction B1 Question Answer Marks

Question 16

16 The diagram shows a wave.



Which dimensions describe the properties of the wave?

- A** P = amplitude, S = speed
- B** P = speed, Q = wavelength
- C** R = amplitude, Q = wavelength
- D** R = amplitude, S = wavelength

40 The table gives data about four planets in our Solar System.

	Uranus	Earth	planet X	planet Y
mean surface temperature / °C	−195	15	−65	−140

What are planets X and Y?

	planet X	planet Y
A	Mars	Saturn
B	Mars	Neptune
C	Mercury	Saturn
D	Mercury	Neptune

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Mark scheme — Answer: C [1]

May/June 0625/21 MCQ

Question 16

- 16** Which statement about waves is correct?
- A** Waves do not transfer either energy or matter.
 - B** Waves transfer both energy and matter.
 - C** Waves transfer energy without transferring matter.
 - D** Waves transfer matter without transferring energy.

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Mark scheme — Answer: C [1]

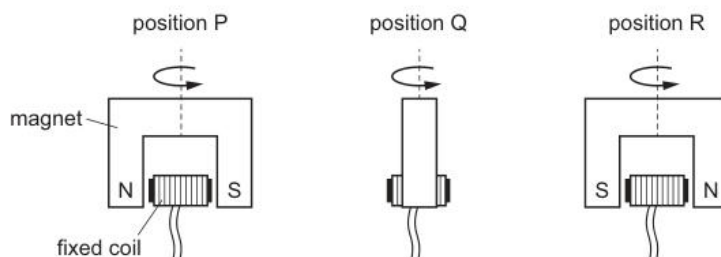
May/June 0625/22 MCQ

Question 17

- 17** In which type of wave is the direction of vibration parallel to the direction of travel?
- A** electromagnetic waves
 - B** seismic P-waves
 - C** seismic S-waves
 - D** water waves

- 31 The diagram shows a simple a.c. generator. The magnet rotates around a fixed coil wound round a fixed iron core. The electromotive force (e.m.f.) output is a sine wave, a series of crests, zeros and troughs.

The magnet is shown in three positions, P, Q and R, during its rotation. The magnetic field in the coil is zero at Q but changing most rapidly.



Which statement about the induced e.m.f. in the coil is correct?

- A The magnitude of the e.m.f. at position Q is larger than that at position P.
 - B The magnitude of the e.m.f. at position Q is zero.
 - C The magnitude of the e.m.f. at position Q is smaller than at position R, but not zero.
 - D The magnitude of the e.m.f. at position R is larger than at position P.
- 32 A 24 V power supply and a 100% efficient transformer are used to light a 6.0 V, 12 W lamp.
- What is the current in the power supply?
- A 0.50 A
 - B 2.0 A
 - C 3.0 A
 - D 8.0 A
- 33 The scattering of particles by a thin gold foil provided scientists with evidence for the nuclear atom.
- Which particles were scattered by the gold nuclei in the thin foil?
- A α -particles
 - B β -particles
 - C neutrons
 - D protons
- 34 A nucleus X has 17 protons and 18 neutrons.
- Which notation is correct for this nucleus?

- A ${}^{17}_{18}\text{X}$
- B ${}^{17}_{35}\text{X}$
- C ${}^{18}_{17}\text{X}$
- D ${}^{35}_{17}\text{X}$

Question 18

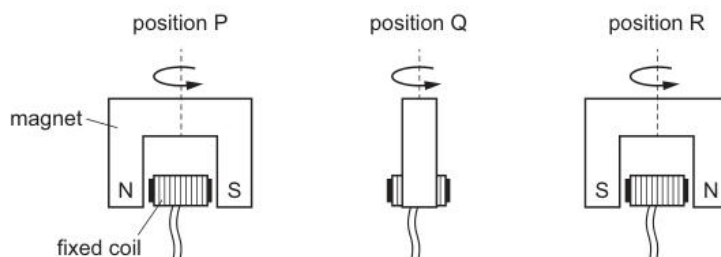
18 Waves in a ripple tank are diffracted as they pass through a narrow gap.

What can be done to make the spreading due to diffraction greater?

- A** Decrease the frequency of the waves and keep the speed constant.
- B** Decrease the speed of the waves and keep the frequency constant.
- C** Increase the frequency of the waves and keep the speed constant.
- D** Increase the frequency of the waves and decrease the speed.

- 31 The diagram shows a simple a.c. generator. The magnet rotates around a fixed coil wound round a fixed iron core. The electromotive force (e.m.f.) output is a sine wave, a series of crests, zeros and troughs.

The magnet is shown in three positions, P, Q and R, during its rotation. The magnetic field in the coil is zero at Q but changing most rapidly.



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- What is the current in the power supply?
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 - C 3.0 A
 - D 8.0 A
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 - B β -particles
 - C neutrons
 - D protons
- 34 A nucleus X has 17 protons and 18 neutrons.
- Which notation is correct for this nucleus?

- A ${}^{17}_{18}\text{X}$
- B ${}^{17}_{35}\text{X}$
- C ${}^{18}_{17}\text{X}$
- D ${}^{35}_{17}\text{X}$

Mark scheme — Answer: A [1]

May/June 0625/23 MCQ

Question 17

17 A seismic wave has a frequency of 5.0 Hz and a speed of 10 km/s.

What is the wavelength of the wave?

- A** 0.20 km **B** 0.50 km **C** 2.0 km **D** 50 km

Mark scheme — Answer: C [1]

Question 18

18 Sound waves will diffract when going through a gap or passing the edge of a barrier.

Which size of wavelength gives the most diffraction in each case?

	passing through a gap	passing the edge of a barrier
A	small wavelength	small wavelength
B	small wavelength	large wavelength
C	wavelength similar to gap size	small wavelength
D	wavelength similar to gap size	large wavelength

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[Turn over

Mark scheme — Answer: D [1]

Question 22

22 Which colours of visible light are in the correct order of increasing wavelength?

- A** red → yellow → blue
B orange → green → violet
C indigo → green → red
D blue → green → violet

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Mark scheme — Answer: C [1]

May/June 5054/21 Theory

Question 4

2

- 1 A ball is released from rest at point A and moves along a smooth track ABCDE as shown in Fig. 1.1. The ball is shown at point A and as it passes point B.

The ball is always in contact with the track and air resistance is negligible.

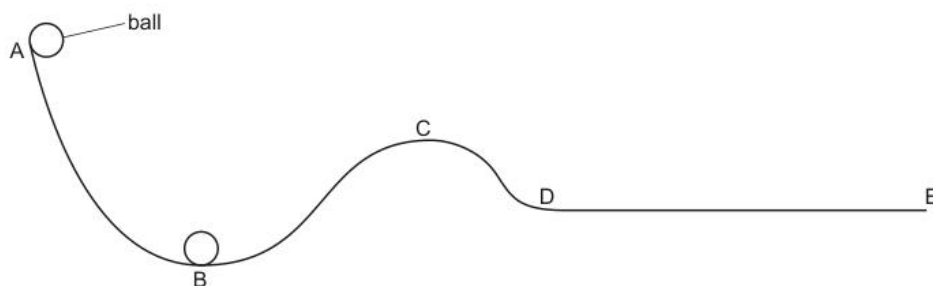


Fig. 1.1

Fig. 1.2 shows the distance–time graph for the ball as it moves from A to E.

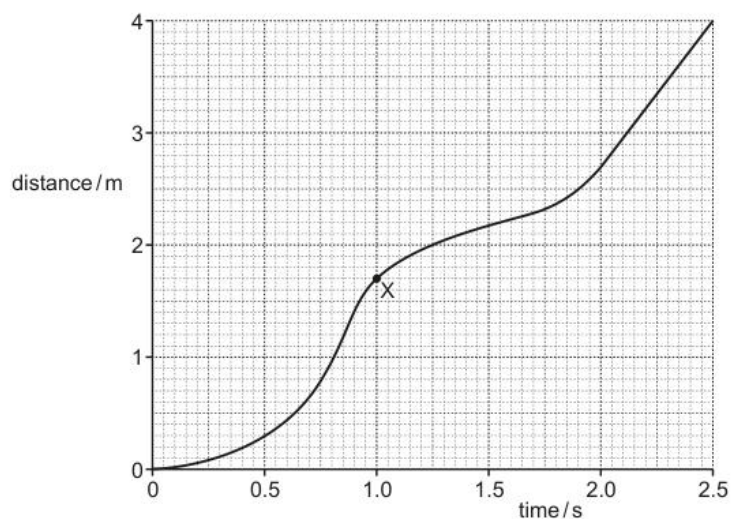


Fig. 1.2

(a) (i) On Fig. 1.2, mark:

- with the letter P **one** point where the ball accelerates
- with the letter Q **one** point where the ball has constant speed.

[2]

- (ii) Determine the speed of the ball at point X on Fig. 1.2.

Show your working.

speed = m/s [3]

- (b) The speeds of the ball at A, B, C and D are v_A , v_B , v_C and v_D respectively.

Arrange these four speeds from slowest to fastest.

slowest —————→ fastest

--	--	--	--

[1]

- (c) Work is done to transfer energy between energy stores as the ball moves from A to B.

Name the force involved in the work done and describe the energy transfer.

.....

.....

.....

..... [3]

- (d) The track at point B is circular in shape.

On Fig. 1.1, draw an arrow to show the direction of the resultant force on the ball at point B.

[1]

[Total: 10]

- 2 A small aircraft takes off from the horizontal deck of a ship.

Before taking off, the aircraft is held in place by a holdback bar.

When the holdback bar is released, the aircraft is pulled along the deck by a steam-powered piston as shown in Fig. 2.1.

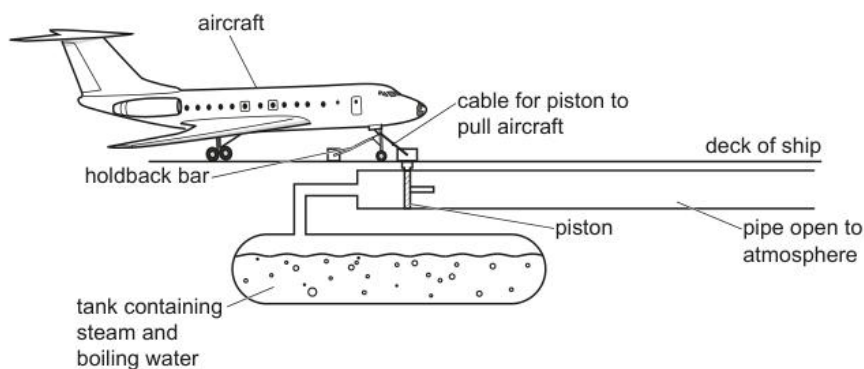


Fig. 2.1 (not to scale)

The steam exerts a high pressure on the piston.

- (a) (i) Explain, using ideas about particles, how the steam creates a pressure on the piston.

.....

.....

.....

..... [2]

- (ii) Explain why the pressure on the piston increases as the temperature of the steam increases.

.....

.....

.....

..... [2]

- (b) When the pressure is high enough, the holdback bar is released. The steam pushes the piston along the pipe shown in Fig. 2.1.

The piston has a cross-sectional area of 0.30 m^2 .

The pressure of the steam in the tank is $2.1 \times 10^6 \text{ Pa}$.

Atmospheric pressure is $1.0 \times 10^5 \text{ Pa}$.

Determine the resultant force on the piston caused by the pressure difference.

Show your working.

force = N [2]

- (c) The force calculated in (b) causes the aircraft to accelerate to a maximum speed of 28 m/s from rest.

The mass of the aircraft is $3.0 \times 10^4 \text{ kg}$.

- (i) Calculate the momentum of the aircraft at the maximum speed.

momentum = kg m/s [2]

- (ii) Calculate the time taken for the aircraft to reach the maximum speed.

time = s [2]

[Total: 10]

- 3 A pan containing ice at -15°C is placed on a gas heater as shown in Fig. 3.1.



Fig. 3.1

- (a) Thermal conduction occurs in the metal from which the pan is made.

By referring to particles, describe the process of thermal conduction in a metal.

.....

.....

..... [2]

- (b) Fig. 3.2 shows how the temperature of the contents of the pan varies with time.

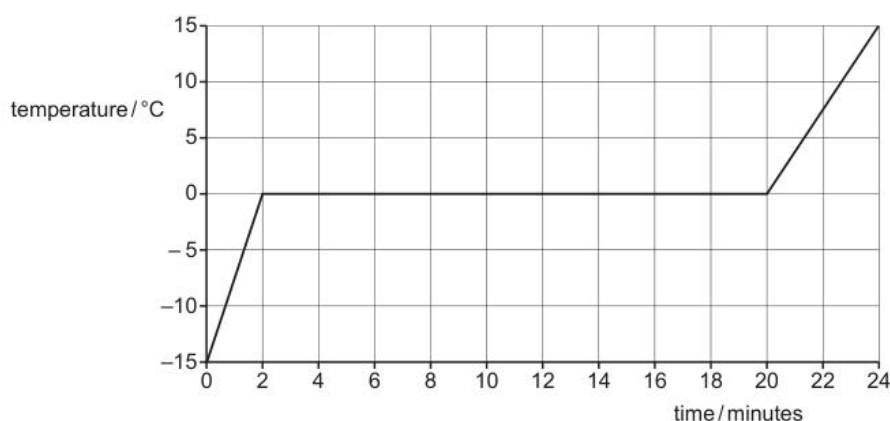


Fig. 3.2

Mark scheme — 4(a)(i) number of (complete) waves (passing a point) per second B1 4(a)(ii) ($V =$) $f\lambda$ in any form C1 330 m / s A1 4(a)(iii) ANY one from cleaning, prenatal and other medical scanning, sonar, calculation of depth or distance B1 4(b)(i) longitudinal wave - oscillation backwards and forwards / in direction of travel of wave B1 transverse wave - oscillation at right angles to (direction of travel of) wave B1 diagram of each type B1 4(b)(ii) earthquake (waves) or other correct source which is a single source that generates both types of wave, e.g. lightning B1 4(c) long wavelengths bend around hill and short wavelengths do not bend B1 (due to) diffraction B1 Question Answer Marks

Question 17

17 Water waves are reflected at a plane surface.

Which property of the waves is changed by the reflection?

- A** direction of propagation
- B** frequency
- C** speed
- D** wavelength

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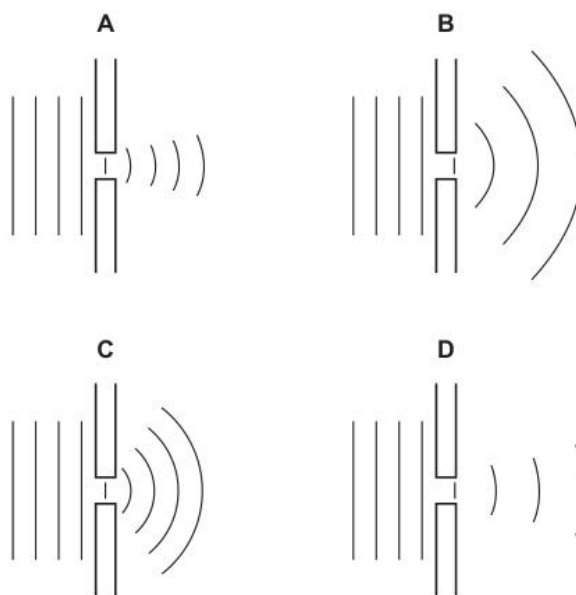
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Mark scheme — Answer: A [1]

Question 19

19 Plane water waves approach a narrow gap in a barrier.

Which diagram shows the diffraction pattern that would occur?



Mark scheme — Answer: C [1]

Question 22

22 Radio waves are electromagnetic waves.

What is the wavelength of a radio wave of frequency $2.0 \times 10^5 \text{ Hz}$?

- A 0.00067 m B 0.0015 m C 670 m D 1500 m

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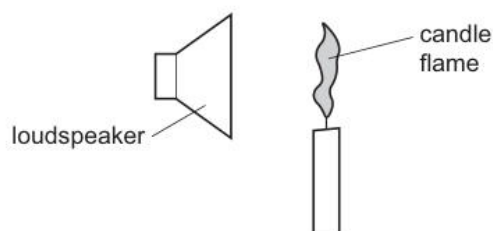
[Turn over

Mark scheme — Answer: D [1]

Question 23

23 A candle flame is placed in front of a loudspeaker.

The loudspeaker produces a sound wave that causes air particles to vibrate. The vibrating air particles make the candle flame vibrate in the same direction as the air particles.



Which row shows the direction of vibration of the candle flame and the nature of sound waves?

	direction of vibration	nature of sound waves
A		longitudinal
B		transverse
C		longitudinal
D		transverse

Mark scheme — Answer: C [1]

Oct/Nov 0625/22 MCQ

Question 2

- 2 A boy takes 30 minutes to cycle a distance of 8.0 km. He then walks a further distance of 2.0 km in 15 minutes.

What is his average speed?

- A 4.5 km/h B 5.6 km/h C 12 km/h D 13 km/h

7

- 16 Which statement about the transfer of thermal energy is correct?
- A All metals conduct thermal energy equally well.
B Convection can only occur in solids or liquids.
C Convection occurs in liquids because hot liquid is more dense than cold liquid.
D The radiation that transfers thermal energy is a type of electromagnetic radiation.
- 17 What is the number of wavefronts per second that pass a fixed point?
- A the amplitude of the wave
B the frequency of the wave
C the speed of the wave
D the wavelength of the wave

- 18 An earthquake under the ocean can produce a dangerous water wave called a tsunami.

The table gives typical data for a tsunami.

ocean depth	velocity	wavelength
m	km / h	km
7000	948	285
2000	504	151

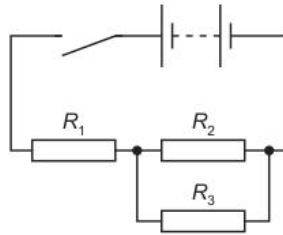
A student suggests three conclusions about the data.

- 1 The deeper the ocean, the faster the wave.
- 2 The longer the wavelength, the faster the wave.
- 3 The faster the wave, the higher the frequency.

Which conclusions are correct?

- A 1, 2 and 3 B 1 and 2 only C 1 and 3 only D 2 and 3 only

- 29 The diagram shows an arrangement of resistors in series and parallel.



Which equation is used to calculate the combined resistance of the resistors?

- A $R_1 + \frac{R_2 + R_3}{R_2 \times R_3}$
- B $R_1 + \frac{R_2 \times R_3}{R_2 + R_3}$
- C $R_1 + \frac{1}{R_2 + R_3}$
- D $R_1 \times \frac{R_2 + R_3}{R_2 \times R_3}$
- 30 An electrical engineer has made a pair of components P and Q.
- The resistance of P decreases as the temperature rises. The resistance of Q increases as the temperature rises.
- The total resistance of P and Q in series remains constant when the temperature changes. The two components are connected in series with a power source supplying a constant current.
- Which statement is correct?
- A If a temperature change causes the potential difference (p.d.) across P to double, the p.d. across Q will always halve.
- B The difference between the p.d. across P and the p.d. across Q does not change with temperature.
- C The p.d. across component P increases as the temperature rises.
- D The sum of the p.d. across P and the p.d. across Q does not change with temperature.

Mark scheme — Answer: D [1]

Question 17

- 17 What is the number of wavefronts per second that pass a fixed point?
- A the amplitude of the wave
 - B the frequency of the wave
 - C the speed of the wave
 - D the wavelength of the wave

Mark scheme — Answer: B [1]

Question 18

- 18 An earthquake under the ocean can produce a dangerous water wave called a tsunami.

The table gives typical data for a tsunami.

ocean depth	velocity	wavelength
m	km / h	km
7000	948	285
2000	504	151

A student suggests three conclusions about the data.

- 1 The deeper the ocean, the faster the wave.
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- 3 The faster the wave, the higher the frequency.

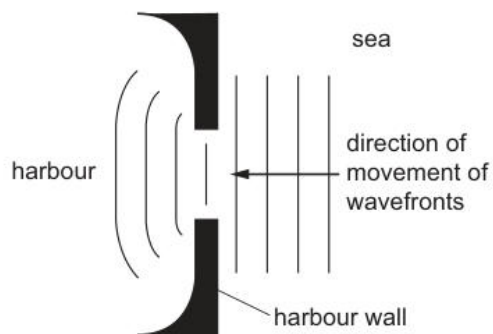
Which conclusions are correct?

- A 1, 2 and 3 B 1 and 2 only C 1 and 3 only D 2 and 3 only

Mark scheme — Answer: B [1]

Question 19

19 The diagram shows what happens to wavefronts at sea when they enter a harbour.



What causes the change in the shape of the wavefronts?

- A** diffraction
- B** dispersion
- C** reflection
- D** refraction

Mark scheme — Answer: A [1]

Oct/Nov 0625/23 MCQ

Question 8

- 8 Useful energy is obtained from different energy resources.

Four examples of energy resources are listed.

- water waves
- geothermal
- wind
- water behind hydroelectric dams

How many of these energy resources depend on radiation from the Sun?

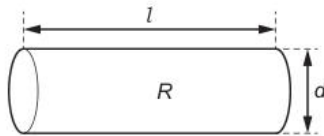
- A** 1 **B** 2 **C** 3 **D** 4

- 26 A battery moves 3.0 C of charge round a complete circuit. The battery transfers 12 J of energy during this process.

What is the electromotive force (e.m.f.) of the cell?

- A 0.25 V B 4.0 V C 12 V D 36 V

- 27 A cylinder of conducting putty has length l , diameter d and resistance R . The putty is now moulded into a cylinder of diameter $2d$ that has the same volume.



By which factor does the resistance of the putty cylinder decrease?

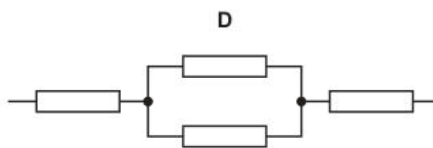
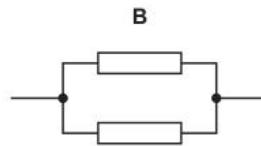
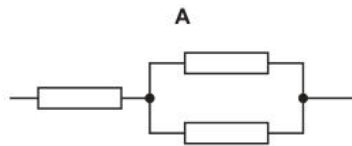
- A 2 B 4 C 8 D 16

- 28 Which statement describes how a diode operates in an electrical circuit?

- A When the temperature increases, its resistance increases.
 B When the current becomes too high, it melts and cuts off the current.
 C It only allows current in one direction.
 D It uses a low-current circuit to switch on a circuit carrying a much larger current.

- 29 A number of identical resistors of resistance $10\ \Omega$ are arranged in circuits.

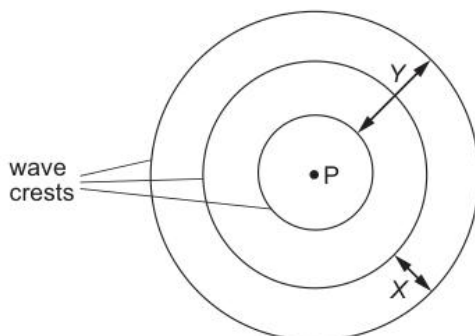
Which arrangement of resistors will give an overall resistance of $25\ \Omega$?



Question 17

17 A vertical stick is dipped up and down in water at P.

In two seconds, three wave crests spread out on the surface of the water.



Which statement is correct?

- A Distance X is the amplitude of the waves.
- B Distance Y is the wavelength of the waves.
- C Each circle represents a wavefront.
- D The frequency of the waves is 3 Hz.

Mark scheme — Answer: C [1]

Question 18

18 Seismic waves from the epicentre of an earthquake take 100 s to reach a point 300 km away. When they arrive, they cause the ground to vibrate with a frequency of 2.0 Hz.

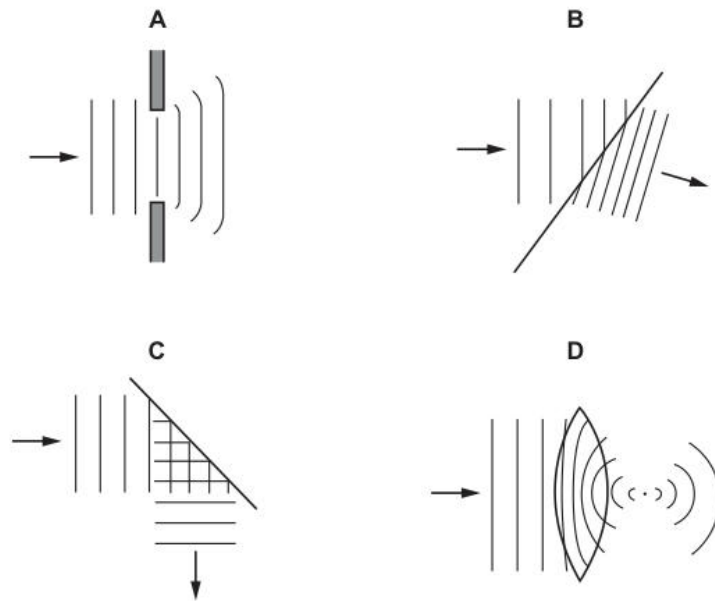
What is the wavelength of these seismic waves?

- A 0.67 km
- B 1.5 km
- C 3.0 km
- D 6.0 km

Mark scheme — Answer: B [1]

Question 19

19 Which diagram illustrates diffraction of water waves?



Mark scheme — Answer: A [1]

Oct/Nov 5054/21 Theory

Question 5

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- 5 Ultrasound is a longitudinal wave which cannot be heard by humans as the frequency of ultrasound is too high.

- (a) Ultrasound waves **cannot** travel in a vacuum.

Explain why.

.....

.....

..... [2]

- (b) Describe how a longitudinal wave differs from a transverse wave.

.....

.....

..... [2]

- (c) An ultrasound transmitter used for a medical scan produces an ultrasound wave of frequency 8.4 MHz.

In soft human tissue, ultrasound travels at 1500 m/s.

- (i) Calculate the wavelength of this ultrasound wave in soft human tissue.

wavelength = m [3]

- (ii) The ultrasound wave passes from the soft tissue into bone where the speed of the ultrasound wave is greater than 1500 m/s.

State what happens to the frequency and to the wavelength of the ultrasound wave as it passes into the bone.

frequency

wavelength [2]

[Total: 9]

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[Turn over]

Mark scheme — 5(a) ultrasound / sound / it is transmitted by vibrating matter / particles B1 no matter / particles in a vacuum / no medium B1 5(b) (in a longitudinal wave,) vibration (of particles) is parallel to the propagation direction B1 in a transverse wave, vibration (of particles) is perpendicular to the propagation direction B1 5(c)(i) (■ =) v / f or $1500 / 8.4$ ■ 106 C1 1.8 ■ 10N C1 1.8 ■ 10–4 (m) A1 5(c)(ii) frequency: does not change B1 wavelength: increases B1 5054/21 Question Answer Marks

Question 19

19 The frequency of a wave is doubled. The speed of the wave does **not** change.

What happens to the wavelength of the wave?

- A It becomes four times as large.
- B It does **not** change.
- C It doubles.
- D It halves.

Mark scheme — Answer: D [1]

Question 20

20 Which statement about ultrasound is correct?

- A It has a higher frequency than audible sound and it is a longitudinal wave.
- B It has a higher frequency than audible sound and it is a transverse wave.
- C It has a lower frequency than audible sound and it is a longitudinal wave.
- D It has a lower frequency than audible sound and it is a transverse wave.

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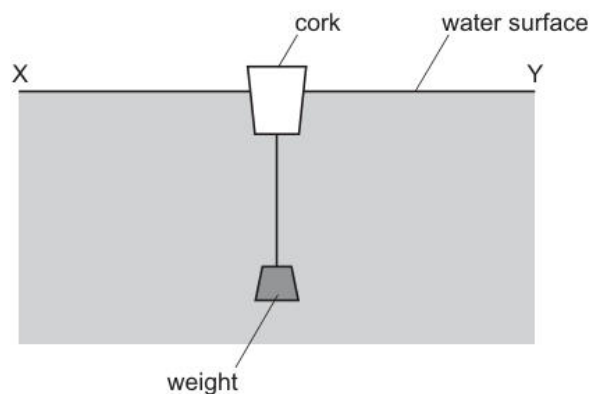
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[Turn over

Mark scheme — Answer: A [1]

Question 17

- 17 The diagram shows a cork with a weight attached so that the cork floats upright in water.



Transverse waves travel across the water from X to Y.

In which direction do the waves make the cork move?

- A → ← right and left
- B ↑ ↓ up and down
- C → only to the right
- D ← only to the left

Mark scheme — Answer: B [1]

Question 21

- 21 A beam of white light is split into a spectrum of seven colours.

Which name is given to this process?

- A diffraction
- B dispersion
- C reflection
- D refraction

Mark scheme — Answer: B [1]

Question 23

- 23** A sound wave travels through the air and strikes a solid block of steel.

Some of the sound is reflected from the steel, and some is transmitted through the steel.

How does the speed and the wavelength of the sound in steel compare to the speed and the wavelength of the sound in air?

	speed in steel	wavelength in steel
A	less than in air	less than in air
B	less than in air	greater than in air
C	greater than in air	less than in air
D	greater than in air	greater than in air

Mark scheme — Answer: D [1]

May/June 0625/22 MCQ**Question 17**

- 17** Which wave will diffract the most when passing through a gap that is 0.85 m wide?

	wave speed m/s	wave frequency / Hz
A	300	280
B	300	400
C	350	280
D	350	400

Mark scheme — Answer: C [1]

Question 19

- 19** Which row describes a seismic S-wave?

	type of wave	direction of vibration
A	longitudinal	parallel to the direction of travel of the wavefront
B	longitudinal	perpendicular to the direction of travel of the wavefront
C	transverse	parallel to the direction of travel of the wavefront
D	transverse	perpendicular to the direction of travel of the wavefront

Mark scheme — Answer: D [1]

May/June 0625/23 MCQ

Question 17

17 Which row correctly describes light waves?

	wave type	direction of vibration
A	longitudinal	parallel to direction of wave travel
B	longitudinal	perpendicular to direction of wave travel
C	transverse	parallel to direction of wave travel
D	transverse	perpendicular to direction of wave travel

Mark scheme — Answer: D [1]

Question 18

18 A narrow beam of light consists of two waves of different wavelengths.

The beam of light hits a sharp edge.

Which effect will be seen beyond the sharp edge?

- A** Both waves continue along the same straight line as before.
- B** Both waves bend sharply and continue along a new straight line.
- C** The wave of longer wavelength bends round the sharp edge more than the wave of shorter wavelength.
- D** The wave of shorter wavelength bends round the sharp edge more than the wave of longer wavelength.

Mark scheme — Answer: C [1]

Question 23

23 What is a compression?

- A** a point in a longitudinal wave where the particles of the medium are close together
- B** a point in a longitudinal wave where the particles of the medium are far apart
- C** a point in a transverse wave where the particles of the medium are close together
- D** a point in a transverse wave where the particles of the medium are far apart

Mark scheme — Answer: A [1]

May/June 5054/21 Theory

Question 5

- 5 (a) Fig. 5.1 is a diagram showing the arrangement of air particles as a longitudinal wave passes through them.



Fig. 5.1

- (i) On Fig. 5.1, mark the centre of a compression with the letter C, and mark the centre of a rarefaction with the letter R. [1]
- (ii) Describe the difference between a compression and a rarefaction.

 [1]
- (b) In a ripple tank, a water wave is produced by a wooden bar moving up and down on the surface of water.
- (i) The wooden bar makes 45 complete oscillations in 1.0 minute.
 Calculate the frequency of the wave produced.

frequency =Hz [1]

- (ii) The frequency of the water wave is increased by moving the wooden bar up and down more quickly.
 State what happens to the speed and what happens to the wavelength of the wave produced.
 speed
 wavelength [2]





(iii) The crests of the water wave move into the shallow region shown in Fig. 5.2.

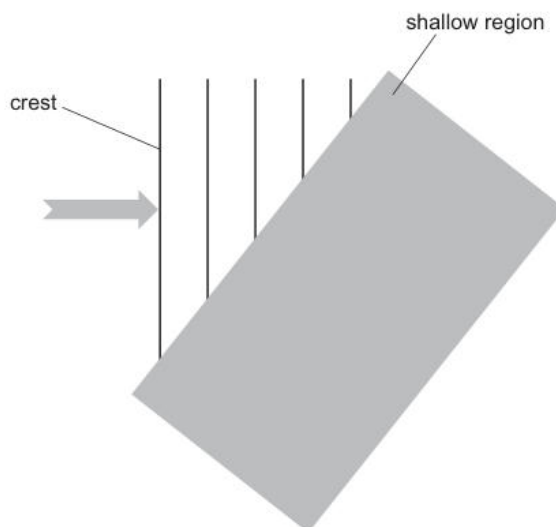


Fig. 5.2

On Fig. 5.2, draw the crests in the shallow region.

[2]

(c) Describe what is meant by the diffraction of a water wave.

.....

.....

..... [1]

[Total: 8]



Mark scheme — 5(a)(i) C and R marked correctly B1 5(a)(ii) compressions are where pressure is high / particles close together / density is high B1 5(b)(i) 0.75 (Hz) B1 5(b)(ii) speed: no change B1 wavelength: decreases B1 5(b)(iii) at least 3 wavefronts in the shallow region parallel to each other and on correct side of normal M1 refracted towards the surface on correct side of surface and join wavefronts shown A1 5(c) spreading out / bending at an edge / obstacle or spreading out after passing through a (small) gap / hole B1 5054/21 Question Answer Marks

Question 4

- 4 A sound wave from a vibrating tuning fork is displayed on the oscilloscope shown in Fig. 4.1. The sound wave is detected using a microphone.

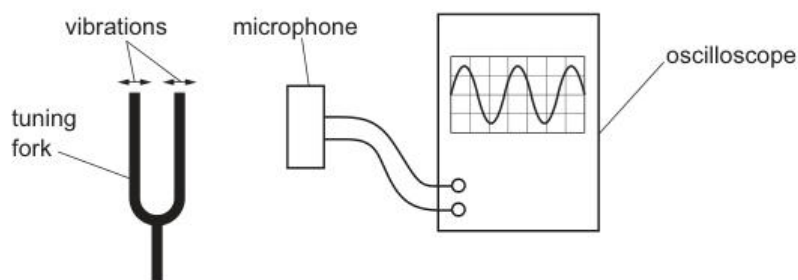


Fig. 4.1

- (a) Describe the motion of the air particles as the sound wave passes from the tuning fork to the microphone.

.....

.....

..... [2]

- (b) Fig. 4.2 shows the trace seen on the oscilloscope when the tuning fork vibrates.

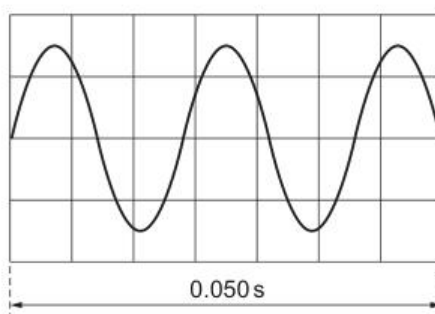


Fig. 4.2

The frequency of the sound is constant.

- (i) State what is meant by 'frequency'.

.....

..... [1]

- (ii) The trace shown on Fig. 4.2 is produced in 0.050 s.

Calculate the frequency of the sound wave.

frequency = Hz [2]





7

- (iii) A student near the tuning fork observes the trace and listens to the sound. He notices that the amplitude of the trace decreases.

State how the sound heard by the student changes.

..... [1]

- (iv) The tuning fork is replaced by a new tuning fork that produces a sound wave of half the frequency. The controls on the oscilloscope are not changed.

On Fig. 4.3, draw the trace obtained with the new tuning fork. The trace obtained with the original tuning fork is shown.



Fig. 4.3

[1]

- (c) Fig. 4.4 shows three adjacent wavefronts of the sound wave as the wavefronts move towards a gap in a wall created by an open door.

Another student, standing at P on the other side of the wall, hears the sound.

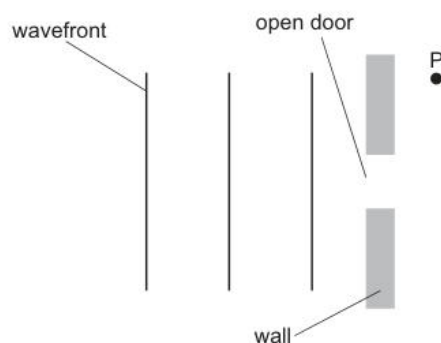


Fig. 4.4

- (i) On Fig. 4.4, draw the three wavefronts after they pass through the gap. [2]

- (ii) State the name of the process involved as the wavefronts pass through the gap and reach P.

..... [1]

[Total: 10]



Mark scheme — 4(a) motion of particles: backwards and forwards / to and fro / closer and further apart B1 (oscillation) in the direction of travel of the wave / to the right / in direction of microphone / longitudinal / horizontal / parallel to vibrations of tuning fork or mention of compressions and rarefactions B1 4(b)(i) number of oscillations per second or number of wave(length)s (passing a point) per second B1 4(b)(ii) 2.5 (waves) seen or (1 wave takes) 0.02 (s) or ($f =$) $1/T$ or (number of) waves / time C1 50 (Hz) A1 4(b)(iii) quieter / less volume B1 4(b)(iv) half frequency shown and at least one complete oscillation seen B1 4(c)(i) (at least) three correctly curved wavefronts centered on gap B1 same wavelength as on left and some curving in correct direction B1 4(c)(ii) diffraction B1 5054/22 Question Answer Marks

Question 1

- 1 A student determines the diameter of a metal ball.

Which apparatus is needed to get an accurate measurement?

- A a ruler and a protractor
- B a ruler only
- C a protractor and a set square
- D a ruler and two rectangular blocks

- 16** Two metal cans are identical, except that one has a shiny silver outer surface and the other has a dull black outer surface. They each have the same mass of hot water at the same temperature sealed inside them. They are both in a vacuum in the darkness of outer space.

How does the temperature of the water in each one change?

- A** The temperature of the water does **not** change.
 - B** The water in the black can cools more slowly than that in the silver can.
 - C** The water in the silver can cools more slowly than that in the black can.
 - D** They both cool down at the same rate.
- 17** A ray of light reflects from a plane mirror.
- The angle between the incident ray and the surface of the mirror is 35° .
- Which statement is correct?
- A** The angle of incidence is 35° .
 - B** The angle of incidence is 55° .
 - C** The angle of reflection is 70° .
 - D** The angle of reflection is 110° .
- 18** Parallel waves in a ripple tank are diffracted as they pass through a gap in a barrier.

Three changes that can be made to this arrangement are listed.

- 1 Decrease the wavelength of the waves.
- 2 Increase the amplitude of the waves.
- 3 Decrease the size of the gap.

Which changes will cause the shape of the diffracted waves to be less curved?

- A** 1, 2 and 3 **B** 1 and 3 only **C** 1 only **D** 2 and 3 only

- 33 $^{239}_{94}\text{Pu}$ is an isotope of plutonium.

The nucleus can absorb a neutron and then split to form two other nuclei.

Two nuclei that form are gold, $^{207}_{79}\text{Au}$, and phosphorus, $^{31}_{15}\text{P}$.

In this reaction, neutrons are also produced.

What is the number of neutrons produced when the nucleus splits?

- A 1 B 2 C 4 D 5

- 34 Three statements about alpha (α) particles and beta (β) particles are listed.

- 1 The magnitude of the charge on an α -particle is twice the charge on a β -particle.
- 2 The mass of an α -particle is large compared with the mass of a β -particle.
- 3 The kinetic energy of an α -particle is small compared to the kinetic energy of a β -particle.

Which statements help to explain the different ionising effects of α -particles and β -particles?

- A 1, 2 and 3 B 1 and 2 only C 1 and 3 only D 2 and 3 only

- 35 A beta-particle is a fast-moving electron.

Which statement explains how beta-particles are emitted from an atom?

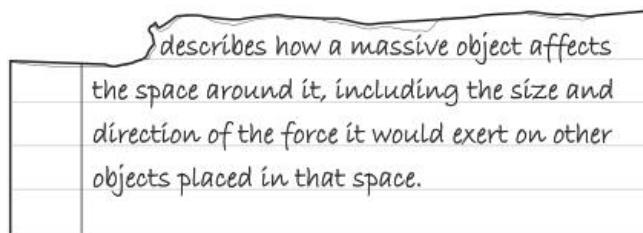
- A An electron is emitted as a beta-particle from an inner electron shell of the atom.
 B An electron is emitted as a beta-particle from an outer electron shell of the atom.
 C A neutron changes into a proton and a beta-particle is emitted from the nucleus.
 D A proton changes into a neutron and a beta-particle is emitted from the nucleus.

- 36 Which properties must a radioisotope have for its radiation to make it suitable to kill bacteria in food?

	half-life	type of radiation emitted
A	less than one minute	γ only
B	several hours	α only
C	several hours	γ only
D	several thousand years	α only

Question 3

- 3 A piece of paper torn out of an exercise book is shown.



What is being described?

- A gravitational charge
- B gravitational field
- C gravitational mass

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[Turn over

Mark scheme — Answer: C [1]

Oct/Nov 0625/22 MCQ

Question 17

17 An earthquake-monitoring station records the arrival of 16 complete wavelengths of an earthquake wave in a time of 20 s.

The speed of the earthquake wave is 6.0 km/s.

What is the wavelength of the earthquake wave?

- A** 1.3×10^{-4} m **B** 2.1×10^{-4} m **C** 4.8×10^3 m **D** 7.5×10^3 m

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[Turn over

Mark scheme — Answer: D [1]

Question 18

18 Which statement is correct?

- A Seismic S-waves are longitudinal.
- B Sound waves are longitudinal.
- C The amplitude is always greater than the wavelength in a longitudinal wave.
- D The vibration is at right-angles to the direction of propagation in a longitudinal wave.

Mark scheme — Answer: B [1]

Question 23

23 Sound travels through air as a series of compressions and rarefactions.

Which statement correctly compares a compression with a rarefaction?

- A In a compression, the wavelength is longer than in a rarefaction.
- B In a compression, the wavelength is shorter than in a rarefaction.
- C In a compression, the density of the air is greater than in a rarefaction.
- D In a compression, the density of the air is lower than in a rarefaction.

Mark scheme — Answer: C [1]

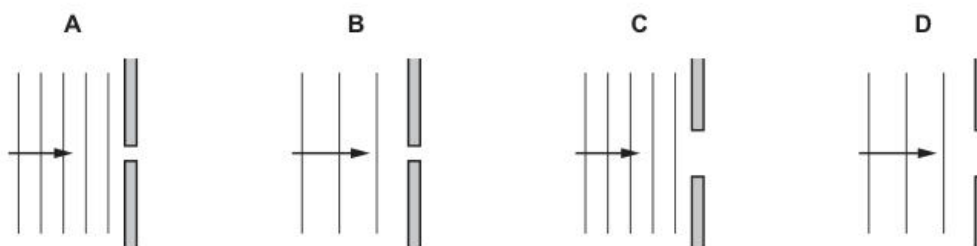
Oct/Nov 0625/23 MCQ

Question 18

18 When water waves pass through a gap, they diffract.

The diagrams show wavefronts approaching a gap.

In which diagram will the diffraction be least?



25 Why does a plastic rod become negatively charged when it is rubbed with a cloth?

- A The rod gains electrons.
- B The rod gains protons.
- C The rod loses electrons.
- D The rod loses protons.

26 There is an electric current in a circuit with a battery. This question refers to the positive and negative terminals of the battery.

Which row correctly describes the directions of the conventional current and the electron flow in the circuit?

	direction of conventional current	direction of electron flow
A	from the negative to the positive terminal	from the negative to the positive terminal
B	from the negative to the positive terminal	from the positive to the negative terminal
C	from the positive to the negative terminal	from the negative to the positive terminal
D	from the positive to the negative terminal	from the positive to the negative terminal

27 Which electrical quantity is defined in terms of the energy supplied by a source in driving charge round a complete circuit?

- A current
- B electromotive force
- C potential difference
- D power

28 A torch has a simple circuit with a 3.0 V battery and a lamp. There is a 20 mA current in the lamp.

How much energy is transferred to the lamp in 5.0 minutes?

- A 0.30 J B 18 J C 60 J D 0.30 kJ